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CliniCause: An LLM-Guided Framework for Deep Proxy-State Prediction and Causal Effect Estimation in Multivariate Irregular ICU Time Series

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תקציר

רשומות רפואיות מיחידות טיפול נמרץ כוללות מדידות רבות הנאספות בזמנים לא אחידים ובתדירות המשתנה בין מטופלים ומשתנים. זמינותן החלקית של המדידות ודפוסי החסר עשויים לשקף הן את מצבו של המטופל והן את תהליך הטיפול. עבודה זו בוחנת מסגרת חישובית העוקבת במפורש אחר מקורות הראיות ומשלבת הפקת ידע קליני וסיבתי בסיוע מודל שפה גדול, הגדרה דטרמיניסטית של מצבי־פרוקסי, חיזוים מתוך סדרות זמן רפואיות לא־סדירות וניתוחים מתוקננים של תמותה באשפוז בהנחיית גרפים סיבתיים. מטרת העבודה היא לבחון את השילוב בין שלבי התהליך ואת המעברים ביניהם, ולא לספק תיקוף קליני, המלצת טיפול או מערכת המוכנה להטמעה.

המסגרת נבחנה בנפרד על נתוני PhysioNet 2012 ועל נתוני MIMIC-III. בשלב התכנון שימש מודל שפה גדול ככלי מסייע להצעת אונטולוגיות של מצבי־פרוקסי, משפחות של כללי החלטה, התייחסויות לדפוסי חסר וגרפים סיבתיים המותאמים לכל מאגר. ההצעות שנבחרו במסגרת הפרויקט קודדו לאחר מכן ככללים וכגרפים דטרמיניסטיים. ארבעה מודלים נלמדים לסדרות זמן — STraTS, GRU, GRU-D ו־TCN — אומנו לחזות את התוויות שהפיקו כללי ההחלטה. האיגוד ששימש בניתוחים הסיבתיים הסופיים כלל את ארבעת מקורות התוויות החזויות האלה ומקור אחד המבוסס על הכללים. מודל השפה לא השתתף כאומד בזמן הריצה. מצבי־הפרוקסי אינם אבחנות שאומתו בבדיקת תיקים רפואיים, והגרפים הסיבתיים לא נלמדו מנתוני המטופלים ולא עברו תיקוף קליני.

באומדני CausalForestDML הראשיים על העוקבות המקוריות התקבל אומדן ממוצע חיובי בכל תשעת מצבי־הפרוקסי שנבחנו ב־MIMIC-III. ב־PhysioNet התקבל אומדן חיובי בתשעה מתוך עשרת המצבים, ואילו אומדן ההלם היה שלילי. CausalForestDML ו־LinearDML הסכימו בכיוון בכל 19 ההשוואות בין מאגר למצב־פרוקסי. שלושת האומדים הסכימו בכיוון ב־18 מתוך 19 ההשוואות; CausalPFN נשמר כאומד חקרני בלבד בשל מגבלות בתיעוד המקור המתודולוגי ובתמיכה האבחונית הזמינה עבורו. ניתוחי ההתאמה שימשו כתמיכה תיאורית באמצעות השוואת תוצאות בזוגות מותאמים, ולא כהוכחה להשפעת התערבות. תוצאות שני המאגרים דווחו בנפרד ולא אוגמו.

תרומתה המרכזית של העבודה היא באינטגרציה שקופה של תהליך המשתרע מהצעות תכנון בסיוע מודל שפה, דרך כללים וגרפים המקודדים בקוד מקור, ועד לחיזוי ולניתוח תצפיתי. האומדנים תלויים בתקפות מצבי־הפרוקסי, בנכונות הגרפים, בסדר הזמנים, באיכות ההתאמה לגורמים מערפלים שנמדדו, בחפיפה בין הקבוצות, בהנחות המודלים ובהיעדר ערבול בלתי נמדד. לפיכך העבודה אינה קובעת שמצבי־הפרוקסי גרמו לתמותה או ששינוים ישנה תוצאות קליניות. תיעוד הראיות משפר את העקיבות המספרית, אך אינו מוכיח שחזור חישובי מלא. נדרשים בהמשך תיקוף קליני, תכנונים סיבתיים חזקים יותר, תיקוף חיצוני והשלמת הפרובננס.

Abstract

Intensive care records contain irregularly timed and incomplete measurements whose missingness may reflect both patient condition and clinical practice. This thesis evaluates a transparent computational workflow linking large-language-model-assisted design, deterministic construction of clinically inspired proxy states, prediction from irregular medical time series, and directed-acyclic-graph-guided observational analyses of in-hospital mortality. Its goal is methodological integration and traceability, not clinical validation, treatment recommendation, or deployment.

Evaluations were conducted separately in the PhysioNet 2012 and MIMIC-III cohorts. At design time, a large language model assisted in proposing dataset-specific proxy-state ontologies, decision-rule families, missingness considerations, and causal graphs; selected proposals were encoded as deterministic rules and graphs. Four learned time-series models—STraTS, GRU, GRU-D, and TCN—were evaluated for predicting rule-derived labels. The archived causal aggregate combined predictions from those four models with one rule-derived source. The language model was not a runtime estimator. The proxy states are not chart-adjudicated diagnoses, and the graphs were neither learned from patient data nor clinically validated.

In primary original-cohort CausalForestDML analyses, mean model-estimated effects were positive for all nine MIMIC-III proxy states and for nine of ten PhysioNet states; PhysioNet shock was negative. CausalForestDML and the secondary LinearDML comparator agreed in direction in all 19 dataset–proxy-state comparisons. All three estimators agreed in 18 of 19; CausalPFN remained exploratory because its methodological provenance and diagnostic support were more limited. Matching supplied descriptive matched-record outcome comparisons, not evidence of intervention effects. Dataset results were reported separately and not pooled.

The contribution is an integrated, inspectable workflow connecting language-model-assisted proposals to source-encoded rules and graphs, then tracing proxy constructs through prediction and observational analysis. Estimates remain conditional on proxy validity, graph and temporal-ordering correctness, adequate control of measured confounding, overlap, model assumptions, and the absence of important unmeasured confounding. They therefore do not establish that proxy states caused mortality or that changing them would improve outcomes. Clinical review, external validation, stronger causal designs, and fuller computational provenance remain necessary.

Keywords

English Keywords

Irregular clinical time series; intensive care; electronic health records; LLM-assisted knowledge elicitation; proxy states; programmatic labeling; weak supervision; multi-label prediction; directed acyclic graphs; observational causal inference; double machine learning; causal forests; sensitivity analysis; MIMIC-III; PhysioNet 2012.

מילות מפתח

סדרות זמן קליניות לא סדירות; טיפול נמרץ; רשומות בריאות אלקטרוניות; חילוץ ידע בסיוע LLM; מצבי פרוקסי; תיוג תכנותי; פיקוח חלש; חיזוי רב-תוויתי; גרפים מכוונים חסרי מעגלים; הסקה סיבתית תצפיתית; למידת מכונה כפולה; יערות סיבתיים; ניתוח רגישות; MIMIC-III; PhysioNet 2012.

Contents

Abbreviations

Abbreviation	Meaning in this thesis
ICU	Intensive Care Unit.
EHR	Electronic Health Record.
DAG	Directed Acyclic Graph.
LLM	Large Language Model; used in this thesis for design-time knowledge elicitation, not runtime patient-level inference.
STraTS	Self-Supervised Transformer for Sparse and Irregularly Sampled Multivariate Clinical Time Series.
GRU	Gated Recurrent Unit.
GRU-D	Gated Recurrent Unit with Decay.
TCN	Temporal Convolutional Network.
DML	Double Machine Learning.
CATE	Conditional Average Treatment Effect; reported here only as a model-estimated quantity under the stated assumptions.
HTE	Heterogeneous Treatment Effect.
AUROC	Area Under the Receiver Operating Characteristic Curve.
AUPRC	Area Under the Precision–Recall Curve.
minRP	The maximum over thresholds of the minimum of precision and recall.
SOFA	Sequential Organ Failure Assessment.
ARDS	Acute Respiratory Distress Syndrome.
CausalPFN	Exploratory model name used in the project; its primary methodological source and expansion are unresolved.

Notation and Symbols

Symbol	Meaning in this thesis
i	Index of a normalized analysis record.
n_i	Number of observed measurement events for record i.
t_{ij}	Elapsed time in minutes for event j of record i.
v_{ij}	Observed variable name for event j of record i.
x_{ij}	Numeric value recorded for event j of record i.
$\mathcal{V}$	Set of observed measurement-variable names.
$\mathcal{O}_i$	Long-format measurement-event object for record i.
L_{ik}^{rule}	Rule-derived binary proxy label for record i and proxy state k.
$\hat{p}_{ik}^{(m)}$	Probability for proxy state k exported by model m for record i.
$\hat{L}_{ik}^{(m)}$	Binary predicted proxy label corresponding to $\hat{p}_{ik}^{(m)}$.
$\widetilde{L}_{ik}$	Aggregated predicted proxy-label representation; in the archived final causal runs, deterministic voting combined one rule-derived source with four aligned binary model-prediction sources.
A_i	Selected binary proxy-state exposure for record i.
Y_i	In-hospital mortality outcome for record i.
W_i	Observed adjustment variables selected from the project-specified DAG and available columns.
X_i	Effect-modifier features used to describe heterogeneity.
$Y_i(a)$	Conceptual potential outcome under exposure value a; its interpretation requires the assumptions stated in the methods.

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Chapter 1

Introduction

1.1 Motivation: Irregular ICU Time-Series Analysis

Intensive-care medicine produces a particularly demanding form of longitudinal data. Electronic health records contain laboratory values, vital signs, treatments, administrative fields, and outcomes, but these quantities are neither observed on a common clock nor collected with a common purpose. Measurements occur at nonuniform times; variables are sampled at markedly different frequencies; many values are absent at most time points; and the intensity of measurement may itself reflect clinical concern, workflow, or resource use. The resulting records are high dimensional and heterogeneous across patients, care trajectories, and institutions. Irregular medical time-series methods therefore face more than a standard sequence-modeling inconvenience: they must represent the timing, absence, and changing availability of observations while preserving a defensible analytical link to the question being asked [1, 2].

The conventional convenience of a fixed, complete measurement matrix can conceal these properties. Regular resampling, imputation, or summary aggregation may be useful engineering choices, but each makes choices about temporal resolution, which observations count, and how absences are represented. A bedside monitor, a laboratory order, and a static admission field need not convey the same temporal or clinical information. Moreover, the care process can affect both the values that are available and the chance that they are recorded. A representation that ignores timing or observation patterns may discard clinically relevant structure; a representation that exploits them can improve prediction while also encoding care-process signals. This tension makes irregular ICU data a methodological problem spanning statistical representation, workflow design, and interpretation rather than solely a model-selection problem.

For ICU analysis, missingness and irregularity matter at every layer of a study. They influence which variables a representation can retain, which patterns a prediction model can exploit, which records satisfy a cohort definition, and whether a rule can assign an analytical label. They also complicate causal adjustment: an absent measurement can represent no measurement, a measurement outside an available window, a source-data limitation, or a clinically informative decision not to measure. Handling masks, time gaps, or sparse events can improve a predictive representation, but it does not by itself remove the interpretive consequences of the observation process. The thesis consequently treats irregularity and informative measurement as motivations for careful representation and explicit limitations, rather than as local causal findings.

Two complementary needs follow. First, an analysis must accommodate irregular longitudinal measurements rather than force all observations into an unqualified regular grid. Architectures for sparse clinical series and missingness-aware recurrent models illustrate different ways to use values, time intervals, and observation patterns [3, 4]. Second, the analysis needs interpretable variables that can connect a rich event stream to later questions about prognosis and adjusted mortality contrasts. In this thesis, that intermediate layer consists of *clinically inspired proxy states*: rule-derived analytical constructs that summarize evidence for selected clinical patterns from available measurements.

This wording is deliberate. A rule-derived proxy label is not a chart-adjudicated diagnosis, ground truth, or a validated phenotype. A positive tag records that the project-specific source rules were satisfied for the available data; a negative tag does not establish absence of a clinical condition. Nevertheless, a transparent proxy-state layer can make the passage from irregular measurements to a structured downstream analysis inspectable. It supplies common targets for multi-label prediction and a documented set of proxy-state exposures for observational analyses, while retaining a visible link to the measurements and rules from which each construct was derived. Rule-based electronic phenotyping and weak-supervision research provide useful context for this design choice, but do not validate the local proxy definitions or their clinical meaning [5, 6].

Prediction and causal analysis then answer different questions. Predicting a rule-derived proxy label asks whether an irregular-time-series model can reproduce that label from the available inputs. Estimating an adjusted mortality contrast for a proxy-state exposure asks a separate observational question about modeled differences in outcome under explicit assumptions. Strong performance on the first task does not establish the clinical validity of the label, causal validity of the second analysis, or actionability of the exposure. The latter task requires a stated causal model, project-specified directed acyclic graphs (DAGs), exposure-specific adjustment logic, attention to support and overlap, sensitivity and permutation diagnostics, and traceable evidence. Even with these elements, adjusted estimates remain conditional on the adequacy of the proxy measurement, graph, temporal ordering, observed covariates, and model specification [7, 8, 9].

The framework is exercised on PhysioNet 2012 and MIMIC-III, two public ICU time-series resources that provide complementary settings for this problem [10, 11]. Both contain longitudinal ICU measurements and in-hospital mortality information, yet they differ in era, scale, available variables, data contracts, measurement processes, proxy-state definitions, and project graphs. These differences make cross-dataset evaluation useful for examining the portability of an analytical workflow, but they preclude treating similarly named constructs as automatically equivalent or pooling numerical estimates without further justification. The thesis therefore uses the datasets to test a shared methodological sequence while preserving dataset-specific definitions and interpretation boundaries.

This problem setting also explains why the thesis moves from data contracts to proxy states, prediction, and only then to adjusted analysis. The same apparent clinical pattern may have different observable correlates when variable availability, sampling frequency, or coding conventions change. If a downstream analysis hides those transformations, a reader cannot determine whether a difference arose from the source data, the rule-derived label, the prediction export, or the effect-estimation model. Making these handoffs explicit is an engineering discipline with scientific consequences: it permits the analysis to be audited, identifies where limitations enter, and prevents an apparently seamless output from being interpreted as a seamless clinical fact.

1.2 Thesis Gap and Objective

Prior work offers important components for this setting. Irregular-time-series research provides representations and predictive methods for sparse, asynchronously observed medical data. Electronic phenotyping provides strategies for constructing analytical variables when direct expert adjudication is unavailable [5]. Data programming formalizes programmatic heuristic labeling [12], while Snorkel provides related weak-supervision and downstream discriminative-training context [6]. Evaluations of medical LLM capabilities provide a bounded precedent for design-time clinical-knowledge elicitation [13]; work on LLM-derived causal-graph priors separately motivates their use as proposals rather than validated graphs [14]. Causal-inference research supplies DAG-based reasoning, double machine learning, heterogeneous-effect estimation, and diagnostics for observational studies [15, 16, 17]. ICU-focused reviews further emphasize the difficulty of defining an actionable question, managing confounding, and distinguishing predictive patterns from treatment-effect claims in observational clinical records [18, 19].

The gap addressed here is an integration gap, not an assertion that any of these components is absent from the literature. In practice, design-time knowledge elicitation, irregular-time-series prediction, proxy-state construction, and causal-effect estimation are often developed or evaluated separately. Connecting them requires explicit interfaces: LLM-generated proposals must be distinguished from selected source-encoded designs; source-specific preprocessing must yield compatible records; rule-derived labels must be distinguishable from model predictions and algorithmic aggregates; prediction exports must be normalized before downstream use; and causal analyses must state how their exposures, covariates, graphs, diagnostics, and populations relate. Without these interfaces, a model can appear technically successful while its design assumptions, labels, analysis rows, adjustment variables, or interpretation change silently between stages.

The objective of this thesis is therefore *to construct and evaluate an evidence-tracked framework that combines LLM-assisted clinical and causal knowledge elicitation, deterministic proxy-label construction, deep prediction of proxy states from irregular ICU time series, and DAG-guided causal-effect estimation of in-hospital mortality.* "Evidence-tracked" here means that the thesis distinguishes LLM design proposals, selected source-coded behavior, checked numerical artifacts, and unresolved provenance or validation requirements. It is a methodological and engineering contribution accompanied by an empirical evaluation; it is not a claim of clinical validation, deployment readiness, or full clean-checkout computational reproducibility.

The framework has two distinct layers. At design time, a structured LLM-assisted protocol elicited proposed proxy-state ontologies, decision-rule families, missingness considerations, and dataset-specific DAGs; project-selected designs were then encoded in deterministic tagger and graph source files. During analytical execution, the sequence is dataset-specific preprocessing; deterministic generation of rule-derived proxy labels; multi-label proxy-state prediction; prediction normalization and algorithmic aggregation; exposure-specific adjustment under the project-specified DAGs; matching and heterogeneous-effect estimation; and robustness and provenance evaluation. The LLM is not invoked in this executed patient-level pipeline. The proxy states are not validated clinical diagnoses; the DAGs are not learned from data; no unconditional causal effect is claimed; and many illness-state proxies do not define well-specified interventions. The workflow does not provide clinical recommendations, bedside decision support, or prospective evidence of benefit.

An integrated workflow is useful precisely because its interfaces can be inspected rather than assumed. Dataset-specific preprocessing preserves the distinction between source records and analysis-ready objects. Rule-based construction makes the origin of each proxy label visible. Prediction normalization distinguishes probabilities from thresholded labels, and algorithmic aggregation distinguishes a deterministic vote from clinical consensus. Project-specified DAGs and exposure-specific adjustment sets make the causal assumptions operationally visible, while matching and model-estimated CATE summaries provide different forms of observational evidence. Finally, robustness and provenance checks separate a checked numerical statement from stronger claims about causal identification, clinical meaning, or rerun reproducibility. The framework's purpose is not to erase these distinctions, but to retain them throughout the analysis.

This framing has practical consequences for how the results should be read. A checked prediction metric concerns the archived evaluation of a model against rule-derived labels, not an independently verified clinical endpoint. A matching output is a descriptive matched-pair outcome difference, not automatically a treatment effect. A mean model-estimated CATE is an aggregate of fitted conditional estimates under the project assumptions, not an unconditional population quantity. Sensitivity, permutation, and support diagnostics contribute evidence about particular failure modes, but none can substitute for a justified intervention definition, an adequate causal graph, or clinical review. By placing these distinctions in the framework itself, the thesis avoids treating provenance, statistical modeling, and clinical interpretation as afterthoughts to an otherwise complete pipeline.

LLM-assisted clinical and causal knowledge elicitation is therefore a substantive design-time stage of CliniCause. The structured protocol produced proposals for the proxy-state ontology, binary decision rules, missingness reasoning, and dataset-specific DAGs; the project selected designs and encoded them in active deterministic source code. This role does not make the LLM a validated medical expert, establish that every proposal was implemented unchanged, or validate the proxy constructs or graphs. Nor was the LLM an executed patient-level estimator: it did not preprocess records, generate runtime predictions, select adjustment variables dynamically, or estimate causal effects. The distinction between design provenance and executable authority makes this contribution inspectable without treating generated proposals as clinical or causal evidence.

1.3 Research Questions and Contributions

The main research question is:

> How can LLM-assisted clinical and causal knowledge elicitation, deterministic proxy-label construction, and deep modeling of irregular ICU time series be integrated into an evidence-tracked framework to support DAG-guided adjusted effect estimation of in-hospital mortality?

The word “support” is central. The thesis investigates how a linked workflow can organize and qualify retrospective evidence; it does not claim to prove that proxy-state exposures cause mortality or that manipulating a proxy would improve outcomes. The main question is addressed through seven secondary research questions:

SRQ-1: Which data contracts are needed to connect PhysioNet 2012, MIMIC-III, the irregular-time-series prediction workflow, and the causal-analysis pipeline?

SRQ-2: How were LLM-assisted proposals for dataset-specific proxy ontologies, decision rules, missingness considerations, and DAGs selected, encoded, and traced to active source?

SRQ-3: How do STraTS and the included sequence-model baselines perform on multi-label proxy-state prediction across MIMIC-III and PhysioNet?

SRQ-4: How are predicted proxy states normalized and aggregated before downstream analysis?

SRQ-5: What observed adjustment sets are selected by the dataset-specific, project-specified DAGs for proxy-state exposure analyses?

SRQ-6: What matching, model-estimated CATE, sensitivity, and permutation evidence is available for the prespecified proxy-state exposures?

SRQ-7: What limitations follow from proxy labels, intervention definition, overlap, measurement error, model dependence, provenance, and unmeasured confounding?

The primary contribution is an end-to-end, evidence-tracked framework linking LLM-assisted clinical and causal knowledge elicitation, irregular ICU preprocessing, deterministic proxy-label construction, deep proxy-state prediction, algorithmic aggregation, DAG-guided adjustment, matching, heterogeneous-effect estimation, and robustness diagnostics. Its value is the explicit connection between these layers: prompt and source artifacts expose how a proposed construct becomes an implemented rule, while a shared proxy-state interface makes it possible to inspect how that construct moves from rule derivation to prediction and then to an observational exposure analysis.

Secondary methodological contributions are a traceable LLM-guided design layer for dataset-specific proxy definitions, decision rules, and DAG proposals; explicit dataset-specific data contracts between the predictive and causal repositories; source-coded, exposure-specific DAG adjustment logic; and a multi-estimator analysis comprising CausalForestDML, LinearDML, and exploratory CausalPFN. The framework also combines matching, sensitivity, permutation, and support diagnostics so that estimator outputs are interpreted alongside their available empirical checks. CausalForestDML is the primary model-estimated causal-analysis estimator, LinearDML is a secondary comparator, and CausalPFN is retained as an exploratory complementary estimator because it lacks the complete diagnostic support available for the DML estimators.

The empirical contribution is a checked comparison of four learned prediction models across the two datasets, original-cohort adjusted-effect analyses for every prespecified dataset-specific proxy-state exposure, cross-estimator directional comparison, separate outcome-downsampling robustness analyses, and cross-dataset evaluation without numerical pooling. The thesis also contributes reproducibility and evidence infrastructure: checked numerical tables, a results manifest, checksums, a source packet, a decision register, corrected result figures, and explicit tracking of unresolved provenance. These materials support numerical traceability, but they do not establish that the analysis can be rerun from a clean checkout or that every historical producing setting has been recovered.

Together, these contributions are intended to be cumulative rather than interchangeable. An LLM-generated proposal alone would not establish an implemented or clinically valid design. A predictive comparison alone would not explain how its targets become downstream analytical variables. A causal estimator alone would not document how irregular measurements, proxy definitions, or graph-selected covariates entered the analysis. Conversely, an auditable interface does not turn a rule-derived label into a clinical diagnosis or a graph-guided contrast into a well-defined intervention effect. The contribution hierarchy therefore places the integrated framework first, its LLM-assisted design and shared proxy/data-contract layers next, empirical comparisons and estimator triangulation after them, and evidence-tracking infrastructure alongside every layer. This ordering keeps the thesis focused on a transparent methodological connection while leaving clinical validation and stronger causal questions as necessary future work.

Table 1.1: Contribution hierarchy, evidence supplied by the thesis, and principal boundaries.

Contribution	Evidence supplied by the thesis	Principal boundary
End-to-end methodological framework	Linked preprocessing, proxy, prediction, aggregation, graph, estimation, and diagnostic chapters.	Integration is demonstrated as a retrospective research workflow, not clinical validation.
LLM-guided design layer	Preserved prompts and run exports linked at schema level to deterministic tagger and graph sources.	Model settings, human selection, line-level traceability, clinical review, and graph correctness remain incomplete.
Shared proxy-state interface	Source-defined rule labels, prediction exports, aggregation, and proxy-state exposure analyses.	Constructs are rule-derived proxies, not chart-adjudicated diagnoses or ground truth.
Cross-dataset predictive evaluation	Checked comparison of four learned models on MIMIC-III and PhysioNet.	Rankings are dataset specific; target validity and export lineage remain qualified.
DAG-guided multi-estimator analysis	Dataset-specific graphs, adjustment logic, matching, CausalForestDML, LinearDML, and exploratory CausalPFN.	Graph correctness, intervention definition, overlap, and unmeasured confounding are not established.
Robustness and diagnostic framework	Matching, sensitivity, permutation, and outcome-downsampling analyses.	Diagnostics qualify results; they do not confirm causal identification.
Evidence tracking and reproducibility infrastructure	Checked tables, manifest, checksums, source packet, decision register, and corrected figures.	Numerical traceability is stronger than clean-checkout computational reproducibility.

The completed empirical record provides a concise orientation for the chapters that follow. The checked predictive comparison included the four learned models STraTS, GRU, GRU-D, and TCN. STraTS led the four archived test metrics in MIMIC-III, whereas GRU-D led them in PhysioNet; this dataset-dependent pattern is not a claim of universal model superiority. In original-cohort adjusted analyses, CausalForestDML and LinearDML agreed in effect direction for all 19 dataset–exposure comparisons, while all three estimators agreed in direction for 18 of 19 comparisons. The exploratory CausalPFN analysis reproduced the prevailing direction in nearly every original-cohort comparison, indicating promise as a complementary estimator while lacking the complete diagnostic support available for the DML estimators. Matching and robustness diagnostics provide qualified supporting evidence, and outcome-downsampled analyses are treated as a separate robustness population rather than pooled with the original cohorts.

These findings are deliberately directional and high level. They do not provide exposure-level effect claims, significance conclusions, or treatment priorities. They instead motivate a careful reading of the later methods, results, and discussion: agreement across fitted estimators is useful triangulation, but it does not validate the proxy states, establish that the project DAGs are correct, eliminate unmeasured confounding, establish positivity, or demonstrate bedside utility. The appropriate interpretation is an evidence-bounded account of what an integrated retrospective workflow can contribute and what further clinical, causal, and provenance work remains necessary.

1.4 Thesis Organization

The remainder of the thesis follows the analytical chain from context to interpretation. Chapter 2 reviews irregular medical time-series modeling, LLM-assisted knowledge elicitation, clinical proxy construction, weak supervision, and observational causal-inference concepts that frame the study. Chapter 3 separates the design-time knowledge-construction layer from the executed analytical layer and formalizes the study units, prediction and causal questions, proxy-state exposures, outcome, and assumptions. Chapters 4–6 build the data, proxy, and prediction layers: Chapter 4 describes dataset-specific preprocessing and data contracts, Chapter 5 documents LLM-assisted clinical proxy-state and decision-rule construction, and Chapter 6 presents predictive modeling of rule-derived proxy labels.

Chapters 7–9 define how the downstream observational analyses are governed. Chapter 7 describes the project-specified causal graphs, adjustment logic, matching, and heterogeneous-effect estimators. Chapter 8 specifies robustness, sensitivity, permutation, and validation design. Chapter 9 records the experimental matrix, original and outcome-downsampled populations, and reproducibility boundary. Chapter 10 then reports the checked numerical results using the approved estimator and population hierarchy. Chapter 11 interprets the research-question answers, cross-dataset patterns, and limitations. Finally, Chapter 12 summarizes the thesis contributions and identifies future work needed to strengthen clinical validity, causal interpretation, reproducibility, and translation.

Chapter 2

Background and Related Work

This chapter places the thesis at the intersection of five methodological traditions. Critical-care data research supplies the two ICU data sources and exposes the consequences of irregular observation. Sequence-modeling research offers several ways to represent sparse measurements for prediction. Electronic-phenotyping and weak-supervision research explains how useful analytical labels can be constructed when direct adjudication is unavailable. Research on LLM medical knowledge and causal-graph priors motivates a carefully bounded design-time elicitation role. Causal-inference research provides a language for specifying questions, encoding assumptions, selecting adjustment variables, estimating heterogeneous contrasts, and diagnosing limited support or sensitivity to omitted variables. Each tradition addresses a necessary part of the problem, but none by itself supplies the complete path from irregular events to an auditable observational analysis. The chapter therefore emphasizes the connections and boundaries among them; project-specific implementation details begin in Chapter 3.

2.1 ICU EHR Datasets and Irregular Sampling

2.1.1 Critical-care EHR time series

ICU electronic health records combine physiological measurements, laboratory results, administered treatments, clinical observations, and administrative information. These sources do not share a natural sampling clock. Vital signs may be recorded repeatedly, laboratory tests are ordered at variable intervals, and static admission fields may appear only once. Observation times are therefore nonuniform both within a variable and across variables, while different patients can have markedly different measurement density and duration. Reviews of irregular medical time series distinguish irregular spacing within a sequence from heterogeneous sampling rates across variables, and emphasize that either feature complicates models built for complete, fixed-dimensional sequences [1].

Missingness in this setting is not a single phenomenon. A value may be absent because a test was not ordered, a device was not connected, a source table did not contain the measurement, the observation fell outside the analysis window, or preprocessing rejected it. Clinical concern and care processes can influence whether and when a variable is measured, so the observation pattern may contain predictive information [2, 4]. It would nevertheless be too strong to treat every absence as informative or clinically meaningful. Observation patterns can mix physiology, workflow, resource availability, and data-quality mechanisms; exploiting them predictively does not identify which mechanism generated them.

Several further properties make ICU EHR data methodologically demanding. Illness trajectories are heterogeneous, measurement values are high dimensional, physiological and administrative records coexist, and identifiers may refer to a person, admission, ICU stay, or challenge record rather than to the same unit. Repeated stays can create dependence if they are mistaken for independent people. Cohort entry, time zero, and the interval over which measurements are summarized can also change the scientific meaning of a record. These are not merely data-cleaning details: they determine what information is available to a model and whether variables can be interpreted as preceding an outcome or exposure.

Regular resampling and imputation can create a convenient matrix, but they necessarily choose a time resolution and a rule for unobserved cells. Sparse-event representations instead preserve observed values and times, while missingness-aware recurrent models attach masks and elapsed-time information to a regularized sequence. Neither strategy is universally preferable. The relevant choice depends on the source data, measurement density, task, and the distinction between modeling the value process and modeling the observation process. This thesis therefore treats irregular sampling as both a representation problem and an interpretation boundary.

2.1.2 PhysioNet/Computing in Cardiology Challenge 2012

The PhysioNet/Computing in Cardiology Challenge 2012 was organized around patient-specific prediction of in-hospital mortality from early ICU information. It combined admission descriptors with time-stamped vital-sign and laboratory series and provided a structured training-and-evaluation setting for mortality classifiers and risk estimators [10]. Its challenge format makes it a useful compact benchmark-style source: the prediction target, observation window, and evaluation setting were defined for a shared research task rather than left to every investigator to derive independently.

That role must be separated from this thesis's local processing. The source paper documents the challenge data and rules, but it does not establish that a later repository reproduces every inclusion rule, missing-value convention, split, or score. Nor does a challenge record automatically correspond to the same unit as an ICU stay in another database. The thesis uses PhysioNet 2012 as one source on which to exercise a common workflow; exact local data contracts and preprocessing decisions are described in Chapter 4, without assuming full reproduction of the original competition.

2.1.3 MIMIC-III and benchmark research

MIMIC-III is a deidentified, single-center critical-care database integrating bedside observations, laboratory data, medications, procedures, coded information, notes, and outcomes over an extended data era. It was made available to qualified researchers subject to training and a data-use agreement, enabling reproducible critical-care research while retaining controlled-access obligations [11]. Unlike a fixed prediction challenge, MIMIC-III is a relational clinical database from which researchers must define the study unit, cohort, variables, time windows, and outcomes.

Benchmark work on MIMIC-III illustrates how a database can be converted into standardized clinical time-series tasks and reusable baselines [20]. The benchmark is a particular extraction and task definition, not the database itself. Consequently, citing the benchmark supports the value of reproducible task construction but does not imply that this thesis uses its cohorts or tasks unchanged. Local preprocessing may draw on a benchmark style while defining different proxy-label targets and downstream analysis objects.

2.1.4 Dataset differences relevant to this thesis

Table 2.1 summarizes the two sources conceptually. PhysioNet 2012 functions primarily as a bounded mortality-challenge resource derived for a shared task, whereas MIMIC-III provides a broader database that requires more extensive extraction and linkage. The sources differ in scale, era, available variables, identifier structure, measurement process, and preprocessing burden. A variable with a similar name can differ in unit, acquisition practice, aggregation, or missingness. Likewise, a similarly named proxy phenotype can be produced from different evidence and should not be assumed to represent an identical construct.

Table 2.1: Conceptual roles of the ICU datasets used in this thesis.

Dataset	Research role	Longitudinal structure	Strength for this thesis	Important limitation
PhysioNet 2012	Bounded critical-care mortality challenge and compact benchmark-style source.	Admission descriptors plus time-stamped early-ICU measurements organized as challenge records.	Provides a clearly framed setting for exercising irregular-time-series prediction and downstream interfaces.	A local pipeline need not reproduce every original challenge rule, split, or preprocessing convention.
MIMIC-III	Broad controlled-access critical-care database supporting many investigator-defined studies.	Relational admission, stay, event, intervention, and outcome tables requiring cohort construction and linkage.	Tests whether the workflow can operate on a richer EHR source with different variables and care-process information.	Any benchmark or local extraction is only one view of the database and carries site-, era-, and definition-specific choices.

The purpose of using both sources is methodological portability, not population pooling. A shared analytical sequence can be evaluated while preserving dataset-specific cohorts, identifiers, preprocessing, proxy rules, and graphs. This distinction also prevents cross-dataset differences from being interpreted automatically as biological differences: they may arise from the data source, care process, construct definition, observation pattern, or model interface.

2.2 Irregular Time-Series Representation Models

2.2.1 Representation challenges and design choices

Predictive models for irregular clinical time series differ first in what they consider an input. One family regularizes observations onto a sequence of time bins and supplies imputed values, masks, or time gaps. Another works directly with observed events, representing measurement identity, value, and time without filling every unobserved cell. Within these representations, temporal dependence can be modeled by recurrent state updates, temporal convolution, or attention. These choices encode distinct assumptions about which temporal relationships should be easy to learn and how the observation process should enter the model [1].

No representation removes the need to define the task and preprocessing contract. Dense sequences require a bin width, aggregation rule, and treatment of missing cells. Sparse events require a variable vocabulary, continuous or discrete encodings of time and value, and a policy for very long event streams. Positional or temporal encodings provide order information to attention mechanisms, while masks can prevent padded or future positions from entering a computation. Self-supervised pretraining can exploit unlabeled records, but its utility depends on the pretext task and the relation between pretraining and supervised populations. The four learned models reviewed below are the predictive families compared by the thesis; their published forms are conceptual precedents rather than proof that every local adaptation is identical.

2.2.2 Recurrent baselines: GRU and GRU-D

The gated recurrent unit (GRU) updates a hidden state sequentially using reset and update gates. Conceptually, the update gate balances retention of the previous state against incorporation of a candidate state, while the reset gate controls how strongly earlier information contributes to that candidate. This gating provides a flexible general sequence baseline with fewer gating components than a conventional long short-term memory unit [21]. A standard GRU can learn from a suitably prepared clinical sequence, but it does not inherently know how much time elapsed between observations or why a variable is absent. Its handling of irregularity therefore depends on the input representation supplied to it.

GRU-D modifies recurrent modeling for multivariate series with missing values. It uses observation masks and elapsed-time information, and learns decay mechanisms that move missing inputs toward empirical reference values and attenuate hidden states as gaps increase [4]. This formulation makes the missingness pattern and time since observation explicit rather than treating imputation as an invisible preprocessing step. It is one principled response to irregular sampling, not a universal solution: learned decay imposes its own structure, the observation process may differ across sites, and predictive use of missingness does not correct causal bias from measurement decisions.

2.2.3 Temporal convolution: TCN

Temporal convolutional networks apply one-dimensional convolution in temporal order. Causal convolution prevents future positions from contributing to an earlier output, dilation expands the receptive field without requiring a kernel at every lag, and residual blocks stabilize deeper models. Because convolutions for many positions can be computed in parallel, a TCN offers a different computational and inductive bias from a recurrent state update [22]. The canonical TCN study is a general sequence-modeling comparison rather than an ICU-specific method paper. In an irregular ICU setting, the model still requires an appropriate regular sequence representation, so resampling, aggregation, masks, and time features remain part of its effective design.

2.2.4 Sparse-event transformer modeling: STraTS

STraTS is designed for sparse and irregular multivariate clinical series. It treats an observed measurement as a triplet of time, variable, and value; continuous embeddings of time and value are combined with a variable embedding, and transformer-style attention contextualizes the observed events. This avoids constructing a fully imputed variable-by-time matrix and preserves fine-grained observation times. The method also introduces a self-supervised forecasting objective for pretraining representations before supervised prediction [3].

This design is well matched to sparse event streams, but it does not make preprocessing irrelevant. Event selection, normalization, truncation, static features, and the variable vocabulary still determine what the transformer sees. Moreover, the published method and the thesis's supervised multi-label adaptation are distinct objects. The paper supports the triplet representation, attention mechanism, and pretraining rationale; Chapter 6 defines the local target heads, training, and export contract.

2.2.5 Relationship among the five included families

The comparison in Table 2.2 is best understood as a comparison of inductive biases rather than a ladder of sophistication. GRU supplies a general recurrent baseline; GRU-D makes missingness and elapsed time explicit within recurrence; TCN emphasizes multiscale local-to-long-range temporal filters; and STraTS applies attention to observed event triplets with a pretraining option. Explicit missingness handling is central to GRU-D, implicit in the absence and timing of STraTS events, and otherwise dependent on the prepared inputs. Likewise, pretraining is a defined part of STraTS but not an assumed property of every model.

Table 2.2: Predictive model families included in the thesis comparison.

Model	Core temporal mechanism	Input and irregularity handling	Role in this thesis	Primary reference
STraTS	Transformer-style contextualization and attention pooling.	Sparse observed time–variable–value triplets; timing and absence are retained without a fully imputed grid; self-supervised pretraining is supported.	Sparse-event focal model with a later local multi-label adaptation.	[3]
GRU	Gated recurrent hidden-state updates.	Requires a prepared sequence; irregular time and missingness are not inherently resolved by the ordinary recurrent unit.	General recurrent baseline.	[21]
GRU-D	Recurrent updates with trainable input and hidden-state decay.	Uses masks and elapsed-time features to model missingness explicitly.	Missingness-aware recurrent baseline.	[4]
TCN	Dilated temporally ordered convolutions with residual blocks.	Operates on a regular sequence representation whose binning, masks, and aggregation must be defined.	Convolutional sequence baseline.	[22]

Architecture suitability can vary with measurement density, missingness, preprocessing, label construction, sample size, and the supervised task. A model can also exploit site-specific observation patterns rather than transferable physiology. For these reasons, related work motivates empirical comparison but not a universal ranking. The thesis-specific implementations and results are reserved for Chapters 6 and 10.

2.3 Proxy Phenotyping and Weak Supervision

2.3.1 Electronic phenotyping and rule-derived labels

Electronic phenotyping identifies patients or records with characteristics of interest from routinely collected EHR data. Phenotypes may combine diagnosis or procedure codes, laboratory and vital measurements, medications, text, temporal criteria, deterministic rules, statistical models, or machine learning. The field has consequently developed from manually specified rule sets toward hybrid and learned approaches, while retaining the need to validate the construct for its intended use [5].

Rule-based phenotypes remain attractive when transparency and inspectability matter. An investigator can trace a positive label to named variables, thresholds, and time conditions; the same rule can be rerun; and domain reviewers can identify implausible clauses. A rule can also provide a practical label when manual adjudication is unavailable. Concrete respiratory-failure phenotyping work illustrates how operational cohort definitions can be constructed from EHR evidence while requiring validation against the intended clinical concept [23].

Transparency is not equivalence to clinical truth. Thresholds can be sensitive to units and preprocessing, required variables may be missing, and a rule developed at one site may not transport to another. Temporal ambiguity can cause measurements before, during, and after a clinical episode to be collapsed into one label. Codes and treatments may reflect billing or care processes as well as physiology. Rule outputs can therefore be reproducible yet noisy, and a negative label can mean insufficient evidence rather than absence of the condition. Validation must be tied to the intended use: a cohort-retrieval rule, prediction target, and causal exposure may require different evidence.

2.3.2 Clinical-definition sources and approximation

Clinical consensus documents provide important conceptual grounding for organ dysfunction and acute syndromes, but their definitions are usually multicomponent and context dependent. Sepsis-3 links sepsis to infection-associated organ dysfunction and gives specific clinical operationalizations; KDIGO defines acute kidney injury using creatinine and urine-output changes; and the ISTH work proposes clinical and laboratory scoring for disseminated intravascular coagulation [24, 25, 26]. SOFA and the Berlin ARDS definition likewise supply authoritative terminology for organ dysfunction and respiratory syndromes [27, 28].

These sources should not be used to upgrade a partial rule into a formal diagnosis. A project rule may use measurements related to a definition, draw on one threshold family, or approximate part of a syndrome while omitting duration, baseline change, imaging, intervention, or contextual requirements. The appropriate claim is therefore that a proxy family is *clinically informed by* or *draws on concepts from* these sources. Whether each local rule captures its intended construct requires rule-level clinical review and, ideally, chart-adjudicated evaluation. The two definition sources without local PDFs are used here only for this limited conceptual role; no detailed local implementation claim depends on them.

2.3.3 Weak supervision and label aggregation

Weak supervision addresses settings in which accurate hand labels are scarce but imperfect programmatic sources are available. Data programming expresses domain heuristics, distant supervision, or other weak sources as labeling functions and models their noisy, potentially conflicting outputs to create training data for downstream discriminative models [12]. Snorkel implements this broader paradigm as a system with a learned generative label model that estimates source accuracies and dependencies without requiring every training example to be manually labeled [6]. The central lesson for this thesis is that label-source quality and correlation matter: adding sources does not guarantee that shared bias cancels.

A deterministic majority vote is not the same method. Voting assigns the class supported by a specified number of binary sources and treats their contributions according to the fixed vote rule. A learned weak-supervision label model instead estimates latent source-quality or dependence parameters and can weight conflicting sources differently. This thesis uses deterministic majority voting, not the Snorkel generative model. The output is algorithmic aggregation, not clinical consensus; correlated voters trained on the same rule-derived labels can preserve or amplify common error.

2.3.4 Implications for this thesis's proxy states

The thesis uses *clinically inspired proxy state*, *rule-derived proxy label*, *proxy phenotype*, and *weak clinical label* to distinguish analytical constructs from verified diagnoses. Repository identifiers beginning with `LAT_` are historical software names. They do not establish that a formally identified latent clinical variable exists. This distinction is essential because the same column interface is used across several stages without making the generating mechanisms equivalent.

Proxy-state validity can be separated into at least five questions. *Implementation validity* asks whether code executes the documented rule. *Schema consistency* asks whether identifiers and columns align across artifacts. *Predictive learnability* asks whether models can estimate rule-derived labels from time-series inputs. *Clinical construct validity* asks whether the labels correspond to the intended clinical concept. *Causal measurement validity* asks whether a proxy can serve as a sufficiently accurate and temporally coherent exposure for the stated causal question. Repository evidence supports the first three more directly than the last two.

The shared proxy layer has three connected roles in this thesis: it supplies supervised prediction targets, provides binary inputs for algorithmic aggregation, and defines downstream proxy-state exposure representations. A common interface makes handoffs auditable and permits the same construct name to be traced across stages. It also creates a compounded error path: rule design can mismeasure a construct, prediction can reproduce or add classification error, aggregation can preserve correlated error, and DAG-guided causal analysis can then condition on or contrast a mismeasured exposure. Predictive accuracy against rule labels demonstrates learnability of those labels; it is not clinical validation and does not establish causal measurement validity.

2.3.5 LLM-Assisted Clinical and Causal Knowledge Elicitation

Advanced LLMs can encode and express substantial medical knowledge under structured evaluation. In MultiMedQA evaluations, PaLM-family models produced medically relevant answers and reasoning, and instruction tuning improved several measured capabilities [13]. The same study's human evaluation exposed clinically important gaps, and Med-PaLM remained inferior to clinicians. This evidence supports the feasibility of using an LLM to generate design proposals; it does not establish reliability for a particular proxy-design task, correctness of every rule, suitability as an autonomous medical expert, or clinical validation of CliniCause outputs.

LLM semantic judgments have also been studied as prior information for causal-graph construction. Darvariu and colleagues evaluate prompt designs and soft LLM-derived priors for causal discovery, reporting the greatest benefit for edge orientation in selected common-sense benchmark settings while also finding that standalone LLM priors are not uniformly beneficial [14]. This is a methodological precedent for treating model output as candidate relations or orientation information, not evidence that the CliniCause DAGs are correct. The local graphs were not learned from patient-level data, the study did not implement the paper's discovery algorithm, and formal adjustment validity still depends on the correctness and completeness of the project-specified assumptions.

The transition from elicited domain reasoning to executable labels has a separate programmatic-labeling precedent. Data programming represents heuristics as inspectable labeling functions that can generate large weakly labeled data sets for discriminative learning [12]. The deterministic CliniCause decision trees play a conceptually related role by translating selected design proposals into source-coded rules that generate proxy-label targets from observed variables. CliniCause does not reproduce the data-programming generative model, its noise-aware loss, or every element of its multi-source framework.

Snorkel further illustrates a workflow in which noisy and correlated rule sources are combined through a learned generative label model before a downstream discriminative model is trained [6]. CliniCause instead uses its own deterministic decision-rule construction and implemented aggregation workflow. Its majority vote is a fixed algorithmic operation, not Snorkel's learned generative label model, and the citation supplies conceptual weak-supervision context rather than implementation equivalence.

Within this thesis, the LLM-assisted elicitation protocol is consequently treated as a substantive design-time component with bounded authority. It produced proposals for proxy-state ontologies, binary decision rules, missingness considerations, and dataset-specific DAGs; the project selected designs and encoded them in deterministic source files. The prompt artifacts document that process, whereas active tagger and graph scripts define implemented behavior. Neither the literature above nor the local prompt record establishes clinical validity, complete human review, or causal identification.

2.4 Causal Diagrams, Identification, HTE, Overlap, and Sensitivity

2.4.1 Observational causal questions and target-trial thinking

Causal analysis begins with a question, not an estimator. Target-trial thinking makes the desired comparison explicit by specifying eligibility, treatment or exposure strategies, assignment, time zero, follow-up, outcome, causal contrast, and analysis plan before observational data are analyzed [8]. This discipline exposes mismatches such as using post-baseline information to define eligibility, comparing prevalent with incident exposure, or allowing eligibility and follow-up to begin at different times. ICU EHR data make these issues acute because clinical state, treatment, measurement, and selection evolve rapidly and are documented through care processes.

The exposure must also correspond to a sufficiently well-defined intervention or contrast. Different ways of producing the same measured state can lead to different outcomes, so a contrast on a state such as body mass index—or, here, an illness-state proxy—may violate consistency when the underlying intervention is unspecified [9]. This concern is not repaired by choosing a flexible estimator. It requires clarity about what changes, when it changes, and which versions of the exposure are being compared.

Retrospective ICU studies additionally face confounding by indication, changing severity, informative measurement, treatment–confounder feedback, selection, and limitations in outcome measurement. A review of observational causal inference in intensive care emphasizes careful question formulation, assumptions, data preparation, and reporting across the analysis workflow [18]. Marginal structural models provide important background for time-varying treatment and confounder processes, but they are not the point/binary exposure estimator implemented in this thesis. Nor does this thesis claim a complete target-trial emulation.

2.4.2 DAGs, backdoor reasoning, and adjustment

A directed acyclic graph (DAG) represents assumed causal relations through nodes and directed edges. Paths into an exposure can encode confounding routes, while colliders require special care because conditioning on them can open otherwise blocked associations. Backdoor reasoning uses the graph to identify covariates that block noncausal paths without adjusting for inappropriate descendants or colliders [7]. A DAG thus makes assumptions inspectable and connects scientific reasoning to adjustment-set selection.

Graphical transparency must not be mistaken for empirical proof. A DAG does not establish that its arrows are correct, and a set that blocks paths in the drawn graph does not guarantee exchangeability in the clinical population. Required confounders may be unavailable in the observed data; measurement can be an imperfect proxy for a graph node; and temporal order must be justified scientifically rather than inferred from an arrow. In this thesis, project-specified DAGs guide observed-variable adjustment at a high level. Their dataset-specific nodes, mappings, and implementation are deferred to Chapter 7.

2.4.3 Double machine learning

Double machine learning (DML) separates the causal parameter of interest from flexible nuisance functions, such as outcome regression and exposure propensity. Machine-learning estimators can fit these functions, but naively substituting regularized predictions into a causal estimating equation can transfer regularization and overfitting bias to the target parameter. DML addresses this problem through Neyman-orthogonal scores, which reduce first-order sensitivity to nuisance-estimation error, and cross-fitting, which estimates nuisance functions on data separate from the observations used to evaluate the score [15].

In a residualization interpretation, covariate-explained components are removed from the outcome and exposure, and the residual association enters the orthogonal score. Cross-fitting rotates training and evaluation folds so observations can contribute efficiently without using the same fitted nuisance error twice. These devices broaden the nuisance models that can be used while retaining inference under stated regularity and rate conditions. They do not eliminate the substantive assumptions of causal inference. DML cannot correct an omitted confounder that was never measured, create overlap where none exists, turn a proxy into an accurately measured exposure, repair an incorrect DAG, or define an otherwise ambiguous intervention.

EconML provides software implementations for heterogeneous-effect estimators, including DML-related workflows, and is useful for documenting implementation context [29]. The software reference is distinct from the theoretical basis: library availability does not validate local nuisance models, cross-fitting behavior, or identification assumptions.

2.4.4 Causal forests and heterogeneous effects

Outcome prediction estimates how outcomes vary with covariates under the observed data distribution; effect estimation asks how an outcome contrast varies between exposure conditions under causal assumptions. A strong prognostic variable need not be an effect modifier, and a subgroup with high baseline risk need not receive greater benefit from an intervention. Keeping prognostic and treatment-effect heterogeneity separate is therefore fundamental.

Causal forests adapt random-forest ideas to heterogeneous treatment-effect estimation. Recursive partitioning seeks regions in covariate space with different treatment contrasts, and aggregation across trees stabilizes local estimates. Under conditions including unconfoundedness and appropriate forest construction, the method provides a nonparametric approach to conditional treatment effects and inference [16]. Generalized random forests extend the adaptive-neighborhood view of forests to parameters defined through local moment equations, providing a broader framework in which heterogeneous-effect estimation can be formulated [17].

These methods are attractive when effect modification is nonlinear or involves interactions, but flexibility does not validate discovered subgroups. Estimated conditional effects can be unstable in sparse regions, depend on nuisance models and tuning, and identify patterns that do not transport. Forest feature importance describes the fitted model rather than a biological mechanism. High-dimensional individual estimates are not automatically clinically actionable.

2.4.5 HTE in EHR and critical care

EHR research is motivated by the possibility that large, heterogeneous observational populations can inform conditional or individualized treatment-effect estimation. The same data introduce confounding, informative missingness, treatment-selection bias, and covariate shift, while individual counterfactual outcomes remain unobserved [19]. Validation is consequently harder than ordinary prediction: a held-out factual outcome cannot directly reveal the accuracy of an individual's unobserved alternative outcome.

Modern HTE reviews emphasize principled separation of hypothesis generation from confirmation, careful performance criteria, and the instability of data-driven subgroups [30, 31]. Critical-care populations add marked variation in baseline risk, disease mechanisms, and competing trajectories. Heterogeneity in observed outcomes or prognosis can be mistaken for heterogeneity of treatment effect, and average trial results can conceal clinically important variation while exploratory subgroup analyses can also produce spurious patterns [32]. These considerations motivate conditional-effect modeling in the thesis but do not imply that its proxy-state analyses validate individualized clinical effects.

The empirical study evaluates CausalPFN in an exploratory role, while its detailed theoretical positioning remains limited pending a verified primary method source.

2.4.6 Overlap and empirical support

Positivity requires that the exposure conditions being compared are possible within relevant covariate strata. Its empirical counterpart, overlap or common support, asks whether comparable exposed and unexposed observations exist in the analyzed sample. When propensity scores approach their boundaries, estimates rely on a small number of observations or extrapolation, producing instability and changing the population for which a comparison is credible. Work on limited overlap formalizes the variance and target-population consequences and motivates explicit attention to the region of adequate support [33].

Overlap is especially important for heterogeneous-effect estimation because local estimates divide the sample into increasingly specific covariate neighborhoods. A global mixture of exposed and unexposed records can coexist with local support failures. Matching rates, failures, and distances can provide indirect evidence about available comparisons, but they depend on the matching representation and algorithm. They do not replace propensity-score distributions, common-support plots, or covariate-balance assessment. Conversely, empirical overlap does not establish exchangeability or validate the DAG. This thesis therefore treats support diagnostics as necessary qualification and does not claim that positivity has been established.

2.4.7 Omitted-variable sensitivity and diagnostic workflows

Even after adjustment, observational estimates can be biased by variables that affect both exposure and outcome but are absent from the model. Omitted-variable-bias sensitivity analysis asks how strongly such a variable would need to relate to the residualized exposure and outcome to change a conclusion. Robustness values provide a compact strength-of-confounding summary, while comparison with observed covariates can make a hypothetical confounder scale more interpretable [34]. Extensions to causal machine learning formulate analogous bias bounds and inference for broader target parameters and flexible nuisance estimation [35].

Sensitivity results remain model- and scale-dependent. Benchmarking is persuasive only when the selected observed covariate is scientifically meaningful, and no observed benchmark necessarily bounds every unmeasured process. A robustness value does not prove that unmeasured confounding is absent. Sensitivity analysis also does not validate a DAG, create support, or repair exposure misclassification. Its role is to quantify how conclusions would respond to specified departures from an identifying assumption.

More broadly, transparent causal workflows separate modeling assumptions, identification, estimation, and attempts to refute or perturb the analysis. DoWhy articulates this model–identify–estimate–refute sequence and implements placebo and other robustness checks as software-supported diagnostics [36]. The value of such checks is procedural: they can expose sensitivity to broken relationships or alternative assumptions. They do not turn every perturbation into a formal randomization test, and the citation does not imply that every DoWhy refuter was used in this thesis.

2.4.8 Positioning of the present study

The literature supplies complementary components rather than one end-to-end solution. Models for irregular series address representation and prediction; phenotyping provides transparent routes from EHR evidence to analytical labels; LLM research motivates bounded knowledge elicitation and graph-prior proposals; data programming and weak supervision explain how domain heuristics can create imperfect training labels and how noisy sources may be combined; DAGs make identification assumptions explicit; DML and forest methods enable flexible adjusted and heterogeneous-effect estimation; and overlap and sensitivity methods expose fragility in support and unmeasured-confounding assumptions. The contribution of this thesis is to integrate and evaluate these components through a shared, evidence-tracked proxy-state interface, not to claim invention of each component.

Integration also compounds risk. Unsupported LLM suggestions can enter selected rule or graph designs; measurement error in source events can affect rule-derived labels; model error can alter predicted labels; aggregation can preserve shared error; a project-specified DAG can be misspecified; local comparisons can lack support; and unmeasured confounding can remain after flexible estimation. Provenance gaps can make it difficult to determine which version of an artifact entered the next stage. LLM-assisted elicitation is a substantive design-time method in this framework, but it was not an executed estimator, a patient-data causal-discovery method, a source of authoritative clinical judgment, or a source of clinical validation. Chapters 3–9 therefore define the actual study objects, source-code implementation, assumptions, and diagnostic hierarchy. This progression preserves the central lesson of the related work: credible integration requires every handoff and limitation to remain visible rather than being absorbed into a single model output.

Chapter 3

Problem Definition and Study Design

3.1 Units, Time Horizons, and Data Objects

This thesis studies retrospective intensive-care records represented as irregular clinical time series. The operational study unit is a time-series record or ICU stay, depending on the source dataset and preprocessing path. PhysioNet 2012 source files use record identifiers, whereas MIMIC-III source tables use ICU-stay and admission identifiers. The implemented pipelines normalize these source-specific identifiers to the canonical join key `ts_id`. This identifier should be read as the analysis-record identifier used by the software; it is not assumed to be a globally unique person identifier across repeated admissions or stays.

For a study unit i, the observed longitudinal data are represented as a finite collection of measurement events

$$\mathcal{O}_i = \{(i, t_{ij}, v_{ij}, x_{ij}) : j = 1, \ldots, n_i\},$$

where i is the normalized `ts_id`, t_{ij} is elapsed time in minutes from record start or ICU admission, $v_{ij} \in \mathcal{V}$ is the observed variable name, and x_{ij} is the numeric value recorded for that variable. In the causal repository this long-format object is stored in `ts` with the required columns `ts_id`, `minute`, `variable`, and `value`. Static or baseline quantities such as age, sex or gender coding, height, weight, and ICU-type indicators are represented within the same event schema as variables observed at or near the beginning of the record when the preprocessing implementation provides them.

The event representation is intentionally irregular. Observations occur at nonuniform times, different variables are measured with different frequencies, and most variables are absent at most possible times. A missing measurement is therefore not interpreted as a normal value. It may reflect a clinical decision not to measure, an unrecorded observation, a source-table limitation, or a preprocessing exclusion. The thesis treats informative measurement and missingness processes as important threats to interpretation, not as causal findings established by the repository.

The primary patient- or stay-level outcome used by the downstream causal repository is `in_hospital_mortality`. In the causal processed artifact, this outcome is stored in `oc`, the outcome table keyed by `ts_id`. The third object in that processed artifact, `ts_ids`, is the sorted collection of normalized identifiers represented in `ts`. Later predictive processing uses a distinct split-aware artifact and must not be conflated with this causal contract.

3.2 Design-Time Knowledge Construction and Executed Analysis

The framework distinguishes a design-time knowledge-construction layer from the patient-level analytical layer. At design time, the official dataset-description link and instructions to research relevant clinical, causal, and time-series literature were supplied to a structured LLM-assisted elicitation protocol. The protocol requested a dataset audit, proxy-state ontology, binary decision-rule families, missingness and measurement-process reasoning, and a dataset-specific DAG. The resulting proposals were project-selected and encoded in deterministic tagger and graph source files. Available evidence does not establish that every proposal was implemented unchanged, a complete line-by-line prompt-output-to-code mapping, or a formal clinician-review chain.

The analytical layer begins only after those designs have been encoded. It preprocesses raw ICU data, applies deterministic decision-tree rules to generate rule-derived proxy labels, trains deep irregular-time-series models to predict those labels, normalizes predicted outputs, and forms the archived final aggregate from one rule-derived source and four predicted-label sources before DAG-guided matching, effect estimation, sensitivity analysis, and permutation diagnostics. No LLM is invoked for patient-level preprocessing, label generation, prediction, adjustment selection, or effect estimation during this executed layer.

Table 3.1: Separation of design-time knowledge construction from executed patient-level analysis.

Layer	Sequence	Evidence boundary
Design time	Official dataset-description link and instructions to research relevant clinical, causal, and time-series literature → structured LLM elicitation → proposed ontology, rules, missingness considerations, and DAG → project selection and source-code encoding.	Prompts document proposals; active source defines implementation; formal clinical review remains incomplete.
Executed analysis	Raw data → preprocessing → deterministic rule-derived labels → deep prediction → prediction normalization and mixed-source aggregation → DAG-guided estimation and diagnostics.	No runtime LLM call; every downstream inference remains conditional on proxy, graph, support, and model assumptions.

3.3 Prediction Task

The prediction task is a multi-label proxy-state prediction problem over irregular ICU time series. For each study unit i and proxy state k, let $L_{ik}^{rule} \in \{0, 1\}$ denote a rule-derived proxy label. These labels are clinically inspired analytical constructs generated from observed data by the deterministic source-code rules selected and encoded after LLM-assisted design elicitation. They are weak clinical labels or proxy phenotypes, not chart-adjudicated diagnoses, manually verified clinical states, or validated phenotypes unless separate validation evidence is supplied in a later chapter.

The supervised predictive workflow estimates a vector of proxy-state labels from the observed time-series representation. Its target matrix is loaded from a latent-label CSV keyed by `ts_id`; the supervised proxy-label targets are not taken from the mortality column in `oc`. For model m, the exported prediction object may contain probabilities $\hat{p}_{ik}^{(m)}$ and corresponding binary predicted labels $\hat{L}_{ik}^{(m)}$. These outputs are model estimates of rule-derived proxy labels rather than direct measurements of patient physiology.

Within the executed layer, five analytical tasks are distinguished. First, deterministic proxy-state construction derives binary rule-derived proxy labels from observed clinical measurements. Second, multi-label proxy-state prediction estimates those rule-derived labels from irregular time-series inputs. Third, the archived final causal runs aggregate one rule-derived table with four aligned binary model-prediction tables into a majority-vote proxy-label table. Fourth, mortality prediction or association analysis evaluates whether proxy-state representations contain prognostic information about `in_hospital_mortality`. Fifth, observational causal-effect estimation treats selected columns from that mixed rule-and-prediction aggregate as modeled exposures and estimates adjusted contrasts under explicit assumptions. Detailed proxy rules, model architectures, estimator algorithms, and numerical results are intentionally deferred to later chapters.

3.4 Causal Question, Exposures, Outcome, Estimands, Assumptions

The downstream causal-analysis setting is retrospective and observational. No intervention is assigned by this study. The repository uses software parameters named `treatment` for selected binary `LAT_*` columns, but many such columns represent illness-state or organ-dysfunction proxy states rather than well-defined administered therapies. In the thesis narrative these variables are therefore described as proxy-state exposures or modeled treatment variables unless a later section explicitly defines a manipulable intervention.

Let $A_i \in \{0, 1\}$ denote a selected proxy-state exposure, Y_i denote the binary outcome `in_hospital_mortality`, W_i denote observed adjustment variables selected from a project-specified DAG and available columns, and X_i denote effect-modifier features used to describe heterogeneity. The dataset-specific DAGs were developed through a project design process that included LLM-assisted clinical and causal elicitation and were then encoded in source code; the implemented graph scripts, not the prompt outputs alone, define the DAG artifacts used by the pipeline. Potential-outcome notation such as $Y_i(a)$ is useful as conceptual language, but causal interpretation requires more than the presence of a graph or an estimator. It depends on assumptions about the causal graph, consistency, exchangeability conditional on observed covariates, positivity or overlap, outcome and exposure measurement, and unmeasured confounding [7, 8, 9].

Accordingly, later estimates should be interpreted as adjusted effect estimates under project assumptions, not as unconditional clinical effects. The existence of a DAG, matching script, or CATE estimator does not prove identification. Proxy states may be useful for structured analysis while still carrying label noise, temporal ambiguity, and intervention-definition limitations.

The main research question is how LLM-assisted clinical and causal knowledge elicitation, deterministic proxy-label construction, and deep modeling of irregular ICU time series can be integrated into an evidence-tracked framework for DAG-guided adjusted effect estimation of in-hospital mortality. The seven secondary questions stated in Chapter 1 cover data contracts; selection, encoding, and traceability of LLM-assisted proxy and DAG proposals; four-model predictive performance; prediction normalization and aggregation; DAG-selected adjustment sets; matching and model-estimated CATE evidence with available diagnostics; and the limitations that bound those findings. These are study questions and design commitments; their answers are reported in Chapters 10 and 11.

Chapter 4

Data and Preprocessing

4.1 PhysioNet 2012 Pipeline

The PhysioNet/Computing in Cardiology Challenge 2012 dataset provides ICU time-series records and associated in-hospital mortality outcomes for a mortality-prediction challenge setting [10]. In this thesis it functions as one of the two ICU time-series sources used to exercise the proxy-state prediction and downstream causal-analysis pipeline. The local repository does not track the raw challenge files or a verified processed-data pickle, so this section describes the implemented preprocessing contract rather than asserting a raw preprocessing cohort total. Chapter 10 separately reports the validated causal-analysis model-row count.

The causal-repository PhysioNet preprocessing implementation traverses the raw `set-a`, `set-b`, and `set-c` folders and reads one patient-record file at a time. It removes rows with missing parameter names, skips records with at most five retained measurement rows, and removes observations with negative values, which the source code treats as missingness indicators. It converts source times from `HH:MM` strings into elapsed minutes, renames `RecordID` to `ts_id`, `Parameter` to `variable`, and `Value` to `value`, and reads outcome files by mapping `In-hospital_death` to `in_hospital_mortality`. After concatenating the source sets, it drops duplicate events and converts the categorical `ICUType` measurement into one-hot-style variables `ICUType_1` through `ICUType_4`.

The resulting causal processed artifact is serialized as three objects: `ts`, `oc`, and `ts_ids`. The `ts` object is a long event table with `ts_id`, `minute`, `variable`, and `value`. The `oc` object contains `ts_id`, `length_of_stay`, `in_hospital_mortality`, and a source subset indicator. The `ts_ids` object is derived from identifiers observed in `ts`. This artifact is the expected input to rule-based tagging, mortality-from-proxy analysis, matching, and CATE estimation in the causal repository.

4.2 MIMIC-III Pipeline

MIMIC-III is a critical-care electronic health-record database used here as the second ICU data source [11]. The implemented workflow follows a clinical time-series benchmark style in that it transforms raw ICU events into stay-level time-series examples suitable for prediction, while adapting the output contract for this thesis pipeline [20]. The repository does not establish a tracked raw-data snapshot, extraction date, or final processed artifact hash.

The active causal-repository MIMIC preprocessing implementation reads broad categories of MIMIC-III source tables: ICU-stay timing from `ICUSTAYS`, demographics from `PATIENTS`, bedside measurements from `CHARTEVENTS`, laboratory measurements from `LABEVENTS`, fluid outputs from `OUTPUTEVENTS`, administered inputs from `INPUTEVENTS_CV` and `INPUTEVENTS_MV`, and mortality/timing information from `ADMISSIONS`. Large event tables are processed in chunks and written to temporary shards before feature rows are collected. The implementation filters to adult ICU stays, removes chart events marked with an error flag, requires valid chart times and numeric or textual values, applies item-ID mappings and range filters for selected variables, converts selected units, and assigns events without direct ICU-stay identifiers to an ICU interval when possible.

For the causal contract, event times are converted to elapsed minutes relative to ICU admission, events are restricted to the early ICU window used by the implementation, age and gender are added as static rows at minute zero, and duplicate (`ts_id`, `minute`, `variable`) events are collapsed by averaging. The helper contract canonicalizes stay identifiers, normalizing numeric-looking identifiers such as `12345.0` to `12345`. It defines the canonical `ts` columns as `ts_id`, `minute`, `variable`, and `value`, defines the canonical `oc` columns as `ts_id`, `length_of_stay`, `in_hospital_mortality`, and `subset`, and intentionally excludes internal MIMIC identifiers such as `HADM_ID` and `SUBJECT_ID` from the exported outcome table.

Inspection found a source-level issue that should be resolved before relying on local MIMIC preprocessing execution: the active script reads `ADMISSIONS.csv` with `DEATHTIME` for early-death filtering, while the canonical outcome helper requires `HOSPITAL_EXPIRE_FLAG` to construct `in_hospital_mortality`. This does not change the documented target contract, but it remains a source-level reproducibility risk.

4.3 STraTS Split-Aware Artifacts Versus Causal Artifacts

The predictive STraTS repository uses a different processed-data contract from the causal repository. Its loader expects a five-object pickle consisting of `events`, `oc`, `train_ids`, `val_ids`, and `test_ids`, where the final three objects explicitly encode train, validation, and test identifiers. The loader accepts stay identifiers named `ts_id`, `icustay_id`, or `ICUSTAY_ID`, canonicalizes them to `ts_id`, and validates the split arrays after the same normalization. This split-aware contract is therefore not interchangeable with the causal repository's unsplit `[ts, oc, ts_ids]` artifact.

In supervised STraTS runs, the latent-label CSV is joined to the selected split identifiers after ID normalization. The loader filters labels to the supervised IDs, raises an error when labels are missing for any supervised `ts_id`, raises an error for duplicate normalized labels, and uses all non-`ts_id` columns in the latent CSV as proxy-label targets. The mortality column in `oc` is not used as the supervised proxy-label target. Prediction export writes `ts_id`, one probability column per proxy state, and one thresholded binary column per proxy state; the inspected wrapper scripts export the `all` split for their prediction CSVs, although final archive-copy provenance remains incomplete.

The predictive loader also implements split-aware leakage controls. It keeps only variables observed in the training split, derives normalization statistics from the training split, computes static-feature normalization from training rows, and validates label alignment before training. For MIMIC-III supervised prediction it restricts events to the first 24 hours. The pretraining path removes test data, uses training-derived variable statistics, uses a 48-hour maximum window for PhysioNet 2012, and permits MIMIC-III observations up to five days while sampling 24-hour input windows. These controls reduce some sources of leakage, but they do not guarantee absence of leakage, nor do they resolve missing raw-data and split-provenance limitations.

The parent router can bridge contracts by reading a causal `[ts, oc, ts_ids]` artifact and constructing a STraTS-style `[ts, oc, train_ids, val_ids, test_ids]` payload through a seeded identifier shuffle. That bridge is an orchestration convenience and does not prove how every archived final STraTS artifact was generated.

Table 4.1: Processed artifact contracts used by the causal and STraTS repositories.

Property	Causal repository	STraTS repository
Processed artifact	`[ts, oc, ts_ids]`	`[events, oc, train_ids, val_ids, test_ids]`
Split representation	Unsplit ID collection	Explicit train/validation/test IDs
Main use	Tagging and causal analysis	Predictive training and export
Identifier convention	Canonical `ts_id`	Loader-normalized `ts_id`
Proxy-label source	Separate latent CSV downstream	Separate latent CSV joined to split IDs

Several provenance limitations remain. Raw PhysioNet and MIMIC-III data are not tracked in Git. The processed pickles used by final runs are external or absent from the audited repository. Some run records contain machine-specific absolute paths, and final cohort counts require verified processed artifacts or manifests. Clean-checkout reproduction therefore requires authorized data access, artifact hashes, and a documented split-generation record.

Chapter 5

Clinical Proxy-State Construction

This chapter describes how an LLM-assisted design protocol and deterministic project source transform heterogeneous, irregular ICU measurements into a smaller set of interpretable binary constructs. The protocol elicited proposed proxy ontologies, decision rules, missingness considerations, and DAGs; project-selected designs were subsequently encoded as tagger and graph source files. The deterministic taggers generate supervised targets for the proxy-state prediction task introduced in Chapter 3, and predicted or aggregated proxy labels become structured exposures for later observational analyses. Their interpretability comes from inspectable source-code rules and stable `LAT_*` column names, not from chart-adjudicated clinical reference labels. They should therefore be read as proxy states: useful analytical representations whose value depends on construct validity, source-variable availability, measurement quality, missingness, and later clinical review.

The chapter is limited to methods and construct definition. It documents the design-provenance artifacts and active rule implementations for PhysioNet 2012 and MIMIC-III, describes available validation artifacts without reporting their numerical contents, and explains how rule-derived proxy labels differ from predicted and aggregated predicted proxy labels. Predictive-model architecture and training mechanics are presented in Chapter 6; final numerical results are reported in Chapter 10.

5.1 Rationale and Terminology

Electronic health-record phenotyping often uses structured measurements, codes, and rules to construct clinically meaningful variables when direct expert labels are unavailable [5]. This thesis uses the same conservative framing. A *proxy state* is a binary construct intended to summarize evidence for a clinically meaningful condition or physiologic state from the available data. A *proxy phenotype* is the same family of construct, with emphasis on phenotype-like grouping rather than direct measurement. A *clinically inspired proxy label* is a weak label based on clinical measurements, thresholds, or rule clauses rather than a chart-adjudicated endpoint. A *rule-derived proxy label* is assigned by deterministic source-code rules. A *predicted proxy label* is produced by a supervised time-series model trained to estimate rule-derived labels.

An *aggregated predicted proxy-label representation* is an algorithmic aggregate used downstream; in the archived final causal runs, its deterministic vote combines one rule-derived source with four aligned binary model-prediction sources. A *binary proxy-state tag* is the realized 0/1 value for one `ts_id` and one proxy-state column.

The repository historically uses the names `latent`, `latent variable`, `latent tag`, and `LAT_*`. Those names are stable implementation vocabulary. The thesis retains the `LAT_*` column names when identifying artifacts, configuration fields, prediction targets, and downstream exposure variables, but uses "proxy state" in explanatory prose. This avoids implying that the implementation has discovered or validated a hidden biologic state.

The proxy-state layer serves four methodological roles. First, it converts selected LLM-assisted design proposals into source-coded, patient- or stay-level analytical constructs. Second, it supplies rule-derived supervised targets for multi-label prediction. Third, it establishes a consistent variable interface for downstream aggregation, mortality, matching, and CATE scripts. Fourth, it preserves a transparent mapping from each `LAT_*` column back to measurement families and rule clauses.

The limitations are equally central. No chart-adjudicated reference labels or clinician-reviewed gold-standard labels were found in the inspected archive. Thresholds approximate clinical ideas but are not equivalent to complete formal clinical definitions. Missing measurements can prevent rule clauses from firing; a negative tag therefore does not prove absence of the condition, and a positive tag does not establish a clinical diagnosis. Source-variable availability differs between PhysioNet and MIMIC-III, so similarly named cross-dataset states are not automatically identical constructs. Weak-supervision literature supports the broad idea that noisy labeling sources and aggregation can be useful, but this project does not implement the Snorkel generative label model [6].

5.1.1 LLM-Assisted Proxy-State, Decision-Rule, and DAG Construction

The proxy-state ontology, decision rules, and dataset-specific DAGs have prompt-provenance artifacts under `thesis-writing/prompts-and-documents/`. Those artifacts document a substantive design-time LLM-assisted elicitation process that requested a dataset audit, clinical and causal ontology, proxy-state definitions, binary decision-tree proposals, missingness and measurement-process reasoning, and a DAG for PhysioNet 2012 and MIMIC-III. The final dataset prompts instantiate a generic prompt template with only the dataset name and official dataset-description URL changed. The final PhysioNet and MIMIC prompt runs were exported on 2026-05-01. Project-supplied metadata identifies the model as ChatGPT 5.4 with extended reasoning, but the PDF metadata does not independently encode that model/version field; the exact system instructions, decoding settings, browsing or research mode, follow-up interactions, and export procedure remain incompletely documented.

The LLM-generated outputs were design proposals subsequently selected and encoded by the project, not clinical validation or source-code authority. The LLM did not learn labels or graphs from patient-level records in this repository, did not execute preprocessing, deterministic labeling, prediction, majority voting, adjustment selection, or causal estimation, and did not produce chart-adjudicated reference labels. The implemented rule-derived proxy labels are defined by the active tagger source files, and the implemented DAG artifacts are defined by the active graph scripts. The prompt outputs explain the design history, whereas any claim about implemented behavior must trace to source code, configuration, and run artifacts.

Table 5.1: Prompt and document artifacts used as design provenance for proxy-state and DAG construction.

Artifact	Dataset	Role	Version status	Thesis relation
Generic prompt template	Generic	Template for staged ontology, rule, missingness, and DAG elicitation.	Final template	Design-time elicitation protocol; not execution or validation.
PhysioNet prompt and run export	PhysioNet	Dataset-specific prompt and exported ChatGPT run.	Final run	Design proposals subsequently selected and source-encoded; not implementation authority.
MIMIC prompt and run export	MIMIC-III	Dataset-specific prompt and exported ChatGPT run.	Final run	Design proposals subsequently selected and source-encoded; not implementation authority.
Earlier prompt-result exports	Both	Earlier prompt outputs.	Superseded	Historical design trace only.
Manager-summary documents	Project	Management summaries of proxy/DAG and clinical CATE review concepts.	Project documents	Do not establish a complete human or clinical review chain.

Earlier prompt-result exports dated 2026-04-29 are superseded design records rather than the final prompt runs. At schema level, the final PhysioNet output proposes the same eleven active `LAT_*` states as the active PhysioNet tagger, and the final MIMIC output proposes the same ten active states as the active MIMIC tagger. Both outputs discuss measurement and missingness processes that also appear in the graph layer. This alignment is high level: the implemented source contains more precise clauses and graph structure, and complete line-by-line correspondence between every LLM output and the active implementation has not been established.

Table 5.2: Traceability from LLM-assisted design artifacts to implemented proxy and DAG artifacts.

Design element	LLM provenance artifact	Implemented source file	Generated artifact	Validation status	Principal limitation
PhysioNet proxy ontology	`physionet-prompt.txt`; `physionet-prompt-running.pdf`	`tagging_latent_variables_physionet.py`	`physionet-latent-tags.csv`	Eleven-state schema alignment; deterministic output present.	No chart adjudication; construct validity and line-level traceability incomplete.
MIMIC proxy ontology	`mimic-prompt.txt`; `mimic-prompt-running.pdf`	`tagging_latent_variables_mimiciii.py`	`latent_tags.csv`	Ten-state schema alignment; deterministic output and descriptive summaries present.	No chart adjudication; input-mode and construct validity remain unresolved.
PhysioNet decision rules	PhysioNet final prompt and run export	`tagging_latent_variables_physionet.py`	PhysioNet proxy-label table and decision-tree artifacts	Active source clauses documented in this chapter.	Source is more precise than prompt narrative; no formal clinical-rule adjudication.
MIMIC decision rules	MIMIC final prompt and run export	`tagging_latent_variables_mimiciii.py`	MIMIC proxy-label, merged-feature, and decision-tree artifacts	Active source clauses and diagnostic summaries documented.	No proof every proposal was implemented unchanged; no formal clinical-rule adjudication.
PhysioNet DAG	PhysioNet final prompt and run export	`physionet2012_causal_graph.py`	`physionet_causal_graph.pkl`; `physionet_causal_dag.png`	Active source reproduces the archived graph artifact.	No clinical edge validation or complete selection-decision record.

Design element	LLM provenance artifact	Implemented source file	Generated artifact	Validation status	Principal limitation
MIMIC DAG	MIMIC final prompt and run export	`mimiciii_causal_graph.py`	`mimic_causal_graph.pkl`; `mimic_causal_dag.png`	Active source reproduces the archived graph artifact.	No clinical edge validation or complete selection-decision record.

Table 5.3: Proxy-state source types used in the thesis pipeline.

Source type	Generation mechanism	Interpretation boundary	Primary evidence
Rule-derived proxy label	Deterministic clinical rules applied to summary features or helper flags.	Transparent weak label; not a verified diagnosis or formal score unless the full score is implemented.	Active tagger source and rule artifacts.
Predicted proxy label	Time-series model probability thresholded at 0.5 for each target column.	Model estimate of a rule-derived proxy label; not a direct clinical measurement.	STraTS export source and prediction CSV schema.
Aggregated predicted proxy-label representation	Deterministic vote across one rule-derived CSV and four aligned binary model-prediction CSVs in the archived final causal runs.	Mixed-source aggregated binary proxy; not clinical consensus and not a probabilistic label model.	Majority-vote logs, source, and causal run summaries.

5.2 PhysioNet Proxy-State Rules

The active PhysioNet implementation is `tagging_latent_variables_physionet.py` in the causal repository's `src/` directory. It loads the processed PhysioNet pickle, pivots the long table by `ts_id` and minute, computes per-record summary columns using minimum, maximum, mean, first observed value, and last observed value, and then applies one decision function per configured `LAT_*` column. The active output order is:

`LAT_CHRONIC_BASELINE_RISK`, `LAT_GLOBAL_SEVERITY`, `LAT_SHOCK`, `LAT_RESPIRATORY_FAILURE`, `LAT_RENAL_DYSFUNCTION`, `LAT_HEPATIC_DYSFUNCTION`, `LAT_COAG_HEME_DYSFUNCTION`, `LAT_INFLAMMATION_SEPSIS_BURDEN`, `LAT_NEUROLOGIC_DYSFUNCTION`, `LAT_CARDIAC_INJURY_STRAIN`, `LAT_METABOLIC_DERANGEMENT`.

The archived `final-results/trees/physionet-tags/physionet-latent-tags.csv` header matches this active column set and contains binary values with no duplicate `ts_id` values in the validated artifact. That artifact-level agreement supports the schema, but it does not by itself prove the producing command, input-artifact hash, source commit, or default-versus-optimized threshold mode.

Missingness is handled conservatively in the active decision helpers. A comparison such as $x < c$ or $x \geq c$ contributes positive evidence only when a non-missing value exists. Missing values therefore do not count as abnormal. The active summarizer does not implement explicit time windows beyond whatever time span is already present in the processed PhysioNet record. It also does not compute a fresh urine sum; urine-related clauses fire only if a compatible precomputed summary column such as `Urine_24h_min`, `Urine_24h_sum`, or `Urine_sum` is present. For oxygenation, the implementation uses `PF_min` when available and otherwise approximates the PaO2/FiO2 ratio from PaO2 and FiO2 summary columns.

By default the script uses the clinically inspired thresholds embedded in the active source. The command-line option `--optimized` switches to externally supplied thresholds loaded from `--thresholds-path`, with unspecified optimized keys falling back to defaults. The existence of this option is not evidence that optimized thresholds generated the archived CSV.

Table 5.4: PhysioNet rule-derived proxy-state definitions from the active tagger.

Proxy-state column	Construct represented	Input measurements and aggregation	Rule structure or threshold family	Grounding and validation status
`LAT_CHRONIC_BASELINE_RISK`	Baseline vulnerability or limited reserve.	Age first value; BMI from height and weight first values; albumin minimum; ICU type first value.	Score at least 2 across age ≥ 75, BMI < 18.5 or ≥ 40, albumin < 3.0, or ICU type in $\{1, 3\}$.	Phenotyping context [5]; exact clauses are project-specific.
`LAT_GLOBAL_SEVERITY`	Multi-domain acute physiologic severity.	Hemodynamic, respiratory, neurologic, renal, hepatic/coagulation, metabolic, cardiac summaries: minima, maxima, and helper flags.	At least 3 abnormal domains, or at least 2 domains plus a critical marker: lactate ≥ 4.0, pH < 7.20, mechanical ventilation, GCS ≤ 8, or MAP < 60. Domain thresholds include MAP < 70, PF < 300, GCS < 15, creatinine ≥ 2.0, bilirubin ≥ 2.0, platelets < 100, lactate > 2.0, and HR ≥ 130.	Broad organ-dysfunction grounding from Sepsis-3 and SOFA [24, 27]; not complete SOFA.
`LAT_SHOCK`	Circulatory shock or hemodynamic instability.	MAP, noninvasive MAP, systolic pressure, lactate, heart rate, urine helper, pH, bicarbonate.	At least 2 abnormal clauses among MAP < 70, SBP ≤ 90, lactate > 2.0, HR ≥ 110, urine < 500, pH < 7.30, or bicarbonate < 18; also positive for low MAP with lactate ≥ 4.0.	Shock/lactate concepts grounded by Sepsis-3 [24]; no vasopressor requirement in PhysioNet rule.

Proxy-state column	Construct represented	Input measurements and aggregation	Rule structure or threshold family	Grounding and validation status
`LAT_RESPIRATORY_FAILURE`	Hypoxemia, ventilatory failure, or ventilatory support.	Mechanical ventilation flag; PF ratio; SaO2; PaO2; respiratory rate; PaCO2; pH.	At least 2 abnormal clauses or mechanical ventilation with PF < 300. Clauses include SaO2 < 92, PaO2 < 60, respiratory rate ≥ 22 or < 8, PaCO2 ≥ 50, or PaCO2 ≥ 45 with pH < 7.30.	Respiratory phenotyping and ARDS/oxygenation grounding [23, 28]; not adjudicated ARDS.
`LAT_RENAL_DYSFUNCTION`	Renal dysfunction.	Creatinine first and maximum; BUN maximum; urine helper; potassium maximum; bicarbonate minimum.	At least 2 abnormal clauses among creatinine ≥ 2.0, creatinine rise ≥ 0.3, BUN ≥ 40, urine < 500, potassium ≥ 5.5, or bicarbonate < 18; creatinine ≥ 3.5 is sufficient.	Renal concepts grounded by KDIGO [25]; not full KDIGO staging.
`LAT_HEPATIC_DYSFUNCTION`	Hepatic injury or reserve.	Bilirubin, AST, ALT, ALP, albumin, platelets.	Weighted score at least 2: bilirubin ≥ 2.0 counts twice; AST or ALT ≥ 200, ALP ≥ 250, albumin < 2.5, or platelets < 100 add one.	Bilirubin/organ-dysfunction grounding from SOFA [27]; transaminase and albumin clauses are project-specific.
`LAT_COAG_HEME_DYSFUNCTION`	Hematologic/ coagulation stress.	Platelets, hematocrit, WBC, platelet first-to-minimum change.	Score at least 2. Platelets < 100 counts twice; platelets < 150, HCT < 25 or > 55, WBC < 4 or > 20, or platelet drop ≥ 50 add one.	Coagulation grounding from ISTH DIC and SOFA [26, 27]; not complete DIC score.

Proxy-state column	Construct represented	Input measurements and aggregation	Rule structure or threshold family	Grounding and validation status
`LAT_INFLAMMATION_SEPSIS_BURDEN`	Systemic inflammation or sepsis-like burden.	Temperature, WBC, HR, respiratory rate, PaCO2, lactate, platelets.	Score at least 3 across temperature > 38.3 or < 36, WBC > 12 or < 4, HR > 90, respiratory rate > 20 or PaCO2 < 32, lactate > 2.0, platelets < 150; also positive when temperature or WBC abnormality anchors lactate stress with score at least 2.	Sepsis context [24]; rule is not formal Sepsis-3 diagnosis.
`LAT_NEUROLOGIC_DYSFUNCTION`	Depressed consciousness or neurologic/metabolic stress.	GCS; sodium; glucose; PaCO2; SaO2; pH.	Score at least 2. GCS ≤ 12 counts twice; GCS < 15, sodium < 130 or > 150, glucose < 70 or > 300, PaCO2 ≥ 50, SaO2 < 90, or pH < 7.25 contribute.	SOFA neurologic context [27]; sedation ambiguity remains unreviewed.
`LAT_CARDIAC_INJURY_STRAIN`	Myocardial injury, ischemia, or severe strain.	Troponin I/T; ICU type; HR; MAP; systolic pressure; lactate.	Score at least 2. Troponin I > 0.1 or troponin T > 0.01 counts twice; cardiac ICU type, HR ≥ 130 or < 50, MAP < 70 or systolic ≤ 90, and lactate > 2.0 add one.	Project-specific cardiac proxy.

Proxy-state column	Construct represented	Input measurements and aggregation	Rule structure or threshold family	Grounding and validation status
LAT_METABOLIC_DERANGEMENT	Acid-base, lactate, electrolyte, or glucose derangement.	pH, bicarbonate, lactate, sodium, potassium, magnesium, glucose, PaCO2.	Score at least 2 across pH < 7.30 or > 7.50, bicarbonate < 18 or > 32, lactate > 2.0, sodium < 130 or > 150, potassium < 3.0 or > 5.5, magnesium < 0.6 or > 1.2, glucose < 70 or > 250, PaCO2 < 32 or > 50; critical pH < 7.20, lactate ≥ 4.0, or potassium ≥ 6.0 is sufficient.	Project-specific metabolic proxy.

The chronic baseline-risk, hepatic, cardiac, and metabolic families remain project-specific rules without complete exact clinical-source support or qualified clinical review. They are therefore presented as analytical proxy definitions rather than formal diagnoses or validated clinical definitions.

Decision-tree pickle files and rendered tree images exist under the inspected archive and repository tree directories. However, the figure-generation path depends on saved callable objects and plotting code, and the active PhysioNet script's CLI tree-save path requires provenance review before the images can be treated as thesis figures. The rules in Table 5.4 therefore come from the active source and CSV schema rather than from unpickled tree artifacts.

5.3 MIMIC Proxy-State Rules and Validation Artifacts

The active MIMIC implementation is `tagging_latent_variables_mimiciii.py` in the causal repository's `src/` directory. Its active column set differs from PhysioNet:

`LAT_CHRONIC_BURDEN`, `LAT_INFLAMMATION_SEPSIS`, `LAT_GLOBAL_SEVERITY`, `LAT_SHOCK`, `LAT_RESPIRATORY_FAILURE`, `LAT_RENAL_DYSFUNCTION`, `LAT_HEPATIC_COAG_DYSFUNCTION`, `LAT_NEUROLOGIC_DYSFUNCTION`, `LAT_METABOLIC_DERANGEMENT`, `LAT_CARDIAC_STRAIN`.

The MIMIC archive header matches this set. Compared with PhysioNet, MIMIC uses a chronic-burden column rather than chronic-baseline-risk, combines hepatic and coagulation evidence into one construct, shortens the inflammation/sepsis and cardiac-strain names, and changes several rule inputs. These naming differences are therefore construct and source-feature differences, not just cosmetic renaming.

The script supports three mutually exclusive input modes. In summary-CSV mode, a precomputed one-row-per-stay table is loaded; accepted identifier columns include `icustay_id`, `patient_id`, or `stay_id`. In canonical-pickle mode, the script reads a PhysioNet-compatible MIMIC payload [`ts`, `oc`, `ts_ids`], canonicalizes `ts_id`, reconstructs summary features from long events, derives GCS minima from `GCS_eye`, `GCS_motor`, and `GCS_verbal`, aggregates first-24-hour urine output and rolling 6-hour ml/kg/hour urine rates when weight is available, and merges optional outcome/context fields from `oc`. Aliases include `MBP` to MAP, `PCO2` to PaCO2, `PO2` to PaO2, `O2 Saturation` to SpO2, `Creatinine Blood` to creatinine, and `Bilirubin (Total)` to bilirubin. In raw-concept mode, the admissions table is required, and optional diagnoses, vitals, labs, urine, vasopressor, ventilation, infection, and troponin-map CSVs are aggregated into the summary table.

Binary helper flags are treated as evidence only when they are positive or truthy. Numeric comparisons require non-missing values. Raw-concept mode fills absent binary helper columns with zero, while pickle mode inserts missing helper columns as missing; in both cases missing numeric evidence does not make a rule positive. Consequently, input mode and concept extraction completeness affect which clauses can fire.

Table 5.5: MIMIC rule-derived proxy-state definitions from the active tagger.

Proxy-state column	Construct represented	Input measurements and aggregation	Rule structure or threshold family	Grounding and validation status
`LAT_CHRONIC_BURDEN`	Baseline chronic burden or vulnerability.	Age; deidentified-old flag; comorbidity count; Elixhauser score; chronic ICD/organ disease helpers; malignancy or immunosuppression; admission type.	Score at least 2 across age ≥ 75 or old-age flag, comorbidity count ≥ 2 or Elixhauser ≥ 5, chronic disease helper, malignancy/immunosuppression helper, or emergency/urgent admission.	Electronic phenotyping context [5]; exact mix is project-specific and input-mode dependent.
`LAT_INFLAMMATION_SEPSIS`	Inflammation, suspected infection, or sepsis-like burden.	Temperature, WBC, lactate, culture/suspected-infection flags, antibiotic/suspected-infection flags.	Score at least 2 across temperature ≥ 38.3 or < 36, WBC > 12 or < 4, lactate > 2.0, culture evidence, or antibiotic evidence; requires infection/inflammation anchoring by culture, antibiotic, temperature, or WBC evidence.	Sepsis context [24]; not formal Sepsis-3 diagnosis.
`LAT_GLOBAL_SEVERITY`	Multi-domain acute physiologic severity.	Circulatory, respiratory, neurologic, renal, metabolic, inflammatory, hepatic/coagulation domains from summaries and helper flags.	At least 3 domains. Clauses include MAP < 70, SBP ≤ 100, vasopressors, SpO2 < 92, PF ≤ 300, mechanical ventilation, GCS < 15, RASS ≤ -3 or ≥ 2, creatinine ≥ 2.0, urine < 500, lactate > 2.0, pH < 7.30, bicarbonate < 18, inflammatory thresholds, bilirubin ≥ 2.0, platelets < 100, or INR ≥ 1.5.	Broad Sepsis-3/SOFA grounding [24, 27]; not complete SOFA.

Proxy-state column	Construct represented	Input measurements and aggregation	Rule structure or threshold family	Grounding and validation status
`LAT_SHOCK`	Shock or hemodynamic instability.	MAP, SBP, sustained hypotension helper, vasopressors, lactate, urine output, pH, base excess.	Score at least 2, or vasopressor use plus tissue hypoperfusion, acidosis, or hypotension. Thresholds include MAP < 65, SBP ≤ 90, lactate > 2.0, urine < 500 in 24 hours or < 0.5 ml/kg/hour over 6 hours, pH < 7.30, base excess ≤ -5.	Shock/lactate grounding [24]; proxy rule remains source-specific.
`LAT_RESPIRATORY_FAILURE`	Respiratory failure or respiratory support.	SpO2, PF ratio, FiO2, mechanical/noninvasive ventilation, respiratory rate, PaCO2, pH.	Score at least 2, or respiratory support plus hypoxemia, oxygenation failure, or ventilatory failure. Thresholds include SpO2 < 92, PF ≤ 300, FiO2 ≥ 0.5, RR ≥ 30 or ≤ 8, and PaCO2 ≥ 50 with pH < 7.30.	Respiratory phenotyping/ARDS context [23, 28]; not adjudicated ARDS.
`LAT_RENAL_DYSFUNCTION`	Renal dysfunction.	Creatinine maximum and delta, BUN, urine output, potassium, bicarbonate, dialysis/CRRT helper.	Score at least 2, or dialysis/CRRT present. Clauses include creatinine ≥ 2.0 or delta ≥ 0.3, BUN ≥ 40, low urine output, potassium ≥ 5.5, or bicarbonate < 18.	KDIGO grounding for creatinine/urine concepts [25]; not full KDIGO staging.
`LAT_HEPATIC_COAG_DYSFUNCTION`	Combined hepatic and coagulation dysfunction.	Bilirubin, AST, ALT, platelets, INR, PT/PTT helpers, albumin.	Score at least 2 across bilirubin ≥ 2.0, AST or ALT ≥ 120, platelets < 100, INR ≥ 1.5 or PT/PTT prolonged, or albumin < 2.5.	SOFA and ISTH DIC context [27, 26]; combined construct needs clinical review.

Proxy-state column	Construct represented	Input measurements and aggregation	Rule structure or threshold family	Grounding and validation status
`LAT_NEUROLOGIC_DYSFUNCTION`	Neurologic dysfunction with sedation caveat.	GCS, GCS component abnormality, RASS, pupil/focal-neurologic helper, sedation/intubation helper.	Score at least 2. GCS ≤ 8 counts twice; GCS ≤ 13, abnormal GCS component, RASS ≤ -3 or ≥ 2, and pupil/focal abnormality contribute. A single sedation/intubation-confounded point is not sufficient.	SOFA neurologic context [27]; sedation ambiguity remains.
`LAT_METABOLIC_DERANGEMENT`	Acid-base, lactate, electrolyte, or glucose derangement.	pH, bicarbonate, base excess, lactate, potassium, sodium, glucose.	Score at least 2 across pH < 7.30 or > 7.55, bicarbonate < 18 or > 35, base excess ≤ -5, lactate > 2.0, potassium < 3.0 or ≥ 5.5, sodium < 130 or > 150, or glucose < 70 or > 250.	Project-specific metabolic proxy.
`LAT_CARDIAC_STRAIN`	Cardiac strain or injury.	Troponin flags and values, CK-MB, arrhythmia helper, HR, hypotension, cardiac care unit or surgery context.	Score at least 2 and requires biomarker or rhythm evidence. Biomarker clauses include troponin-positive flags, troponin T ≥ 0.1, troponin I ≥ 0.4, or CK-MB evidence; rhythm/hemodynamic clauses include arrhythmia, HR > 150 or < 40, HR stress with hypotension, or cardiac-care context.	Project-specific cardiac proxy.

Exact clinical-source support remains incomplete for the MIMIC chronic-burden, metabolic/electrolyte/acid-base, and cardiac-strain rule clauses. These rows are therefore retained only as project-specific proxy definitions pending qualified clinical review.

The MIMIC output directory contains a binary tag CSV, a merged feature table, prevalence and co-occurrence summaries, mortality-association summaries, a JSON validation summary, and a decision-tree pickle. The tag CSV, `latent_tags.csv`, is keyed by `ts_id`. The merged feature table retains the summary features used by the rules together with the resulting tags, making it a diagnostic trace of the construction process. The prevalence table describes how often each proxy label fires; those values are descriptive properties of the labels, not clinical validation. The mortality-by-tag table reports unadjusted mortality association by proxy value, and its risk ratios are not causal estimates. The co-occurrence table reports pairwise association among proxy states. The JSON validation summary stores available-latent metadata, prevalence records, mortality-association records, sanity checks, and interpretation notes. None of these files is a substitute for expert chart-review validation.

5.4 Predicted and Aggregated Proxy Labels

The same `LAT_*` column names can appear in three different source types. A rule-derived proxy label is generated by the PhysioNet or MIMIC tagger rules above. A predicted proxy label is exported by one of the four supervised time-series models described in Chapter 6. An aggregated predicted proxy-label representation is generated by aligning binary voter files and applying a deterministic majority-vote rule. These source types share an interface but not a generation mechanism. Across all twelve archived final causal runs, the vote logs enumerate exactly five sources: the dataset's deterministic rule-derived proxy-label table and thresholded predicted-label tables from GRU, GRU-D, STraTS, and TCN. The logs verify the filenames and downstream handoff, but the external voter files, their hashes, and their checkpoint-to-export mappings are not retained within the run folders.

Prediction exports contain `ts_id`, one probability column named `<label>_prob` per proxy target, and one thresholded binary column named exactly as the target label. The STraTS export code thresholds probabilities at 0.5. Downstream voting does not consume probability columns as votes. The helper `split_predicted_latent_tags.py` separates the probability half and binary-tag half of a combined prediction CSV, while the parent router can also drop `_prob` columns and write a binary-only voter table in configured `LAT_*` order. Export provenance remains incomplete, so this chapter does not claim that every archived prediction CSV is out-of-sample or tied to a verified checkpoint manifest.

The active majority-vote script discovers non-hidden `.csv` files in the input voter directory, sorts them by path string, and treats the first file as the reference column-order file. Every voter file must contain `ts_id`, have no duplicate normalized identifiers, and contain at least one non-identifier column. All non-`ts_id` columns are treated as latent columns; nonbinary values, including unsplit probability columns, cause validation failure. Later voters must have exactly the same latent-column set as the reference, although they are reordered to the reference order before voting. Rows are aligned on the intersection of `ts_id` values shared by all voters, and the output order follows the reference file restricted to that shared intersection.

For record i, proxy state k, and M aligned voter files, write $L_{ik}^{(m)}$ for the binary value from voter m. The implemented rule is

$$\widetilde{L}_{ik} = \mathbb{I}\left(\sum_{m=1}^{M} L_{ik}^{(m)} \geq \left\lceil \frac{M}{2} \right\rceil\right).$$

This is equivalent to the source expression `2 * ones_count >= n_voters`; therefore an even-voter tie is mapped to 1. The output schema is `ts_id` followed by the reference latent columns. The causal orchestrator writes this mixed-source aggregated proxy-label table under the run's `majority_vote/` directory and then passes it to mortality prediction, matching, CATE estimation, sensitivity analysis, and permutation stages.

Majority voting is an algorithmic aggregation of a binary rule-derived source and binary model-prediction sources, not expert consensus. It does not establish clinical truth, remove shared model bias, guarantee independence among voters, or prove ensemble superiority. It can amplify correlated errors because the predicted voters were trained against the same rule-derived targets and the rule source itself participates in the vote. Its interpretation depends on identical construct definitions, aligned identifiers, binary-only inputs, and the exact voter set supplied to each run. The archived logs enumerate that voter set, but the external source files and hashes are not archived with each run.

Chapter 6

Predictive Modeling of Proxy States

This chapter describes the predictive layer that estimates rule-derived proxy-state vectors from ICU time-series records. Its supervision chain is LLM-assisted decision-rule design, project selection and deterministic source encoding, generation of rule-derived proxy labels, and deep-model prediction of those labels. The supervised target is the resulting rule-derived proxy-label matrix loaded from a separate CSV file keyed by `ts_id`; the inputs are irregular measurement histories plus static covariates prepared under the data contracts described in Chapter 4. The models therefore predict analytical proxy labels rather than adjudicated clinical states.

The implementation compares STraTS-family irregular-observation models with recurrent, convolutional, attention-based, and decay-based baselines. This chapter defines the representation, training, evaluation, and export mechanics needed to connect these predictors to the later proxy-aggregation workflow. Final run configurations, selected result artifacts, and numerical model comparisons are intentionally deferred to Chapters 9 and 10.

6.1 Self-Supervised STraTS Pretraining

STraTS represents an irregular time series as a set of observed measurement tokens rather than as a fully imputed grid. The implemented STraTS class constructs each token by adding three learned components: a continuous embedding of measurement time, a continuous embedding of the normalized measurement value, and an embedding of the variable identity. The token sequence is contextualized with a transformer-style block and then pooled with fusion attention to obtain a fixed-dimensional time-series representation [3]. Static covariates are combined with this representation before either the self-supervised forecasting head or the supervised proxy-label head is applied.

The same source file implements both `strats` and `istrats` model types. In the verified implementation, `strats` pools contextualized observation embeddings and passes static covariates through a learned demographic embedding. The `istrats` branch instead pools the initial triplet embeddings and concatenates the raw static-covariate vector. The pretraining command-line guard permits self-supervised pretraining only for `strats` and `istrats`; the recurrent, convolutional, attention-grid, and decay baselines do not have a supported self-supervised pretraining path in the current source.

The pretraining dataset loader reads the split-aware processed artifact and removes held-out test identifiers before constructing unsupervised examples. It builds the variable vocabulary from variables observed in the pretraining train split, derives value means and standard deviations from that train split, and saves the resulting variable list, normalization table, and maximum time horizon to `pt_saved_variables.pkl`. For PhysioNet 2012 the maximum time is 48 hours. For MIMIC-III the loader permits observations up to five days but samples at most a 24-hour input context and clamps implausibly old age values in the same way as the supervised loader.

Each self-supervised example is formed by sampling a forecast anchor from observed timestamps that are not the final timestamp and are at least 12 hours after record start. Observations up to the anchor form the historical context, truncated by the maximum observation count and, for MIMIC-III, by the 24-hour context limit. The target window extends two hours after the anchor. For each variable observed in that future window, the implementation keeps the last observed normalized value and sets the corresponding forecast mask entry to one; variables without future observations are masked out of the loss. Input times are rescaled to the interval $[-1, 1]$ using the dataset-specific maximum time.

The pretraining head maps the combined time-series and static representation to one forecast value per variable. Its objective is masked squared error over variables observed in the sampled future window, so unobserved future variables do not contribute to the loss. The pretraining evaluator materializes batches for each evaluation split, sampling three forecast batches per evaluation chunk when a split is first evaluated, and reports `loss_neg`, the negative masked loss, for checkpoint selection. The best pretraining checkpoint is written as `checkpoint_best.bin`; loss values themselves are not reported in this methods chapter.

Checkpoint-loaded supervised training is implemented by constructing the target supervised architecture, loading the supplied pretraining checkpoint, and copying overlapping tensors into the current model state before calling `load_state_dict`. When `--load_ckpt_path` is supplied, the supervised dataset also loads `pt_saved_variables.pkl` from the same directory to reuse the pretraining variable vocabulary, normalization statistics, and maximum-time metadata. This source path supports warm-started STraTS-family training, but the archived final exports still lack a complete manifest linking each exported CSV to the intended pretraining checkpoint, supervised checkpoint, split artifact, and archive-copy step.

6.2 Supervised Multi-Label Models

The supervised prediction task uses a deterministic rule-derived latent-label CSV as its target matrix. The loader accepts identifier columns named `ts_id`, `icustay_id`, or `ICUSTAY_ID`, canonicalizes them to `ts_id`, and checks consistency when more than one accepted identifier column is present. It then filters the label table to the supervised split identifiers, raises an error for missing labels, raises an error for duplicate normalized identifiers, sorts rows to match the internal supervised identifier order, and treats every non-`ts_id` column as a proxy-label target. The number and order of supervised targets are therefore determined at runtime by the CSV schema, but their semantic status remains rule-derived rather than clinical ground truth.

All supervised backends share a multi-label output contract. During training the model returns binary cross-entropy with logits against the full target vector, using per-target positive-class weights computed from the supervised train split. During evaluation or prediction export, the same model call without labels returns sigmoid probabilities. Static covariates are normalized from training rows and are concatenated, either after a learned demographic embedding or directly in the `istrats` branch, with each model's time-series representation. In scratch supervised training the binary head maps the combined representation directly to the proxy-label logits. In checkpoint-loaded training the source inserts an intermediate forecasting head before the binary head so that overlapping pretraining tensors can be reused.

The STraTS supervised backend uses the irregular observation triplets described in Section 6.1: time, value, and variable embeddings, transformer contextualization, fusion attention, and static-feature combination [3]. The `istrats` branch shares the same triplet construction and fusion-attention module, but its verified pooling and static-feature path differ as described above. The final artifact archive contains STraTS prediction CSVs, while no final iSTraTS prediction artifact was found.

The GRU baseline uses a dense hourly grid rather than the raw irregular token set. For each hour and variable, the loader builds value, observation-mask, and elapsed-time-since-observation channels, mean-fills unobserved values from training data, normalizes the filled values, and concatenates the three channel groups before passing them to a gated recurrent unit [21]. The final recurrent hidden state is concatenated with the static-feature embedding and projected through the shared supervised head.

The GRU-D backend uses a model-specific irregular sequence representation containing observed values, missingness masks, elapsed-time tensors, sequence lengths, and static covariates [4]. The implemented cell updates last observed values, applies learned exponential decay to inputs and hidden states as a function of elapsed time, and includes missingness masks in the recurrent gates. The representation passed to the supervised head is the final valid recurrent state for each sequence combined with the static-feature embedding.

The TCN baseline uses the same dense hourly value, observation-mask, and elapsed-time representation as the GRU baseline, but processes the sequence with temporal convolutional residual blocks [22]. The implementation permutes the dense grid into channel-first form, applies a temporal convolutional network, takes the last temporal embedding, and concatenates it with the static-feature embedding before projection to proxy-label logits.

Table 6.1: Implemented predictive model families and non-numeric evidence status.

Model	Temporal representation and irregularity handling	Static features and pretraining path	Citation and repository evidence status
STraTS	Irregular observation tokens built from time, value, and variable embeddings; observation masks handle padding, and fusion attention pools contextualized tokens.	Learned demographic embedding is concatenated with the pooled representation; self-supervised pretraining is supported.	[3]. Implementation confirmed; wrappers and archived STraTS outputs present; final checkpoint-to-export manifest incomplete.
GRU	Dense hourly grid with value, observation-mask, and elapsed-time channels; unobserved values are mean-filled before normalization.	Learned demographic embedding is concatenated with the final recurrent state; pretraining is not supported by the current guard.	[21]. Implementation confirmed; archived GRU prediction CSVs present; export provenance incomplete.
GRU-D	Irregular sequence of value, mask, elapsed-time, and sequence-length tensors; learned decays and masks enter the recurrent update.	Learned demographic embedding is concatenated with the final valid recurrent state; pretraining is not supported by the current guard.	[4]. Implementation confirmed; archived GRU-D prediction CSVs present; export provenance incomplete.
TCN	Dense hourly grid with value, observation-mask, and elapsed-time channels processed by temporal convolutional residual blocks.	Learned demographic embedding is concatenated with the last convolutional embedding; pretraining is not supported by the current guard.	[22]. Implementation confirmed; archived TCN prediction CSVs present; export provenance incomplete.

The supervised evaluator computes metrics independently for each rule-derived target column in the requested split. Target columns with only one observed class in that split are skipped. For the remaining targets, it computes AUROC, area under the precision-recall curve, and the maximum value of the pointwise minimum of precision and recall. The reported split-level values are simple averages over the non-degenerate target columns, with loss and negative loss also recorded. During supervised training, checkpoint selection uses the validation value of `auprc + auroc`; this validation score should not be interpreted as a test score. These metrics quantify learnability or reproducibility of rule-derived targets, not clinical construct validity or correctness of the LLM-assisted decision-rule design. A model can reproduce both useful rule structure and systematic rule error.

Training mechanics are shared across the supervised backends at the top-level script. The script constructs a deterministic seed by adding the run index to the configured seed, uses AdamW for trainable parameters, clips gradient norms at 0.3, evaluates at the configured validation schedule, and saves `checkpoint_best.bin` when the validation selection metric improves. If an `eval_train` split is evaluated, it is the first 2000 training examples in the current supervised split object, which makes it a diagnostic subset rather than a separate external evaluation set.

6.3 Prediction Export and Normalization

Prediction export is requested with `--save_pred_csv_path`. The export routine can run after a training run or in an export-only invocation with no training steps scheduled. Immediately before export, the script reloads `checkpoint_best.bin` from the current output directory only if that file exists. The exported split is selected by `--predict_split`, whose accepted values are `train`, `val`, `test`, and `all`; in the `all` case the exporter iterates over all supervised identifiers in the train, validation, and test split object. The inspected wrapper scripts and archived export logs support `predict_split=all` for the final prediction CSVs. The logs also record `seed=2023`, and archived `checkpoint_best.bin` files can be hashed; nevertheless, the records do not supply split identifiers, a processed-artifact hash, or an independently verified checkpoint-to-export and archive-copy manifest.

For every exported row, the exporter removes labels from the batch, obtains probability outputs from the model, thresholds each probability at 0.5, and writes a combined CSV. The first column is `ts_id`. It is followed by one probability column named `<label>_prob` for each target and then one thresholded binary prediction column named exactly as the target label. These binary columns are model-predicted proxy labels, not rule-derived labels and not clinical adjudications.

Table 6.2: Prediction export schema before downstream voter normalization.

Column group	Naming pattern	Meaning	Source of value
Identifier	`ts_id`	Normalized supervised record identifier.	STraTS dataset loader and split object.
Probability outputs	`<label>_prob`	Sigmoid model probability for a proxy-label target.	Supervised model called without labels.
Binary predictions	`<label>`	Thresholded model-predicted proxy label.	Probability output thresholded at 0.5.

The combined prediction CSV is not the same file shape required by downstream voter aggregation. The helper `split_predicted_latent_tags.py` validates that `ts_id` is the first column, expects an even number of non-identifier columns, splits the first half into probability columns and the second half into binary tag columns, and writes separate probability and absolute-tag CSVs. The majority-vote input contract is stricter: each voter file must contain `ts_id` plus binary proxy columns, with all voter files sharing the same latent-column set after validation. The parent router can collect STraTS outputs, drop probability columns, enforce the approved latent ordering, and write normalized voter CSVs for later aggregation. The semantics of the aggregated proxy-label representation belong to Chapter 5; the archived final vote combines one rule-derived table with predicted-label tables from GRU, GRU-D, STraTS, and TCN. Here the important point is the predictive handoff contract. Majority voting is algorithmic aggregation, not expert consensus.

Chapter 7

Causal Graph and Effect-Estimation Methodology

7.1 Dataset-Specific DAGs

The causal component of this project does not learn a causal graph from the observational data. It uses two LLM-assisted, project-specified directed acyclic graphs (DAGs), one for PhysioNet 2012 and one for MIMIC-III, as explicit design assumptions for selecting adjustment variables and organizing later effect-estimation outputs. A structured clinical and causal elicitation process generated graph proposals that were selected and encoded as active graph-construction source code. Work on LLM-derived soft priors provides a methodological precedent for using semantic judgments as candidate causal relations or orientation information [14], but it does not validate these local graphs or imply that its discovery algorithm was implemented here. The diagrams should therefore be read as source-encoded causal hypotheses, not as clinically validated or statistically discovered ground truth.

This separation is important because the main exposure variables in the causal stage are binary proxy states rather than randomized treatments. In the causal code, each selected proxy state is treated as a binary exposure A, the outcome is in-hospital mortality Y, and the graph is used to identify candidate pre-exposure background or latent proxy variables for adjustment. The thesis follows the graphical language of Pearl's DAG and backdoor framework while retaining the cautions around observational, proxy-defined exposures and well-defined interventions discussed by Hernan and Taubman [7, 9]. These cautions are especially relevant here because several variables named as "treatments" in the code are clinically closer to patient states or care-process proxies than to assigned therapeutic interventions.

The LLM has no estimator-time role in this chapter's workflow. It is not called to inspect patient records, orient edges dynamically, select adjustment variables, or choose estimators during execution. The active graph scripts define the implemented nodes and edges, and the matching and CATE source code determines the implemented adjustment workflow from those encoded graphs and the columns available in each analysis dataframe.

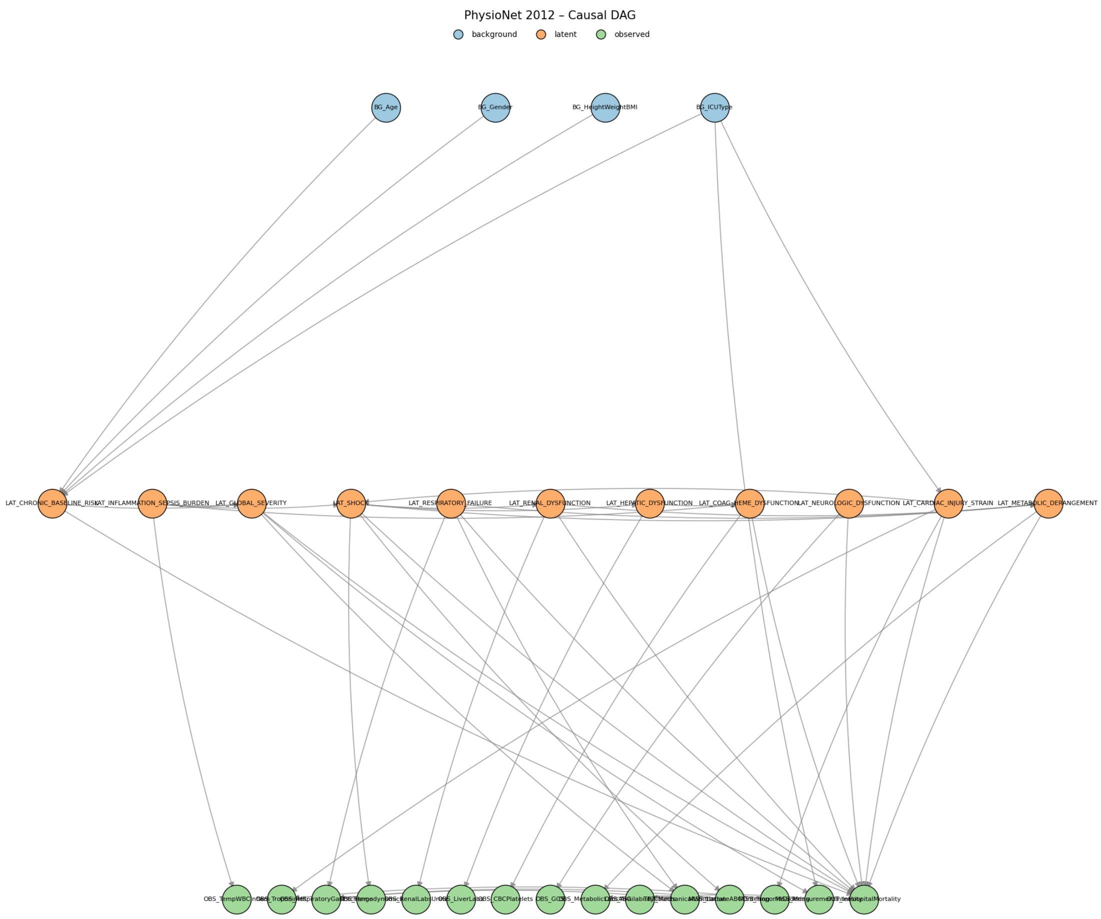

Figure 7.1: Project-specified PhysioNet 2012 DAG used by the causal pipeline. The figure is a thesis-local byte-identical copy of the canonical archived graph artifact. The active PhysioNet graph-construction source reproduced that artifact byte-for-byte in the audited environment, while the exact historical producing command remains unrecovered. Arrows encode assumed causal directions rather than statistically learned relationships; exact node and edge definitions remain source-code definitions.

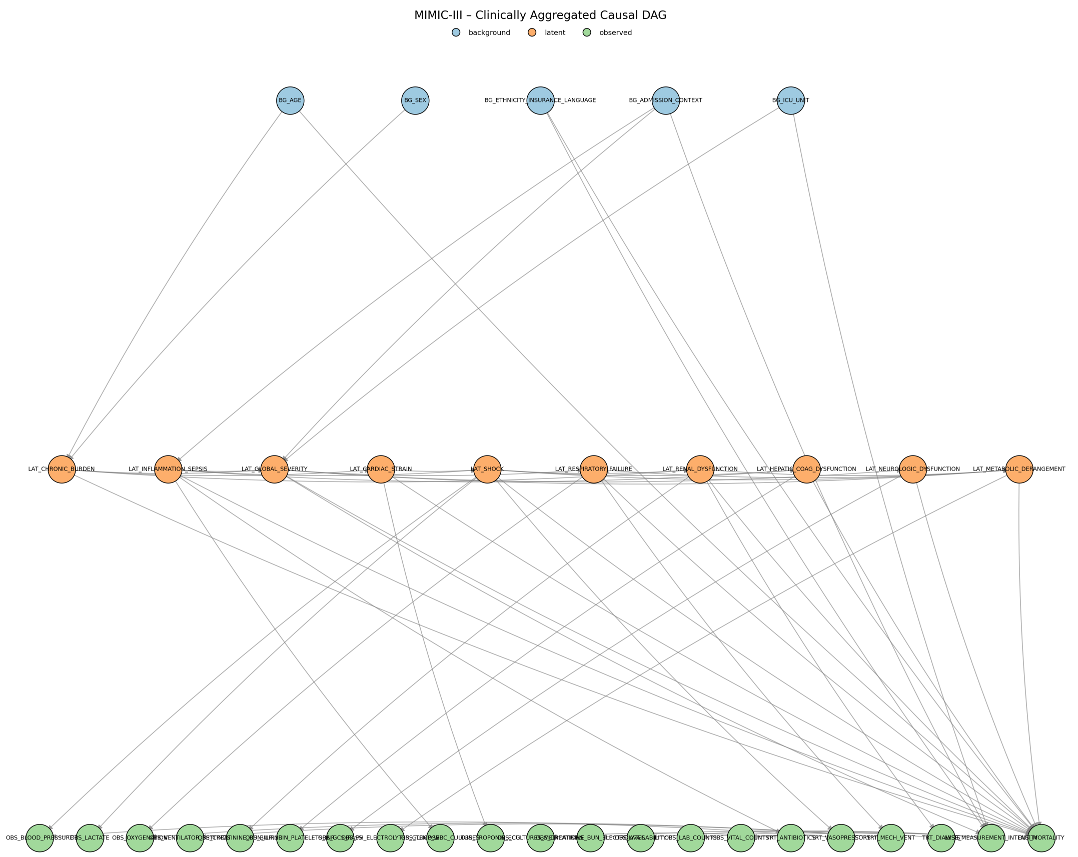

Figure 7.2: Project-specified MIMIC-III DAG used by the causal pipeline. The figure is a thesis-local byte-identical copy of the canonical archived graph artifact. The active MIMIC graph-construction source reproduced that artifact byte-for-byte in the audited environment, while the exact historical producing command remains unrecovered. Arrows encode assumed causal directions rather than statistically learned relationships; exact node and edge definitions remain source-code definitions.

Table 7.1: Implemented DAG node families used by the causal pipeline.

family	role in graph	PhysioNet 2012 implementation	MIMIC-III implementation
Background	Baseline variables allowed as adjustment candidates if mapped to dataframe columns.	Four nodes: age, gender, height/weight/BMI, and ICU type.	Five nodes: age, sex, ethnicity/insurance/language, admission context, and ICU unit.
Latent or proxy state	Binary project proxy states. Selected non-chronic states are analyzed as exposure variables; eligible pre-exposure latent nodes may also enter adjustment sets.	Eleven nodes, including chronic baseline risk, inflammation/sepsis burden, global severity, shock, respiratory failure, renal dysfunction, hepatic dysfunction, coagulation/hematologic dysfunction, neurologic dysfunction, cardiac injury/strain, and metabolic derangement.	Ten nodes, including chronic burden, inflammation/sepsis, global severity, cardiac strain, shock, respiratory failure, renal dysfunction, hepatic/coagulation dysfunction, neurologic dysfunction, and metabolic derangement.
Observed measurements	Measurement aggregates and manifestations. These document how clinical constructs relate to observed data but are excluded from adjustment-set candidates.	Ten observed aggregate nodes, including inflammatory, cardiac, respiratory, hemodynamic, renal, liver, hematologic, neurologic, metabolic, and availability-count measurements.	Fifteen observed aggregate nodes, including vital, laboratory, medication, procedure, ventilator, culture, and availability/count nodes.

family	role in graph	PhysioNet 2012 implementation	MIMIC-III implementation
Treatment or care process	Care-process nodes retained in the graph to represent downstream clinical response or measurement context. These are not treated as adjustment candidates.	Mechanical ventilation node.	Antibiotics, vasopressors, mechanical ventilation, and dialysis nodes.
Missingness or measurement intensity	Process nodes for ordering and measurement availability. These are graph variables but not adjustment candidates.	Lactate/ABG ordering, troponin ordering, and measurement intensity.	Measurement intensity, with outgoing links to availability, laboratory-count, and vital-count nodes.
Outcome	Mortality endpoint.	`OUT_InHospitalMortality`.	`OUT_MORTALITY`.

The PhysioNet graph contains 30 nodes and 45 directed edges. The implemented exposure list contains the ten non-chronic latent proxy states in the PhysioNet configuration: global severity, shock, respiratory failure, renal dysfunction, hepatic dysfunction, coagulation/hematologic dysfunction, inflammation/sepsis burden, neurologic dysfunction, cardiac injury/strain, and metabolic derangement. Chronic baseline risk is retained as an upstream baseline proxy and effect-modifier candidate rather than as an analyzed exposure.

The MIMIC graph contains 36 nodes and 57 directed edges. The implemented exposure list contains nine non-chronic latent proxy states: global severity, inflammation/sepsis, shock, respiratory failure, renal dysfunction, hepatic/coagulation dysfunction, neurologic dysfunction, cardiac strain, and metabolic derangement. Chronic burden is retained as an upstream baseline proxy and effect-modifier candidate. The MIMIC graph contains richer administrative and care-process structure than the PhysioNet graph, but the downstream adjustment implementation only uses graph nodes that can be mapped into the analysis dataframe.

Qualified clinical review of the graph arrows and a complete record of the human decisions applied to the LLM-assisted proposals remain unavailable. The graph artifacts and active source structure are recorded, but the exact producing source version, configuration, and command are not fully recovered.

The archived final run summaries and majority-vote logs establish the downstream proxy-label lineage at the stage boundary. In all twelve runs, the vote combines five enumerated inputs: one deterministic rule-derived proxy-label table and four thresholded prediction tables from GRU, GRU-D, STraTS, and TCN. Each run writes `majority_vote/latent_tags_majority_vote.csv` and passes that mixed rule-and-prediction aggregate to mortality prediction, matching, and CATE estimation. The logs preserve the external absolute filenames, but not the voter-file bytes, hashes, or checkpoint-to-export mappings; those parts of the lineage therefore remain incomplete.

7.2 Adjustment-Set Logic

For each dataset, treatment-like proxy state, and outcome pair, the causal pipeline derives an adjustment set from the project DAG and the columns available in the analysis dataframe. The implementation first maps dataframe variables to graph nodes and graph nodes back to dataframe columns. Background and latent graph nodes are eligible only if they can be represented by available columns. Observed measurement nodes, care-process nodes, missingness nodes, and the outcome node are excluded from candidate adjustment sets.

Let G denote the implemented project DAG, A the binary proxy-state exposure, and Y the in-hospital mortality outcome. The code constructs a graph $G_{\underline{A}}$ by removing outgoing edges from A. This is a project helper for finding adjustment candidates; it should not be confused with the usual intervention graph notation that removes incoming edges into A. The initial candidate pool is:

$$\mathcal{C}(A, Y) = \mathcal{A}_{\text{allowed}} \cap \mathrm{An}_G(A) \cap \mathrm{An}_{G_{\underline{A}}}(Y) \setminus \mathrm{De}_G(A) \setminus \{A, Y\},$$

where $\mathcal{A}_{\text{allowed}}$ is the set of available graph nodes whose node type is background or latent. The implementation then enumerates backdoor paths between A and Y in the undirected skeleton whose first directed edge points into A. A path is treated as blocked if an adjusted non-collider is present or if an unadjusted collider has no adjusted descendant. This is an encoded graphical test over the project DAG, not proof that all real-world confounding has been controlled.

The preferred helper is the safe expanded adjustment mode. It begins with the full candidate pool, removes colliders on the encoded backdoor paths and removes their descendants, then checks whether the remaining pool blocks all encoded backdoor paths. If it succeeds, this expanded pool is returned. If it does not block all paths, the pipeline falls back to a greedy minimal helper that starts from the candidate pool and removes variables in sorted order only while all encoded backdoor paths remain blocked. The resulting set is inclusion-minimal under the implementation order; it is not guaranteed to be the unique minimal adjustment set.

Table 7.2: Implemented adjustment-set logic and thesis interpretation.

implementation step	source behavior	thesis interpretation
Graph and column aliasing	Maps selected dataframe columns such as age, sex/gender, ICU type, admission context, and chronic proxy states to project graph nodes.	Adjustment is limited by what the analysis dataframe actually contains; graph nodes without available mapped columns cannot be adjusted for.
Allowed nodes	Keeps only graph nodes typed as background or latent and available in the dataframe.	Measurement, care-process, missingness, and outcome variables are deliberately excluded from candidate adjustment sets.
Candidate pool	Intersects allowed nodes with ancestors of exposure and ancestors of outcome in the outgoing-edge-removed graph, then removes exposure descendants and the exposure/outcome nodes.	Candidate variables are source-defined pre-exposure or upstream variables under the project DAG.
Backdoor-path check	Enumerates encoded backdoor paths and applies collider/non-collider blocking rules.	The check assesses the implemented graph, not unmeasured real-world mechanisms outside the DAG.
Safe expanded set	Removes colliders and descendants of colliders before testing whether the remaining pool blocks paths.	Main adjustment mode, intended to reduce collider-conditioning risk.
Minimal fallback	Greedily removes sorted candidate variables while preserving path blocking.	A deterministic implementation fallback, not a claim that no alternative sufficient set exists.
Identifiability flag	Records whether available mapped variables block all encoded backdoor paths in the project DAG.	A project-DAG availability diagnostic; it should not be described as definitive causal identification in the clinical population.

Table 7.3: Causal assumptions required for interpreting the estimator outputs.

assumption	methodological requirement	repository support	unresolved limitation
Graph correctness for the target question	The DAG must correctly represent the relevant causal structure for the exposure contrast and mortality outcome.	Dataset-specific graphs were developed through LLM-assisted elicitation, project selection, and active source encoding.	The arrows were not learned from data, and the human clinical review record remains incomplete.
Sufficient measured adjustment	Available adjustment variables must block the relevant open backdoor paths for each proxy exposure.	The helper tests path blocking within the implemented DAG using mapped background and latent columns.	Graph variables may be unavailable, and unmeasured or misspecified confounding outside the project DAG remains possible.
Positivity or overlap	Comparable covariate strata must contain both exposed and unexposed records.	Matching outputs report match availability and match rates.	Dedicated treatment-specific overlap diagnostics and thresholds remain to be approved.
Consistency and well-defined exposure	Each binary proxy-state contrast must have a sufficiently stable meaning across records.	The pipeline applies one binary exposure column at a time.	Disease-state and severity proxies are not necessarily assignable interventions, so the causal estimand remains unresolved.
Proxy-state measurement	The binary exposure should measure the intended construct with sufficiently limited construct and classification error.	Final archived analyses use a deterministic aggregate of one rule-derived source and four predicted-label sources trained against the rule-derived targets.	No code path verifies construct validity or removes proxy-state measurement error; clinical and chart-review validation remains incomplete.

assumption	methodological requirement	repository support	unresolved limitation
Outcome measurement	The processed `in_hospital_mortality` field must be consistently and correctly constructed from the outcome artifact.	Both estimator paths merge this processed outcome column by record identifier and require it to be present.	Source checks establish schema availability, not endpoint validity or cross-dataset construction equivalence.
Temporal ordering and avoidance of post-exposure adjustment	Adjustment variables must be measured or conceptually occur before the relevant proxy-state contrast; post-exposure adjustment should be avoided.	Protection depends on the project DAG, source node classification, descendant filtering, and allowed-node filtering.	Repository graph ordering does not prove patient-record timing; actual variable timing and the correctness of the proxy-state construction window remain unverified.
Nuisance- and effect-model adequacy	LinearDML and CausalForestDML require sufficiently adequate nuisance and effect models, regularity conditions, and appropriate cross-fitting behavior for their intended interpretation.	The source configures EconML estimators, random-forest nuisance models, and estimator-provided fitting behavior.	Source configuration alone does not verify model adequacy, regularity conditions, calibration, or valid cross-fitting performance in these data.
No interference	One record's proxy-state exposure should not affect another record's outcome under the working unit-level interpretation.	Estimation treats records as separate units.	The repository does not empirically test interference or dependence across admissions, patients, units, or care processes.

assumption	methodological requirement	repository support	unresolved limitation
Sampling and target population	The analysis sample must support the stated target population, or reweighting or another transport argument must justify a sampled target.	The optional outcome-downsampling condition is deterministic given the configured seed and is applied consistently before treatment iteration.	Original and outcome-downsampled analyses correspond to different empirical populations; no weighting or target-population correction is documented.

7.3 Matching and Heterogeneous-Effect Estimators

The causal pipeline applies two families of estimators to the mixed-source majority-vote proxy-label exposures: a transparent matched-pair baseline and model-based heterogeneous-effect estimators. Both families use the same broad analysis frame. Patient-level aggregated proxy labels are merged with the canonical in-hospital mortality outcome and baseline covariates. For a selected exposure $A_i \in \{0, 1\}$, outcome Y_i, adjustment variables W_i, and effect modifiers X_i, the target working quantity is a contrast in predicted or matched mortality outcome between $A_i = 1$ and $A_i = 0$, conditional on the source-defined covariate information. Because the exposures are proxy states and the data are observational, these contrasts require cautious interpretation.

Measurement error can propagate through this entire chain. An unsupported design proposal can enter a selected rule or edge; a deterministic rule can misclassify the intended construct; a deep model can reproduce that error or add prediction error; aggregation can preserve correlated error; and the resulting exposure misclassification can alter adjustment, matching, and effect estimates. An LLM-assisted graph is not intrinsically more correct than a manually proposed graph, and predictive agreement with source-encoded labels does not repair graph or exposure-measurement assumptions.

An optional analysis condition, controlled by `DOWN_SAMPLE`, down-samples mortality outcome classes rather than exposure groups. All outcome-positive rows are retained. When the outcome-negative class is larger and the positive class is nonempty, outcome-negative rows are sampled without replacement until their count equals the number of outcome-positive rows; the configured random seed controls both this sample and the subsequent row shuffle. This occurs once before treatment-specific adjustment selection and estimation, including matching and DML/PFN fitting. It is not propensity trimming, treatment balancing, matching, or a correction for confounding, and it does not improve causal identification by itself. The procedure changes the analyzed population and observed outcome prevalence. The thesis therefore treats original cohorts as primary and outcome-downsampled cohorts as robustness analyses only, with no pooling between them.

Outcome-based downsampling can change nuisance-model estimation, the distribution of covariates and proxy exposures, patient-level fitted effects, the population over which `mean_cate` is averaged, and the availability and composition of matched pairs. It also changes the sample outcome rate and therefore affects `normalized_CATE`, whose denominator is that rate. Effect summaries from original and outcome-downsampled analyses should not be compared as though they estimated the same population quantity without an approved sampling and estimand argument. Outcome-based case-control sampling does not necessarily invalidate every estimator, but interpretation here requires estimator-specific methodological justification and an explicit target-population argument.

The matching estimator is a deterministic baseline. For each proxy exposure, the pipeline drops rows with missing exposure or outcome, coerces the exposure and outcome to binary variables, imputes numeric adjustment variables with medians, and converts non-binary numeric adjustment variables into binary columns using sample-median thresholds. Already-binary variables, including ICU-type indicators, remain binary. The matcher then greedily pairs exposed and unexposed patients by Hamming distance over these binary adjustment columns, increasing the allowed distance up to the configured maximum. By default, controls are not reused. The per-pair contrast is

$$\Delta_j = Y_{j,A=1} - Y_{j,A=0},$$

and the matching summary reports the mean and standard deviation of these matched-pair differences, the number of matched pairs, the match rate, the final allowed Hamming distance, and the observed adjustment columns. This output is useful as a transparent matched descriptive contrast. It should not be labeled as an average treatment effect unless the estimand, graph assumptions, overlap, and matching design are later approved for that interpretation.

For model-based heterogeneous effects, the pipeline includes EconML double-machine-learning estimators and an exploratory CausalPFN branch. The prespecified hierarchy treats CausalForestDML as primary, LinearDML as the secondary comparator, and CausalPFN as exploratory. Double machine learning estimates nuisance outcome and treatment models and then estimates treatment effects after residualization, reducing sensitivity to nuisance-model overfitting under appropriate assumptions [15, 29]. The LinearDML branch uses random-forest nuisance models with a linear final effect model. The CausalForestDML branch uses the same random-forest nuisance-model family and a causal forest final stage, drawing on generalized random-forest and causal-forest ideas for heterogeneous treatment effects [16, 17]. This modeling choice is consistent with the broader use of machine learning for individualized treatment-effect estimation in electronic health record and critical-care settings, while still requiring domain-specific validation [18, 19, 30, 31, 32].

The CATE pipeline saves patient-level CATE values, normalized CATE values, optional 95 percent effect-interval columns when the estimator provides them, feature-importance or coefficient diagnostics when available, model artifacts, and run-level summaries. In the current source, `mean_cate` is the arithmetic mean of patient-level CATE estimates over the model rows. It is an aggregate of the fitted model's conditional estimates and should not automatically be described as the study ATE. The normalized CATE column is computed by dividing CATE by the sample outcome rate when that rate is positive. It is an implementation-defined rescaling for comparison within this project; it is not a risk ratio, relative risk, or percent reduction.

The effect-modifier matrix X is selected from configured effect-modifier columns, typically baseline variables plus chronic and inflammation/sepsis proxy states when present. The adjustment matrix W is the observed confounder set selected from the DAG helper. The implementation can allow overlap between X and W, because configured effect modifiers are not automatically removed from the adjustment set. This is a source-supported behavior and a remaining methodological limitation.

Table 7.4: Implemented causal estimators and interpretation limits.

method	implementation summary	main outputs	interpretation boundary
Greedy Hamming matching	Binary adjustment matrix, median thresholding for non-binary covariates, median imputation, nearest available control within allowed Hamming distance, no default control reuse.	Matched pairs, pair effects, match rate, final distance, and per-treatment/global summaries.	Matched descriptive contrast; not automatically ATE or ATT.
LinearDML	EconML `LinearDML` with random-forest nuisance outcome and treatment models, binary treatment handling, configured W and X.	Patient-level CATE, optional intervals, coefficients when aligned, model artifact, global summaries.	Model-based heterogeneity estimate under project DAG, nuisance-model, overlap, and proxy-exposure assumptions.
CausalForest DML	EconML `CausalForestDML` with random-forest nuisance models and a causal-forest final stage.	Patient-level CATE, optional intervals, feature importance, model artifact, global summaries.	Flexible heterogeneity estimate; feature importance is a model diagnostic, not mechanism proof.

method	implementation summary	main outputs	interpretation boundary
CausalPFN	Exploratory branch using deduplicated confounder and effect-modifier features. It writes CATE CSVs and metadata but skips EconML-specific sensitivity and permutation stages.	Patient-level CATE and normalized CATE; interval columns are unavailable and stored as missing.	Exploratory only; its primary method source, implementation version, uncertainty, and comparable diagnostic support remain unresolved.

Chapter 8

Robustness, Sensitivity, and Validation Design

This chapter records the diagnostic framework surrounding the causal analyses in Chapter 7. It concerns empirical support, omitted-confounding sensitivity, estimator-diagnostic availability, benchmark-confounder comparisons, permutation checks, and run integrity. These procedures supplement rather than replace the project DAG, adjustment assumptions, and exposure definition. In particular, they do not validate the DAG, prove exchangeability, establish that a proxy-state exposure is well defined, or establish clinical actionability. They also do not compensate automatically for measurement error, selection bias, unmeasured confounding, insufficient overlap, an incorrect target-population interpretation, or incomplete run provenance.

8.1 Overlap and Support Diagnostics

The matching implementation provides indirect empirical support evidence under its implemented matching representation. For each proxy-state exposure, the matching summary records the number of treated and control records, number of matched pairs, match rate, final allowed Hamming distance, a sufficient-pair flag, a maximum-distance failure flag, and treatment-level failure information where matching cannot proceed. These fields are useful support indicators because a comparison requiring no or few acceptable pairs merits caution. They are not a propensity-score overlap analysis.

Matching is greedy, one-to-one, and based on a binary representation of the selected confounders. Binary columns are retained and other numeric columns may be median-binarized before Hamming distances are evaluated. Consequently, a larger accepted distance denotes weaker exact similarity in that representation; it is not a calibrated distance in the original covariate space. Failure to obtain enough pairs is a support warning, not a zero effect. Conversely, an adequate matching rate does not prove positivity for the intended target population. The binarization, greedy ordering, allowed-distance rule, and outcome-downsampled analysis frame all affect this diagnostic.

Dedicated propensity-score distributions, common-support plots, covariate balance before and after matching, standardized mean differences, weighting effective sample sizes, treatment-probability histograms, and formal positivity diagnostics were not found in the approved archive. Not every analysis necessarily requires every one of these diagnostics, but their absence limits the strength of overlap claims. Limited overlap is a general concern for observational effect estimation, not evidence that this repository satisfies positivity [33].

Table 8.1: Overlap and support diagnostic design; this table contains no numerical results.

Diagnostic	Repository source	What it assesses	What it does not establish	Result status
Treated versus control counts, pairs, and match rate	`matching/global_summary.csv`; per-treatment summaries	Whether the implemented matcher found comparison pairs	Propensity overlap, covariate balance, or positivity	archived; indirect only
Final Hamming distance and sufficient-pair flags	`matching_causal_effect.py`; matching summaries	Similarity under the binary matching representation and support warnings	Similarity in the original covariate space or a target-population guarantee	implemented; archived
Matching failure reason	Per-treatment summaries; cross-run matching-failure table	Why a matching contrast was not produced	A zero effect or absence of an exposure effect	archived
Propensity, common-support, and balance diagnostics	Approved archive search	Dedicated treatment-probability and pre/post-match balance evidence	Any conclusion about their value in a particular run	not found
Thesis support display	Validated original-cohort matching table in Chapter 10	Pair availability, match rates, accepted distance, and warning or failure status	Empirical positivity or covariate balance	Selected descriptive support summary

8.2 Sensitivity and Robustness Values

The repository separates several evidence layers that should not be collapsed into the statement that an estimator "produced sensitivity results": (1) estimator-method availability recorded during fitting; (2) values called directly from the fitted estimator during training; (3) serialized estimator parameters or residuals; (4) later saved-artifact analysis; (5) a fallback or reconstructed diagnostic; (6) benchmark-confounder comparison; and (7) a rendered sensitivity contour. Robustness-value and omitted-variable-bias sensitivity concepts provide a useful framework for this work [34, 35], but these references do not establish the correctness of local artifacts or assumptions.

During non-PFN fitting, `cate_estimation.py` records whether `robustness_value`, `sensitivity_interval`, `sensitivity_summary`, `summary`, and residual access are available. It records direct-call values, source labels, unavailable-API errors, a serializable sensitivity-parameter snapshot when exposed, residual availability, estimator metadata, artifact schema version, and package/environment fields. The per-treatment model artifact and the run-level control-message CSV preserve these fields. A non-null value must therefore be interpreted together with its recorded source and status; it is not enough to inspect a numerical cell alone. EconML is the relevant software context for the DML and causal-forest branches, distinct from a claim that the software validates local identification [29].

The saved-artifact analysis requires a CATE result directory, then loads available model artifacts and reconstructs the relevant latent-tag and processed-data inputs. It first considers saved training-direct information and loaded-estimator calls, and it records residual-dependent calculations, benchmark-variable availability, per-treatment reports/CSVs, contours, warnings, and run-level aggregation. A report can be partial or failed; archived stage completion alone does not imply complete sensitivity information for every treatment.

For clarity, this thesis uses *estimator-native* for a method called on the fitted estimator, *saved-training direct* for that value preserved by the training artifact, and *saved-artifact recomputed* for a later direct call or reconstruction. A *fallback diagnostic* is a separately labelled calculation using serialized or residual-space information when the desired estimator API or serialized object is unavailable. *Unavailable*, *partial*, *failed*, and *not applicable* are status outcomes, not numerical sensitivity values. A fallback exists to retain diagnostic information across version and serialization limitations, but it depends on its own reconstruction assumptions and cannot be presented as identical to an estimator-native calculation. Any reported sensitivity quantity must therefore retain its source and status classification.

Benchmark-confounder analysis seeks an interpretable scale: it compares a hypothetical omitted-confounding strength with an observed candidate covariate benchmark. The analysis checks candidate-variable availability, records the selected benchmark and source when possible, and records missing or unimplemented benchmark paths rather than treating them as null evidence. Such a comparison can expose sensitivity; it cannot show that an observed benchmark bounds every possible unmeasured confounder.

The CausalPFN branch does not implement the same EconML residual, interval, sensitivity, or permutation workflow. Its missing diagnostics are therefore not zero sensitivity or robustness. The citation and validation limitation for CausalPFN stated in Chapter 7 remains in force.

Table 8.2: Sensitivity-diagnostic design and interpretation boundary.

Diagnostic type	Source stage	Applicable estimators	Required saved object	Output form	Interpretation boundary
Method availability	CATE fitting	EconML branches	Fitted estimator metadata	Boolean availability and errors	API availability is not a diagnostic value
Saved-training direct	CATE fitting	EconML branches where callable	Schema-versioned model artifact	Value plus source/error fields	Interpret only as labelled direct training-time evidence
Saved-artifact recomputed	Saved-CATE analysis	Non-PFN when artifact or environment permits	Model artifact and reconstructed inputs	CSV/report source fields	Later recomputation can differ from training-time access
Fallback diagnostic	Saved-CATE analysis	Non-PFN when direct paths fail	Serialized parameters or residual information	Explicit fallback source and warnings	Not identical to estimator-native output
Benchmark-confounder comparison	Saved-CATE analysis	Non-PFN where candidates are available	Candidate covariate and analysis inputs	Benchmark CSV/report	Gives a scale, not a bound on all unmeasured confounding
Sensitivity contour	Saved-CATE analysis	Non-PFN where parameters permit	Parameters or residual reconstruction and plot path	Rendered diagnostic figure	Requires source and status validation before selection
CausalPFN diagnostic status	Orchestration	CausalPFN	N/A for EconML workflow	Skipped/not applicable status	Absence is not a zero diagnostic

No sensitivity contour is selected as main-text evidence. Chapter 10 reports validated availability and status counts rather than combining diagnostically non-equivalent magnitudes.

8.3 Permutation Checks and Reproducibility Validation

The permutation script is a sanity-diagnostic wrapper around repeated CATE estimation. For a treatment permutation, it copies the latent-tag dataframe and shuffles only the currently iterated treatment column. The remaining latent columns, processed outcome data, graph input, dataset configuration, and estimator type are held fixed. Each temporary latent CSV is passed to `cate_estimation.py` in a subprocess, and the temporary directory is removed after the aggregate metrics are extracted.

For an outcome permutation, the script copies the processed dataset structure and shuffles only the configured mortality outcome column in the outcome dataframe. The time-series object, identifiers, latent-tag dataframe, graph input, dataset configuration, and estimator type remain unchanged. One temporary outcome-pickle run yields treatment-level summaries for that trial, after which the temporary input and output directory are removed. Both procedures derive deterministic trial seeds from a base seed, an experiment code, the treatment index where applicable, and the trial index. Trial count, base seed, dataset, estimator, sampling condition, aggregation fields, warnings, and subprocess status must be verified for every reported archived run. The validated non-PFN source table exposes ten trials and seed 42 for every DML dataset–exposure–sampling combination; Chapter 10 reports this only as diagnostic availability.

The script captures nonzero subprocess return codes and summary-schema failures as errors rather than silently using incomplete trial data. It supports the LinearDML and CausalForest-DML paths; CausalPFN is excluded by the orchestrator from saved-CATE analysis and permutations. The resulting aggregate CSVs are diagnostic artifacts, not final effect tables. They can indicate whether similarly structured outputs arise after a key relationship is disrupted. They do not prove that an original effect is causal, that the DAG is correct, that confounding is controlled, that the estimator is calibrated, that the proxy exposure is clinically valid, or that an original result is unbiased. In this limited sense they are negative-control-like disruptions or null-reference exercises, not formal randomization tests. A model–identify–estimate–refute workflow motivates such checks at a general level, but does not show that this repository implements DoWhy [36].

Configuration validation is a separate reproducibility control. The Python and shell validators check the compact dataset-config schema and invoke active scripts with `--validate-config-only`; router tests additionally exercise command parsing and construction. The causal orchestrator writes `run_summary.json` with resolved stage commands, paths, statuses, timestamps, and expected outputs, while CATE and saved-analysis control-message CSVs record field-level provenance. These are different evidence layers: schema/config validation, command-construction validation, stage execution status, artifact existence, and scientific validity. A successful config check does not prove full execution, and a successful stage status does not validate a causal claim.

Table 8.3: Permutation and reproducibility-validation design.

Check	Disrupted element	Held-fixed elements	Output	Primary purpose	Limitation
Treatment permutation	One iterated treatment column in copied latent tags	Other latent columns, outcome data, graph, config, estimator	Aggregate treatment permutation CSV	Sanity check after exposure label disruption	Does not establish identification or calibration
Outcome permutation	Mortality column in copied processed outcome dataframe	Time series, IDs, latent tags, graph, config, estimator	Aggregate outcome permutation CSV	Null-reference exercise after outcome disruption	Does not validate proxy exposure or DAG
Configuration validation	None; configuration only	Compact schema and declared script contracts	PASS or FAIL; resolved field output	Detect invalid or missing config fields	Does not load data or execute analysis
Run-summary stage status	None; archived orchestration record	Stage command, log, expected outputs	`run_summary.json`	Execution and provenance audit	Status is not scientific validity
Artifact-schema check	None; header/status inspection	Selected artifact path and producing stage context	Manifest validation record	Prevent wrong file or wrong granularity use	Does not establish result correctness

Chapter 9

Experimental Design

This chapter defines the experiment families and reproducibility boundary underlying the selected results. It covers datasets, prediction experiments, causal run families, original and outcome-downsampled conditions, configuration resolution, workflow orchestration, software and hardware recording, and artifact requirements. The LLM-assisted proxy, rule, and DAG design stage precedes these experiments and is documented in Chapters 3 and 5; it is not rerun by the patient-level experiment orchestrators. Chapter 4 describes data preparation, Chapter 6 describes predictive models, Chapter 7 describes causal estimators, and Chapter 8 describes diagnostic design. This chapter reports design and provenance rather than empirical findings; the analysis hierarchy is stated below and applied in Chapter 10.

9.1 Prediction Experiment Matrix

The predictive matrix spans PhysioNet 2012 and MIMIC-III, with the four learned model families STraTS, GRU, GRU-D, and TCN. The archive provides supervised training summaries and prediction exports for all four models across the two datasets; STraTS also has pretraining support. Implementation availability, training-artifact availability, prediction-export availability, final-result availability, and provenance completeness are separate labels rather than interchangeable evidence.

The task is multi-label prediction of dataset-specific rule-derived proxy labels generated by the deterministic decision trees. These labels, not majority-vote outputs, provide the deep-model training targets. STraTS artifacts use a split-aware dataset contract; prediction exports provide patient-level probability and thresholded binary predicted-label columns. The predicted columns are normalized and, in the archived final causal runs, participate alongside the rule-derived table in downstream aggregation. The experiment design includes pretraining where supported, supervised fine-tuning, family-specific models, a multi-label loss with class weighting where configured, validation/test evaluation, checkpoint selection, prediction export, random-seed or split settings, and training-fraction controls. These design elements are source-confirmed; they do not by themselves recover the final producing run.

The crucial provenance gaps are processed-pickle hashes, split manifests, checkpoint-to-export mappings, exact producing wrappers/commands, and archive-copy history. A requirements file describes an intended software environment, not proof of the environment that produced an archived training summary or export.

Table 9.1: Predictive experimental matrix and artifact-status design; no predictive metric is reported.

Dataset	Model family	Pretraining support	Input representation	Primary output	Final-artifact status	Provenance limitation
PhysioNet 2012 and MIMIC-III	STraTS	Available for STraTS family	Split-aware irregular time series and proxy-label targets	Multi-label predictions and exported proxy-state columns	Training summary and export archived	Split, checkpoint, command, and copy lineage incomplete
PhysioNet 2012 and MIMIC-III	GRU, GRU-D, TCN	Not the STraTS pretraining path	Dataset-specific irregular-series representation and proxy-label targets	Multi-label predictions and exports	Training summaries and exports archived	Final producing configuration and split/checkpoint linkage incomplete

9.2 Causal Experiment Matrix

The causal archive contains a verified design matrix spanning PhysioNet 2012 and MIMIC-III; CausalForestDML, LinearDML, and CausalPFN; and original and outcome-downsampled conditions. The prespecified estimator hierarchy designates CausalForestDML as primary, LinearDML as secondary, and CausalPFN as exploratory. The twelve archived run families are verified from `experiment_inventory.csv` and each `run_summary.json`, rather than inferred from directory names alone. Run summaries support execution-oriented metadata and workflow statuses, but their referenced numbered configuration CSVs are external absolute paths and are not locally archived.

The runtime orchestrator orders the stages as source-encoded graph construction, majority-vote aggregation, mortality prediction, matching, CATE estimation, saved-CATE analysis, and permutation tests. In every archived final run, the vote logs enumerate one deterministic rule-derived source plus normalized binary predictions from GRU, GRU-D, STraTS, and TCN. The orchestrator does not call an LLM or revise graph edges during execution. It assigns stage-specific output directories and logs, writes commands and status fields to `run_summary.json`, verifies required outputs, and records enabled, failed, or skipped stages. The CausalPFN run summaries record the saved-CATE analysis and permutation stages as skipped because that branch does not use the EconML diagnostic workflow; this is an estimator-specific limitation rather than an execution failure.

Outcome downsampling operates on mortality outcome classes, not exposure groups. It changes the empirical analysis population and sample outcome rate before treatment-specific matching and CATE estimation, and can affect matching availability, nuisance fitting, CATE aggregation, and normalized CATE. Original and outcome-downsampled runs therefore should not be treated as estimates of the same population quantity without an approved sampling, weighting, or transport argument. The prespecified hierarchy treats original cohorts as primary and outcome-downsampled cohorts as robustness analyses only.

Within each causal run, the exposure loop evaluates multiple dataset-specific binary proxy-state columns. Inclusion of any exposure in a later Results chapter requires treatment-column availability, graph-node availability, valid binary values, adjustment and support diagnostics, artifact completeness, and approved estimand wording. The exposure list is a design dimension, not a list of approved causal claims.

Table 9.2: Causal experimental matrix. The table records run-family design, not effect estimates.

Dataset	Estimator	Sampling condition	Graph	Exposure source	Diagnostics expected	Archived run status	Configuration provenance
PhysioNet 2012 and MIMIC-III	CausalForestDML	Original cohort	LLM-assisted, project-specified graph	Five-source rule-and-prediction majority vote	Matching, saved-CATE analysis, and permutations	Archived run family with stage records	Numbered producing config path recorded; local file missing
PhysioNet 2012 and MIMIC-III	LinearDML	Original cohort	LLM-assisted, project-specified graph	Five-source rule-and-prediction majority vote	Matching, saved-CATE analysis, and permutations	Archived run family with stage records	Numbered producing config path recorded; local file missing
PhysioNet 2012 and MIMIC-III	CausalPFN	Original cohort	LLM-assisted, project-specified graph	Five-source rule-and-prediction majority vote	Matching; CATE; EconML diagnostics not applicable	Archived run family; later diagnostic stages skipped	Numbered producing config path recorded; local file missing
PhysioNet 2012 and MIMIC-III	CausalForestDML, LinearDML, CausalPFN	Outcome-downsampled cohort	LLM-assisted, project-specified graph	Five-source rule-and-prediction majority vote	Same family-specific checks, interpreted for a different empirical population	Archived run families with stage records	Numbered producing config path recorded; downsampling setting needs manifest confirmation

9.3 Computational Environment and Reproducibility

Reproducibility requires distinguishing versioned source from local modifications. A complete record should identify the source versions of the main and nested repositories and disclose any uncommitted producer changes; a commit identifier alone cannot reproduce an uncommitted producer state. Similarly, run summaries record historical absolute producing-machine paths. These provide context but are not portable execution instructions, and their local existence cannot be assumed.

The active CATE code is designed to capture Python and package versions for EconML, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, NetworkX, Optuna, and PyTorch; platform; training timestamp; estimator module/class; artifact schema version; CUDA availability; and device metadata. These fields must be classified as captured by active code, actually present in an archived artifact, missing from an older run, or inferred only from a requirements file. Requirements express an intended environment, not producing-environment proof. CPU, RAM, GPU, operating-system, and cluster-node information must be reported only where stored or separately documented; a machine-specific storage path is not hardware evidence.

Configuration resolution follows a hierarchy of source defaults, dataset configuration CSV, command-line overrides, resolved runtime values, run summary, and saved-artifact metadata. The precedence is script-specific: for example, CATE exposes command-line overrides for selected options, whereas matching reads its downsampling setting from the dataset configuration. The numbered final-run configurations referenced by archived summaries are unavailable locally, so recovered defaults cannot be represented as proof of exact producing settings.

The result manifest records artifact paths and roles, datasets, models or estimators, sampling conditions, available source and configuration fields, hashes, schemas, and status. The accompanying compact reproducibility package indexes the exact archived causal command arrays and the log-recorded predictive arguments, seeds, and export-split setting, while retaining the missing numbered configurations, split identifiers, checkpoint-to-export proof, producing source versions, environments, and archive-copy histories. Thus the records support traceability without establishing complete rerun provenance.

Table 9.3: Reproducibility-status design.

Reproducibility element	Available evidence	Missing evidence	Impact	Required resolution
Source versions	Repository history and nested source state can be recorded	Exact producing version for every archived artifact and reconciliation of local changes	Version records alone may not reproduce an uncommitted producer state	Producer provenance

Reproducibility element	Available evidence	Missing evidence	Impact	Required resolution
Dataset configs	Local compact defaults and archived config paths	Numbered final-run config CSVs and hashes	Exact archived settings remain configuration-incomplete	Configuration recovery
Raw data and processed artifacts	Code contracts and path references	Access record, hashes, and processing manifest	Cohort lineage cannot be reproduced from the archive alone	Data manifest and authorized access
Predictive splits and check-point/export mapping	Training summaries, exports, wrappers	Split IDs and seeds, checkpoint hashes, producing commands, and copy record	Final prediction provenance incomplete	Per-export lineage
Causal run configuration and artifacts	Run summaries, logs, workflow folders, schemas	Resolved configs, input/output hashes, copy history	Execution-supported but provenance incomplete	Per-run lineage
Software environment	Active metadata collection and intended requirements	Producing environment for older final runs where absent	Version compatibility cannot be assumed	Producing-environment record
Hardware environment	Limited device fields where archived	CPU, RAM, GPU, node, and OS details where absent	Hardware claims must remain incomplete	Producing-environment record
Result checksums and prompt/clinical review	Result hashes plus partial source and prompt records	Prompt-run settings and clinical-review record	Review and artifact lineage remain incomplete	Human and clinical review

9.4 Evaluation and Result-Selection Policy

The result-admission policy applied before numerical reporting requires the producing run to be identified; the dataset, estimator or model, and sampling condition to be known; the source artifact to exist; the required schema to be validated; failed statuses to be excluded; source fields to be interpreted correctly; input/output relationships to be documented; fallback diagnostics to be labelled as fallback; the relevant estimand and population to be stated; and unresolved limitations to remain visible. Directory names, copied summaries, or non-null cells alone are insufficient.

Table 9.4: Result-admission policy applied to the validated result sources.

Item	Admission requirement	Evidence to validate
1–3	Identify producing run; identify dataset/model/estimator/sampling condition; locate source artifact	Run summary, manifest, and source path
4–6	Validate schema; exclude failed status; interpret source/status fields	Header/schema record, control CSV, report/log
7–9	Document input/output relationship; label fallback; state estimand and population	Producing command/config, artifact metadata, methods approval
10	Keep unresolved limitations visible	Result manifest and recorded evidence boundaries

Chapter 10

Results

This chapter reports only results admitted through the validated result-source tables and prespecified evidence hierarchy. Original-cohort analyses are primary: CausalForestDML is the primary causal-model estimator, LinearDML is the secondary comparator, and CausalPFN is exploratory. All prespecified dataset-specific proxy-state exposures are retained; none was selected according to its observed result. In all twelve archived causal runs, the implemented majority-vote stage combines one rule-derived proxy-label source with predicted-label sources from GRU, GRU-D, STraTS, and TCN, and passes the resulting aggregate downstream. Detailed robustness outputs are summarized selectively, and outcome-downsampled analyses are treated only as a robustness perspective. Every model-estimated contrast remains conditional on the causal assumptions and proxy-exposure interpretation in Chapter 7.

10.1 Analysis-Population Counts

The original causal-analysis population contained 26,845 MIMIC-III records and 7,993 PhysioNet records (Table 10.1). MIMIC-III therefore contributed 18,852 more records and was approximately 3.36 times as large. These are causal-analysis population counts, not reconstructed raw cohort totals. Pipeline-stage artifacts have different contracts: prediction-export, proxy-label, majority-vote, and causal-analysis counts must not be interpreted as one universally identical cohort count.

Table 10.1: Original causal-analysis populations admitted to the primary analyses. Counts describe model rows in the checked original-cohort runs, not raw-source cohort totals.

Dataset	Pipeline population	Sampling	Records	Scope
MIMIC-III	Causal-analysis model rows	Original	26,845	Primary
PhysioNet 2012	Causal-analysis model rows	Original	7,993	Primary

10.2 Predictive Performance

The checked archive contains eight completed dataset–model test combinations spanning the four learned models STraTS, GRU, GRU-D, and TCN (Table 10.2). Every predictive metric compares model output with deterministic rule-derived proxy-label targets produced by the source-encoded decision trees. On MIMIC-III, STraTS had the smallest archived test loss and the highest archived test AUROC, AUPRC, and minRP. On PhysioNet, GRU-D led the same four archived metrics. Thus the leading model differed by dataset, although within each dataset the leader did not depend on which of these four metrics was used. Rankings among the remaining models did vary by metric; the table therefore does not support a universal model ordering or a claim of statistical superiority.

The numerical rows are test summaries and not clinical-validation values. Their summary and paired log values agreed within the validated tolerance, but the archived train/validation/test split manifest and checkpoint-to-export lineage remain incomplete. No confidence intervals or between-model significance tests were available in the validated sources. Strong agreement with a rule-derived target demonstrates learnability of that target, not validity of the clinical construct or correctness of the LLM-assisted rule design.

Table 10.2: Archived predictive test performance for the four learned models. AUROC, AUPRC, and minRP are macro-averaged across non-degenerate rule-derived proxy-label targets on the archived test split.

Dataset	Model	Test loss	Test AUROC	Test AUPRC	Test minRP
MIMIC-III	STraTS	0.348	0.905	0.869	0.795
MIMIC-III	GRU	0.386	0.881	0.840	0.770
MIMIC-III	GRU-D	0.404	0.884	0.841	0.773
MIMIC-III	TCN	0.414	0.867	0.823	0.758
PhysioNet 2012	STraTS	0.341	0.915	0.877	0.818
PhysioNet 2012	GRU	0.397	0.915	0.897	0.831
PhysioNet 2012	GRU-D	0.331	0.918	0.905	0.835
PhysioNet 2012	TCN	0.477	0.899	0.875	0.811

10.3 Primary CausalForestDML Results

Tables 10.3 and 10.4 report every prespecified original-cohort majority-vote proxy-state exposure from that mixed rule-and-prediction lineage. The reported quantity is the *mean model-estimated CATE over the analyzed sample*, on the model outcome scale. It is neither a risk ratio nor an unqualified population-average causal effect. Exposure prevalence is the proportion of analyzed records carrying the corresponding binary aggregated proxy-state tag. The adjustment-set codes refer to the observed variables archived for each exposure; identifiability with the available nodes remained unresolved for every validated row.

These results do not evaluate the LLM itself. The study contains no LLM ablation, clinician comparison, prompt-sensitivity experiment, alternative-LLM replication, or graph-correctness experiment. The numerical analyses therefore cannot establish that LLM assistance improved the rules or DAGs, nor can they separate design error from rule, prediction, aggregation, or estimator error.

For MIMIC-III, all nine archived means were positive. The largest mean was for `LAT_CARDIAC_STRAIN` (0.220), followed by `LAT_INFLAMMATION_SEPSIS` (0.161). The smallest were for `LAT_METABOLIC_DERANGEMENT` (0.020) and `LAT_SHOCK` (0.021). Figure 10.1 displays this source-exact ordering. It is descriptive and does not establish clinical importance or statistical significance.

Table 10.3: Primary original-cohort MIMIC-III CausalForestDML results (n = 26,845). Prevalence and mean model-estimated CATE are proportions on their archived scales.

Proxy-state exposure	Prevalence	Mean CATE	Adjustment	Status
`LAT_CARDIAC_STRAIN`	0.178	0.220	M1	Identifiability unresolved
`LAT_INFLAMMATION_SEPSIS`	0.472	0.161	M0	Identifiability unresolved
`LAT_HEPATIC_COAG_DYSFUNCTION`	0.241	0.098	M2	Identifiability unresolved
`LAT_RENAL_DYSFUNCTION`	0.363	0.091	M3	Identifiability unresolved
`LAT_GLOBAL_SEVERITY`	0.719	0.062	M4	Identifiability unresolved
`LAT_RESPIRATORY_FAILURE`	0.590	0.036	M5	Identifiability unresolved
`LAT_NEUROLOGIC_DYSFUNCTION`	0.434	0.034	M5	Identifiability unresolved
`LAT_SHOCK`	0.540	0.021	M6	Identifiability unresolved
`LAT_METABOLIC_DERANGEMENT`	0.373	0.020	M7	Identifiability unresolved

The MIMIC adjustment codes are: M0, no observed adjustment variables archived; M1, age, gender, and chronic burden; M2, inflammation/sepsis and shock; M3, chronic burden and shock; M4, age, gender, chronic burden, and inflammation/sepsis; M5, age, gender, chronic burden, global severity, and inflammation/sepsis; M6, age, gender, cardiac strain, chronic burden, and inflammation/sepsis; and M7, global severity, renal dysfunction, respiratory failure, and shock. These are observed adjustment-variable records, not proof that all confounding was controlled.

For PhysioNet, nine archived means were positive and the shock mean was negative. Renal dysfunction had the largest mean (0.120), followed by cardiac injury/strain (0.112) and global severity (0.108). Coagulation/hematologic dysfunction was small and positive (0.011), while

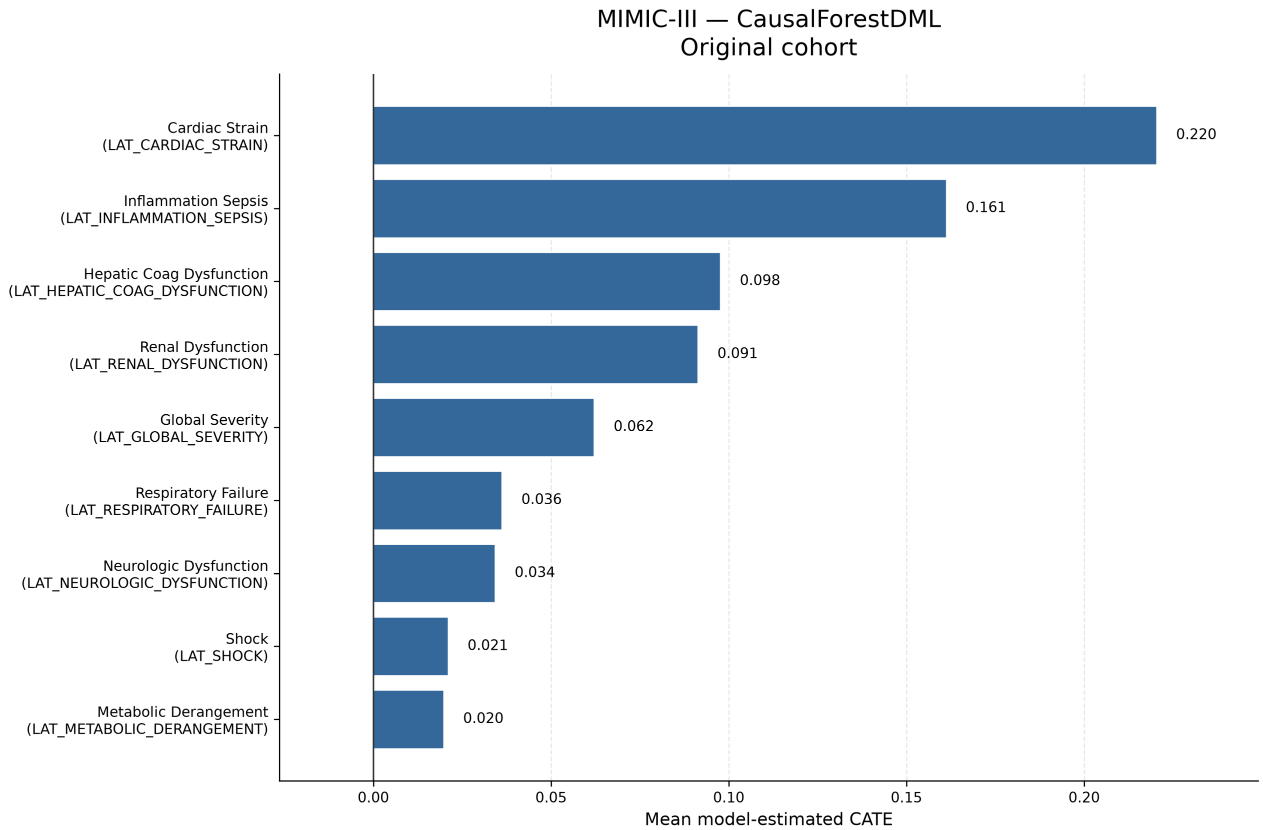

Figure 10.1: Original-cohort MIMIC-III mean model-estimated CATE values from CausalForestDML. All nine prespecified proxy-state exposures are shown. The ordering is descriptive and does not establish statistical significance, clinical importance, or population-average causal effects.

shock was −0.014. Figure 10.2 retains that negative value and the source-exact ordering; the ranking is descriptive only.

Table 10.4: Primary original-cohort PhysioNet CausalForestDML results ($n = 7{,}993$). Prevalence and mean model-estimated CATE are proportions on their archived scales.

Proxy-state exposure	Prevalence	Mean CATE	Adjustment	Status
LAT_RENAL_DYSFUNCTION	0.230	0.120	P1	Identifiability unresolved
LAT_CARDIAC_INJURY_STRAIN	0.355	0.112	P2	Identifiability unresolved
LAT_GLOBAL_SEVERITY	0.777	0.108	P3	Identifiability unresolved
LAT_HEPATIC_DYSFUNCTION	0.142	0.091	P1	Identifiability unresolved
LAT_NEUROLOGIC_DYSFUNCTION	0.659	0.082	P0	Identifiability unresolved
LAT_METABOLIC_DERANGEMENT	0.558	0.078	P4	Identifiability unresolved
LAT_INFLAMMATION_SEPSIS_BURDEN	0.563	0.072	P0	Identifiability unresolved
LAT_RESPIRATORY_FAILURE	0.614	0.065	P0	Identifiability unresolved
LAT_COAG_HEME_DYSFUNCTION	0.395	0.011	P5	Identifiability unresolved
LAT_SHOCK	0.696	−0.014	P6	Identifiability unresolved

The PhysioNet adjustment codes are: P0, no observed adjustment variables archived; P1, ICU-type indicators 1–4, cardiac injury/strain, inflammation/sepsis burden, and shock; P2, ICU-type indicators 1–4; P3, age, gender, weight, ICU-type indicators 1–4, chronic baseline risk, and inflammation/sepsis burden; P4, renal dysfunction, respiratory failure, and shock; P5, inflammation/sepsis burden and shock; and P6, ICU-type indicators 1–4, cardiac injury/strain, and inflammation/sepsis burden. The same qualification about incomplete confounding control applies.

10.4 Matching and Empirical Support

Matching was identical across the three estimator run families within a dataset and sampling condition, so Table 10.5 reports the cross-run-consistent original-cohort rows once, as supporting evidence for the primary CausalForestDML analysis. The numerical contrast is a *descriptive matched-pair outcome difference*. Treated and control record counts were unavailable in the checked table and are left unreported rather than inferred.

Eight of nine MIMIC exposures and seven of ten PhysioNet exposures produced a matched-

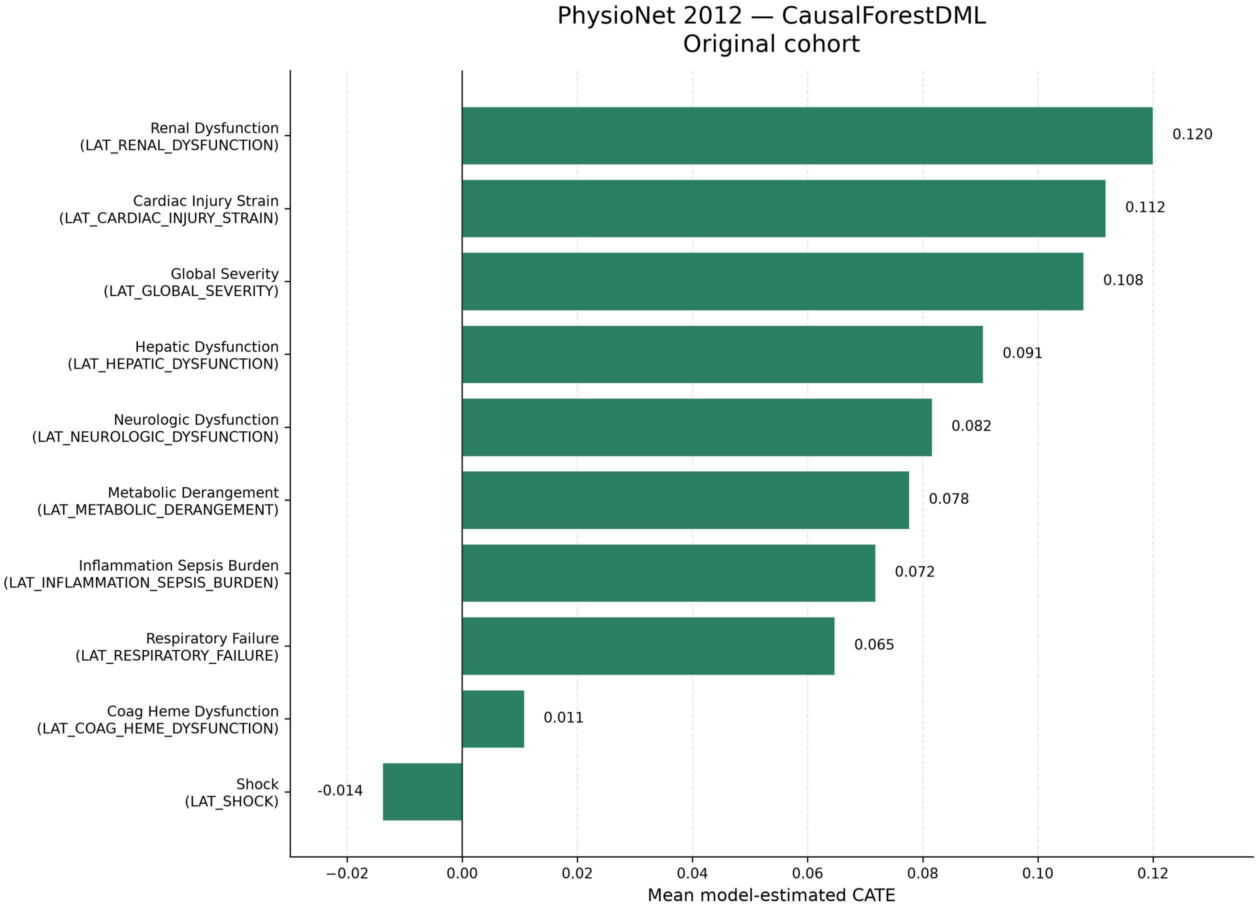

Figure 10.2: Original-cohort PhysioNet mean model-estimated CATE values from Causal-ForestDML. All ten prespecified proxy-state exposures are shown. The ordering is descriptive; the negative shock estimate is retained. The figure does not establish statistical significance, clinical importance, or population-average causal effects.

pair summary. Matching failed for MIMIC inflammation/sepsis and for PhysioNet inflammation/sepsis burden, neurologic dysfunction, and respiratory failure because no binary matching columns remained after preprocessing. A failure is not a zero effect. Among successful rows, final allowed Hamming distance ranged from 0 to 2; increased allowed distance represents weaker exact similarity in the binary matching representation. Three successful rows were flagged as not reaching the configured sufficient-pair criterion: global severity in both datasets and PhysioNet shock.

Table 10.5: Original-cohort descriptive matching and empirical-support summaries. "Difference" is the descriptive matched-pair outcome difference; it is not a model-estimated CATE.

Dataset	Proxy-state exposure	Pairs	Rate	Distance	Difference	Support/status
MIMIC	`LAT_CARDIAC_STRAIN`	4,769	1.000	0	0.345	Sufficient
MIMIC	`LAT_GLOBAL_SEVERITY`	7,537	0.390	1	0.075	Weak: insufficient-pair flag
MIMIC	`LAT_HEPATIC_COAG_DYSFUNCTION`	6,478	1.000	0	0.133	Sufficient
MIMIC	`LAT_METABOLIC_DERANGEMENT`	5,061	0.506	0	0.037	Sufficient
MIMIC	`LAT_NEUROLOGIC_DYSFUNCTION`	6,275	0.539	0	0.027	Sufficient
MIMIC	`LAT_RENAL_DYSFUNCTION`	8,915	0.915	0	0.134	Sufficient
MIMIC	`LAT_RESPIRATORY_FAILURE`	10,998	0.694	2	0.106	Sufficient; distance 2
MIMIC	`LAT_SHOCK`	10,380	0.717	1	0.039	Sufficient
MIMIC	`LAT_INFLAMMATION_SEPSIS`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_CARDIAC_INJURY_STRAIN`	2,840	1.000	0	0.134	Sufficient
PhysioNet	`LAT_COAG_HEME_DYSFUNCTION`	1,763	0.558	0	0.0017	Sufficient
PhysioNet	`LAT_GLOBAL_SEVERITY`	1,782	0.287	1	0.095	Weak: insufficient-pair flag
PhysioNet	`LAT_HEPATIC_DYSFUNCTION`	1,133	1.000	0	0.137	Sufficient
PhysioNet	`LAT_METABOLIC_DERANGEMENT`	3,378	0.757	2	0.073	Sufficient; distance 2
PhysioNet	`LAT_RENAL_DYSFUNCTION`	1,841	1.000	0	0.154	Sufficient
PhysioNet	`LAT_SHOCK`	2,433	0.438	1	0.010	Weak: insufficient-pair flag
PhysioNet	`LAT_INFLAMMATION_SEPSIS_BURDEN`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_NEUROLOGIC_DYSFUNCTION`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_RESPIRATORY_FAILURE`	–	–	–	–	Failed: no binary matching columns

For the 15 successful rows, matching and CausalForestDML shared direction in 14 cases. PhysioNet shock was the exception: its matched-pair difference was positive (0.010), while its model-estimated mean was negative (-0.014). This is a descriptive comparison, not independent causal validation. A high match rate does not establish positivity; the procedure is greedy and representation-dependent, and no dedicated propensity-overlap or balance figure was available.

10.5 LinearDML Comparison

LinearDML and CausalForestDML agreed in mean-effect direction for all 19 original-cohort dataset–exposure pairs (Table 10.6). This agreement does not imply equal magnitude or ranking. In MIMIC, CausalForestDML estimated a larger cardiac-strain mean by approximately 0.033, while LinearDML estimated a larger global-severity mean by approximately 0.018; renal and hepatic/coagulation dysfunction exchanged third and fourth positions. In PhysioNet, the largest absolute differences were for hepatic dysfunction (approximately 0.030) and renal dysfunction (approximately 0.029), and the top-ranked mean changed from renal dysfunction under CausalForestDML to cardiac injury/strain under LinearDML. LinearDML imposes a simpler final effect structure; similarity between these archived means does not show that either estimator is unbiased.

Table 10.6: Original-cohort comparison of mean model-estimated CATE from CausalForestDML and LinearDML.

Dataset	Proxy-state exposure	Forest	Linear	Same direction
MIMIC	`LAT_CARDIAC_STRAIN`	0.220	0.188	Yes
MIMIC	`LAT_INFLAMMATION_SEPSIS`	0.161	0.161	Yes
MIMIC	`LAT_HEPATIC_COAG_DYSFUNCTION`	0.098	0.083	Yes
MIMIC	`LAT_RENAL_DYSFUNCTION`	0.091	0.089	Yes
MIMIC	`LAT_GLOBAL_SEVERITY`	0.062	0.080	Yes
MIMIC	`LAT_RESPIRATORY_FAILURE`	0.036	0.039	Yes
MIMIC	`LAT_NEUROLOGIC_DYSFUNCTION`	0.034	0.032	Yes
MIMIC	`LAT_SHOCK`	0.021	0.021	Yes
MIMIC	`LAT_METABOLIC_DERANGEMENT`	0.020	0.018	Yes
PhysioNet	`LAT_RENAL_DYSFUNCTION`	0.120	0.091	Yes
PhysioNet	`LAT_CARDIAC_INJURY_STRAIN`	0.112	0.122	Yes
PhysioNet	`LAT_GLOBAL_SEVERITY`	0.108	0.107	Yes
PhysioNet	`LAT_HEPATIC_DYSFUNCTION`	0.091	0.060	Yes
PhysioNet	`LAT_NEUROLOGIC_DYSFUNCTION`	0.082	0.076	Yes
PhysioNet	`LAT_METABOLIC_DERANGEMENT`	0.078	0.080	Yes
PhysioNet	`LAT_INFLAMMATION_SEPSIS_BURDEN`	0.072	0.071	Yes
PhysioNet	`LAT_RESPIRATORY_FAILURE`	0.065	0.063	Yes
PhysioNet	`LAT_COAG_HEME_DYSFUNCTION`	0.011	0.004	Yes
PhysioNet	`LAT_SHOCK`	−0.014	−0.027	Yes

10.6 CausalPFN Exploratory Results

CausalPFN CATE execution completed for all 19 original-cohort comparisons. The three estimators agreed in mean-effect direction for 18 of the 19 dataset–exposure comparisons: MIMIC was concordant for 9 of 9 and PhysioNet for 9 of 10 (Table 10.7 and Figure 10.3). The sole exception was PhysioNet `LAT_SHOCK`: CausalForestDML was negative (−0.014), LinearDML was negative (−0.027), and CausalPFN was slightly positive (0.0041).

CausalPFN reproduced the prevailing direction in nearly every comparison, supporting its promise as a complementary estimator within this pipeline. This broad agreement is not complete agreement and does not establish estimator equivalence, superiority, causal validity, or interchangeable uncertainty quantification. Its later sensitivity and permutation stages were intentionally skipped rather than failed, so it lacks the archived robustness-diagnostic family available for the DML estimators. Its role remains exploratory, and the unresolved primary-citation limitation in Chapter 7 remains relevant.

Table 10.7: Original-cohort exploratory CausalPFN mean model-estimated CATE and direction agreement with both DML estimators.

Dataset	Proxy-state exposure	CausalPFN	Forest/Linear signs	All three agree
MIMIC	LAT_CARDIAC_STRAIN	0.259	+, +	Yes
MIMIC	LAT_INFLAMMATION_SEPSIS	0.149	+, +	Yes
MIMIC	LAT_HEPATIC_COAG_DYSFUNCTION	0.097	+, +	Yes
MIMIC	LAT_RENAL_DYSFUNCTION	0.087	+, +	Yes
MIMIC	LAT_GLOBAL_SEVERITY	0.073	+, +	Yes
MIMIC	LAT_RESPIRATORY_FAILURE	0.050	+, +	Yes
MIMIC	LAT_NEUROLOGIC_DYSFUNCTION	0.035	+, +	Yes
MIMIC	LAT_SHOCK	0.032	+, +	Yes
MIMIC	LAT_METABOLIC_DERANGEMENT	0.031	+, +	Yes
PhysioNet	LAT_CARDIAC_INJURY_STRAIN	0.119	+, +	Yes
PhysioNet	LAT_RENAL_DYSFUNCTION	0.111	+, +	Yes
PhysioNet	LAT_HEPATIC_DYSFUNCTION	0.107	+, +	Yes
PhysioNet	LAT_GLOBAL_SEVERITY	0.105	+, +	Yes
PhysioNet	LAT_METABOLIC_DERANGEMENT	0.091	+, +	Yes
PhysioNet	LAT_NEUROLOGIC_DYSFUNCTION	0.077	+, +	Yes
PhysioNet	LAT_INFLAMMATION_SEPSIS_BURDEN	0.067	+, +	Yes
PhysioNet	LAT_RESPIRATORY_FAILURE	0.063	+, +	Yes
PhysioNet	LAT_COAG_HEME_DYSFUNCTION	0.025	+, +	Yes
PhysioNet	LAT_SHOCK	0.0041	−, −	No

10.7 Cross-Dataset Comparison

MIMIC-III's original causal-analysis population was substantially larger: 26,845 versus 7,993 records, a difference of 18,852 and a ratio of approximately 3.36. The larger analysis population can affect estimate stability, feasible model flexibility, and the amount of empirical support available; it does not make the MIMIC estimates automatically correct. The checked outcome rates also differed (0.121 in MIMIC and 0.142 in PhysioNet), but these rates describe the analyzed original-cohort model rows rather than equivalent raw-source cohorts.

Some proxy names suggest nominal alignment, including global severity, shock, respiratory failure, renal dysfunction, neurologic dysfunction, and metabolic derangement. Others are only approximately corresponding project-defined states: MIMIC combines hepatic and coagulation dysfunction while PhysioNet separates them, and the inflammation/sepsis and cardiac constructs use dataset-specific names and rules. Consequently, similarly named states are not assumed to be construct-equivalent. Dataset era, measurement systems, variable availability, proxy rules, DAG structure, empirical support, and estimator behavior also differ. No effects are pooled, averaged across datasets, or interpreted as evidence of biological inconsistency; detailed clinical interpretation is deferred to Chapter 11.

10.8 Robustness and Sensitivity

10.8.1 Outcome-downsampled analyses

The outcome-downsampled analyses retained all outcome-positive records and sampled outcome-negative records, producing 6,486 MIMIC rows and 2,276 PhysioNet rows with sample outcome rates of 0.500. These are different empirical populations from the original cohorts, so their mean estimates are neither pooled with nor treated as estimates of the same population quantity.

Direction was preserved in 55 of 57 matched original-versus-downsampled estimator–exposure comparisons. CausalForestDML retained direction for all 19 comparisons. LinearDML changed direction only for PhysioNet `LAT_COAG_HEME_DYSFUNCTION`, from 0.0041 to −0.0174; CausalPFN changed direction only for PhysioNet `LAT_SHOCK`, from 0.0041 to −0.0413. All 57 paired means changed numerically, so magnitude changes were widespread and not confined to the two sign changes. This does not strengthen causal evidence because the sampled population changed.

Matching availability did not change: MIMIC again had eight successful summaries and one failure, while PhysioNet had seven successful summaries and three failures. Pair counts, match rates, distances, and descriptive differences nevertheless changed under downsampling. Detailed downsampled values remain supplementary.

10.8.2 Sensitivity, permutation, and support diagnostics

For both CausalForestDML and LinearDML, the validated sensitivity family was partial for eight of nine MIMIC exposures and seven of ten PhysioNet exposures in each sampling condition. It failed for MIMIC inflammation/sepsis and for the three PhysioNet exposures whose adjustment records were empty: inflammation/sepsis burden, neurologic dysfunction, and respiratory failure. Available rows combine saved-training-direct values with later recomputation and custom reconstruction from residual parameters; the planned real benchmark extraction is marked unimplemented by design. These source classifications prevent the values from being presented as a single estimator-native family. CausalPFN sensitivity was intentionally skipped and is not equivalent to a zero diagnostic.

Treatment- and outcome-permutation aggregate rows were available for every DML dataset–exposure–sampling combination, with ten archived trials and seed 42 in each row. The validated source table records these as sanity diagnostics and supplies no new p-values. They are not formal randomization tests and do not prove identification. CausalPFN permutations were intentionally skipped.

Table 10.8 summarizes availability rather than diagnostic magnitude. Matching remains indirect, greedy, binary-representation-dependent support evidence. No dedicated overlap or propensity figure, pre/post-match balance table, or empirical proof of positivity was available. The principal warnings are the four exposure-specific matching failures per sampling-condition pair, three successful original rows with insufficient-pair flags, and the absence of an equivalent PFN diagnostic family.

Table 10.8: Robustness and diagnostic availability. Counts are exposure-level statuses, not effect estimates.

Dataset	Estimator	Sampling	Matching	Sensitivity	Permutation
MIMIC	Forest	Original	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	Linear	Original	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	PFN	Original	8 available; 1 failed	Intentional skip	Intentional skip
MIMIC	Forest	Downsampled	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	Linear	Downsampled	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	PFN	Downsampled	8 available; 1 failed	Intentional skip	Intentional skip
PhysioNet	Forest	Original	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	Linear	Original	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	PFN	Original	7 available; 3 failed	Intentional skip	Intentional skip
PhysioNet	Forest	Downsampled	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	Linear	Downsampled	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	PFN	Downsampled	7 available; 3 failed	Intentional skip	Intentional skip

10.8.3 Result provenance boundary

The checked hashes and source paths support arithmetic transcription of the admitted archive rows, but they do not remove the archive's reproducibility limitations. The `final-results/` tree is ignored or untracked in the main repository; exact numbered causal configuration files remain unavailable locally; some producing commits and archive-copy histories are incomplete; predictive split and checkpoint-to-export lineage is incomplete; and raw and processed data cannot be reproduced from the repository alone. These limitations qualify reproducibility claims, not the arithmetic correspondence between the checked rows and the tables above.

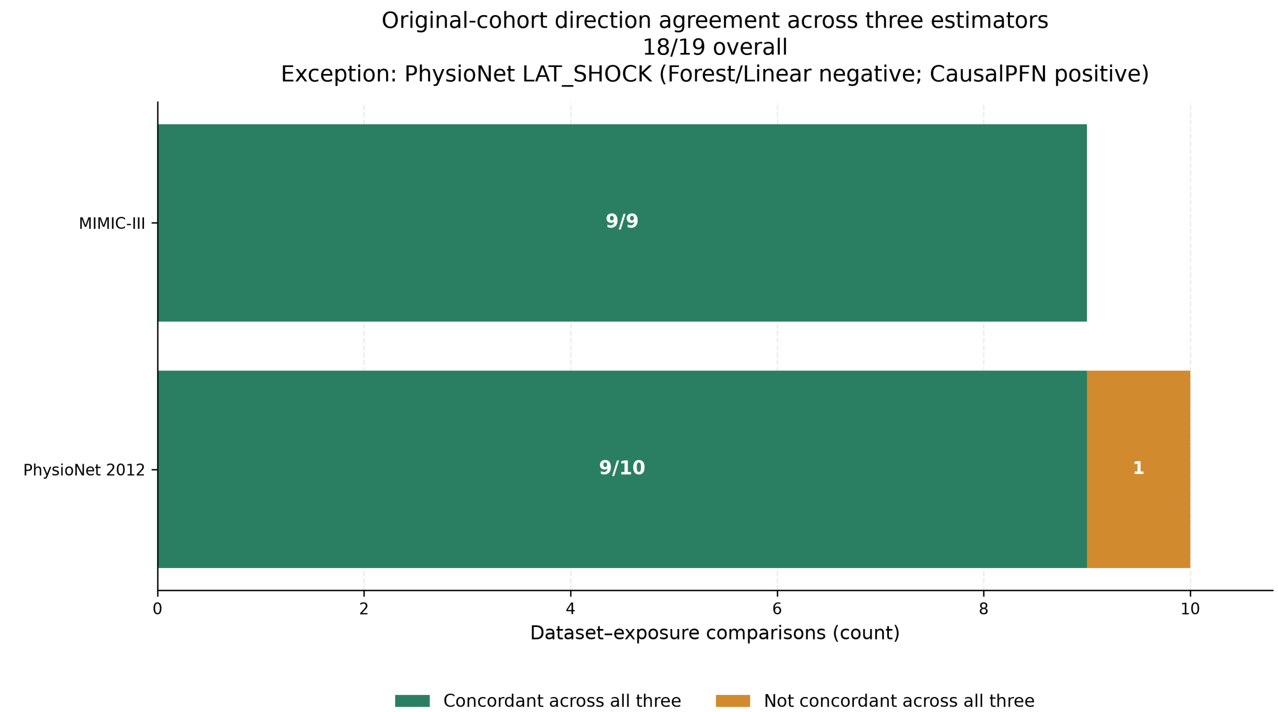

Figure 10.3: Directional concordance of the original-cohort mean model-estimated CATE across CausalForestDML, LinearDML, and exploratory CausalPFN. All three estimators shared the same direction for 18 of 19 dataset–exposure comparisons: 9 of 9 in MIMIC-III and 9 of 10 in PhysioNet. PhysioNet shock was the sole exception. Direction agreement does not imply equal magnitude, uncertainty, estimator equivalence, or causal validity.

Chapter 11

Discussion

This chapter interprets the evidence reported in Chapter 10 within the study design and assumptions established in Chapters 3–9. The central distinction is between what the thesis evidence demonstrates empirically—an implemented, evidence-tracked workflow and stable patterns across several analyses—and what it does not establish clinically or causally. The discussion therefore treats the reported quantities as prediction results, descriptive matched-pair comparisons, or adjusted model estimates under project assumptions, according to their source.

11.1 Answering the Research Questions

11.1.1 Main Research Question

The main research question asked: *How can LLM-assisted clinical and causal knowledge elicitation, deterministic proxy-label construction, and deep modeling of irregular ICU time series be integrated into an evidence-tracked framework for DAG-guided adjusted effect estimation of in-hospital mortality?*

The direct answer is that this thesis demonstrates an evidence-tracked methodological framework with a sequence of explicit interfaces. At design time, a structured LLM-assisted protocol generated proposals for dataset-specific proxy ontologies, decision rules, missingness considerations, and DAGs; project-selected designs were encoded in deterministic source. During execution, dataset-specific irregular measurements are converted into normalized data contracts; the source-coded rules create rule-derived proxy labels; irregular-time-series models predict those labels; and probabilities and binary predicted fields are exported and normalized. In every archived final causal run, the vote then combines one rule-derived table with four model-prediction tables. Source-coded, project-specified DAGs define exposure-specific adjustment logic, and matching and heterogeneous-effect estimators analyze the resulting mixed-source aggregate. Robustness and provenance checks qualify the interpretation. The framework was implemented and empirically exercised across MIMIC-III and PhysioNet 2012 without a runtime LLM call.

This is a feasibility and integration result, not a conclusive solution to the clinical causal question or an empirical evaluation of LLM quality. The proxy states are not chart-adjudicated diagnoses, many exposures describe illness states rather than well-defined interventions, and the LLM-assisted project DAGs are assumed rather than empirically established. The adjusted estimates remain conditional on construct measurement, consistency, exchangeability, temporal ordering, overlap, and model adequacy. The framework therefore supports auditable retrospective analysis and hypothesis generation, but it does not produce clinical treatment recommendations. In summary, the thesis demonstrates the feasibility of linking LLM-assisted knowledge construction, irregular-time-series representation, proxy-state prediction, and assumption-guided adjusted effect estimation in one auditable workflow, while showing that the strength of causal interpretation is constrained chiefly by construct validity, intervention definition, causal-graph validity, empirical support, and provenance.

11.1.2 Data Contracts and End-to-End Pipeline Integration

SRQ-1—data contracts. The pipeline requires at least two distinct processed-artifact contracts. The causal repository uses `[ts, oc, ts_ids]`, with a normalized `ts_id` linking the long event table to the outcome table. The STraTS and baseline workflow uses a split-aware artifact with events, outcomes, and explicit train, validation, and test identifier collections. The router and conversion logic bridge these contracts by normalizing identifiers and constructing the split-aware form when requested. Prediction exports add another interface: `ts_id`, paired probability and binary proxy-state columns, and subsequently binary-only voter files. The answer is therefore *implementation supported*: the repository defines and validates these boundaries, but exact processed-pickle hashes, producing commands, and final split lineage remain incomplete.

SRQ-4—prediction normalization and aggregation. Prediction outputs are normalized into paired probability and binary proxy-state fields, while downstream voter inputs retain the binary fields in an approved order. Majority voting aligns shared normalized identifiers, checks identical binary proxy-state schemas, and applies the implemented tie rule. In every archived final run, the logs enumerate one deterministic rule-derived source and four thresholded prediction sources—GRU, GRU-D, STraTS, and TCN. This supplies a deterministic mixed-source handoff into the causal repository. It is algorithmic aggregation rather than clinical consensus; the logs retain external filenames but do not co-archive the voter-file bytes or hashes, while checkpoint mappings and archive-copy provenance remain incomplete. Moreover, aggregation cannot automatically correct systematic error shared by models trained on the same rule-derived targets, especially when the originating rule source itself participates in the vote.

11.1.3 Proxy-State Construction, Prediction, and Aggregation

SRQ-2—LLM-assisted design selection, encoding, and traceability. The preserved generic and dataset-specific prompts, final run exports, active tagger and graph files, and generated proxy and DAG artifacts establish a traceable design sequence at artifact and schema level. The final prompt outputs align with the active eleven-state PhysioNet and ten-state MIMIC schemas, while deterministic source rules and stable `LAT_*` columns make implemented behavior inspectable. The answer remains *partially traceable and unresolved clinically*: complete accepted/rejected decision records and line-by-line prompt-output-to-code correspondence are unavailable; the two datasets use different ontologies and measurements; threshold rules may misclassify constructs; clinical review remains incomplete; and no chart-adjudicated reference labels were archived. These are LLM-assisted, project-selected proxy designs encoded as deterministic rules, not diagnoses, ground truth, or validated phenotypes. The rule-to-label transition is conceptually related to programmatic labeling, while the aggregation layer does not implement a learned Snorkel label model [12, 6].

SRQ-3—predictive modeling. The validated comparison included the four learned models STraTS, GRU, GRU-D, and TCN. STraTS led the four archived test metrics in MIMIC-III, whereas GRU-D led them in PhysioNet. The leading architecture therefore differed by dataset. This is empirically supported evidence that representation choice is dataset dependent, not evidence for universal model superiority. Handling irregularity or missingness can be useful in several ways, and prior work distinguishes sparse-token representations from missingness-aware recurrent processing [1, 2, 3, 4]. The local comparison does not reproduce the original papers' benchmark tasks exactly.

Several mechanisms may contribute to the dataset-specific pattern: variable availability, measurement frequency, missingness, label prevalence, proxy definitions, preprocessing, dataset era, sample size, and ICU composition. The study did not isolate those mechanisms, so they remain possible explanations rather than findings. In addition, no uncertainty intervals or paired between-model tests were archived, checkpoint-to-export provenance is incomplete, and the prediction targets themselves lack clinical validation. Statistical superiority is therefore not established.

11.1.4 Adjusted Effect Estimation and Estimator Agreement

SRQ-5—DAG-guided adjustment. Dataset-specific, source-coded DAGs select exposure-specific observed adjustment sets from columns available in each analysis frame. LLM-assisted elicitation was a substantive design-time input, while the active graph code is the implementation authority. The graphs were neither learned from the data nor clinically or empirically validated, and the adjustment sets differ across exposures. Several validated rows contain empty observed adjustment sets, while treatment-specific identifiability remains unresolved throughout the result sources. The answer is thus *conditional on assumptions*: a computed set documents how the project applied its graph, but it does not prove exchangeability or the absence of unmeasured confounding [7].

SRQ-6—matching, CATE, sensitivity, and permutation evidence. The approved hierarchy uses original cohorts as primary, CausalForestDML as the primary estimator, LinearDML as secondary, CausalPFN as exploratory, and outcome-downsampled analyses as robustness evidence. In the primary forest results, all nine MIMIC original-cohort mean model-estimated CATE summaries were positive; nine of ten PhysioNet summaries were positive, with shock negative. This pattern supports the narrower conclusion that the project-defined proxy-state exposures contain systematic mortality-related variation after the implemented adjustment procedure, that patterns differ by construct and dataset, and that heterogeneous-effect estimators can be applied to the constructed analysis objects. It does not show that a proxy state caused death, that changing it would change mortality, or that a ranking identifies intervention priorities.

CausalForestDML and LinearDML agreed in direction for all 19 original-cohort dataset–exposure pairs. This complete directional concordance is useful model-form triangulation: the sign pattern was not unique to the nonlinear forest final stage. It does not imply equal magnitudes, orderings, estimands, uncertainty, or correctness, and it does not validate the DAG or rule out residual confounding. Indeed, Chapter 10 shows that magnitudes and within-dataset orderings can differ even when signs agree.

Within this pipeline, CausalPFN showed promise as a complementary estimator because it reproduced the prevailing direction across nearly all prespecified comparisons. It contributed to all-three directional concordance for 18 of 19 original-cohort dataset–exposure pairs despite using a different modeling approach. This positive result is bounded: magnitudes were not identical to DML magnitudes, the archived sensitivity and permutation stages were intentionally absent, and the approved literature corpus lacks a primary method citation. CausalPFN therefore remains an exploratory empirical contribution rather than an estimator with the same diagnostic support as the DML analyses.

PhysioNet shock is the central exception rather than a detail to be averaged away. CausalForestDML and LinearDML were negative, CausalPFN was slightly positive, and the descriptive matching summary was positive. This disagreement is a substantive caution about model, representation, and support sensitivity. It should not be resolved by a vote among estimators, and the negative DML estimates should not be described as a clinically protective effect.

Matching provides a transparent descriptive baseline and an empirical-support warning system. Most prespecified exposures produced summaries, and 14 of 15 successful original-cohort matching summaries shared the primary forest direction. Failures identified cases in which no usable binary matching columns remained; some successful rows carried insufficient-pair warnings; and accepted Hamming distances varied. Stable matching availability across original and downsampled populations supports the reproducibility of the availability pattern, but matching neither proves positivity nor supplies complete balance evidence. A failed comparison is missing support, not a zero estimate, and a successful descriptive matched-pair outcome difference is not automatically an ATT or independent causal confirmation.

Sensitivity and permutation evidence adds layers rather than closure. DML sensitivity outputs were saved-training direct, reconstructed, partial, or failed depending on the exposure, and reconstructed diagnostics are not identical to estimator-native calculations. Benchmark-confounder comparisons supply a scale, not proof that omitted-variable bias is absent. Treatment and outcome permutations provide disruption-based sanity checks for the DML estimators, but the archived procedure is not automatically a formal randomization test and does not establish identification. The corresponding CausalPFN stages were intentionally skipped. Finally, matching fields provide indirect support evidence, but dedicated propensity, common-support, and balance plots were not archived; positivity remains unestablished [33, 34, 35].

11.1.5 Cross-Dataset Interpretation and Robustness

The original causal-analysis populations contained 26,845 MIMIC records and 7,993 PhysioNet records, so the MIMIC analysis population was approximately 3.36 times larger. A larger analysis population can provide more information for nuisance-model fitting and modeled effect heterogeneity, and it may contribute to numerical stability. It does not demonstrate better data quality, make the MIMIC findings more valid, or erase differences in era, variables, measurement systems, missingness, proxy rules, DAGs, or patient composition. Predictive training and causal analysis also use distinct artifact contracts, so their populations should not be assumed identical merely because some row counts align.

Broad methodological replication across both datasets is a strength: the pipeline operated with the same overall sequence of representation, proxy prediction, aggregation, graph-guided adjustment, and estimator comparison. However, similarly named proxy states are not automatically construct-equivalent. MIMIC combines some domains that PhysioNet separates, and the rules and graphs are dataset specific. The deliberate no-pooling policy is therefore essential. Cross-dataset differences are evidence about portability of the workflow and sensitivity to context, not a pooled clinical effect or proof of biological inconsistency.

Outcome-based downsampling remained a contained robustness analysis. It retained outcome-positive records and sampled outcome-negative records, thereby changing the empirical population and outcome prevalence. Original and downsampled means do not automatically target the same population quantity. Nevertheless, direction was preserved in 55 of 57 matched estimator–dataset–exposure comparisons, which indicates broad sign stability under a substantial sampling perturbation and suggests that many sign patterns were not solely driven by the original mortality imbalance. Every paired mean changed numerically, and two PhysioNet comparisons changed sign. The procedure is not formal transport or reweighting; magnitude changes cannot be read as stronger or weaker original-population effects, and directional preservation does not validate causal identification.

11.1.6 Relationship to Prior Work and Methodological Implications

The contribution lies in connecting layers that are often evaluated separately. First, LLM-assisted elicitation converts unstructured clinical and causal reasoning into explicit proposals that can be selected, source-encoded, and audited. Second, a shared proxy-state interface links irregular-time-series prediction and observational effect-estimation workflows. Third, explicit separation of causal and split-aware predictive data contracts prevents silent identifier, split, and cohort conflation. Fourth, source-coded DAGs make adjustment choices inspectable even when their scientific correctness remains an assumption. Fifth, multi-estimator comparison exposes both broad directional agreement and important exceptions such as PhysioNet shock. Matching, sensitivity, and permutation diagnostics reveal support and provenance limitations that a single effect table would conceal, while result manifests and validated source tables materially improve numerical traceability in a complex workflow.

These implications align with the wider emphasis on explicit intervention definitions and target-trial thinking in observational research [8, 9]. They also address difficulties identified in observational ICU causal analysis and machine-learning HTE research: defining the target question, controlling bias, validating heterogeneity, handling covariate shift, and distinguishing predictive variation from actionable treatment-effect variation [18, 19, 30, 31, 32]. The literature supplies methodological context; it does not validate this project's proxy labels, graphs, estimates, or local execution. Accordingly, the framework should be described as an auditable research workflow rather than a generally validated clinical methodology.

11.1.7 Clinical Meaning and Appropriate Use

The proxy states summarize clinically inspired patterns and are analytically useful for organizing retrospective evidence. They may support hypothesis generation and help identify proxy-state contrasts that merit targeted clinical and causal investigation. High prediction performance against rule labels does not establish clinical accuracy, however, and a positive adjusted proxy-state contrast does not imply that clinicians should attempt to manipulate the proxy directly. The system did not undergo prospective validation, external clinical validation, or clinical impact analysis. Its causal quantities are not ready for bedside decision support, and no patient-level treatment recommendation follows from the results.

SRQ-7—principal limitations. The main limitations arise from proxy-label construct validity and measurement error; ill-defined or non-manipulable exposures; unvalidated DAG assumptions and possible unmeasured confounding; temporal ambiguity; incomplete empirical support; model dependence; outcome-class downsampling; dataset heterogeneity; incomplete final-run provenance; and the absence of clinical, fairness, and deployment validation. The answer is therefore *empirically supported with substantial unresolved boundaries*. These limitations determine the strength of every preceding answer and motivate the validity analysis below.

Table 11.1: Evidence-based answers to the research questions.

Question	Evidence-based answer	Strength of answer	Principal boundary
Main RQ	An auditable workflow links LLM-assisted design, irregular measurements, deterministic proxy construction, deep prediction, algorithmic aggregation, project DAGs, and adjusted analyses across both datasets.	Implementation supported	Clinical and causal meaning remains assumption dependent.
SRQ-1	Separate causal and split-aware predictive contracts are connected through normalized identifiers, routing, and explicit export schemas.	Implementation supported	Producing hashes, commands, and split lineage are incomplete.
SRQ-2	Prompt, source, and generated artifacts trace selected LLM-assisted proxy, rule, and DAG designs at schema level.	Partially traceable; unresolved clinically	Human selection, line-level correspondence, chart adjudication, and graph review are incomplete.
SRQ-3	The included model families completed the proxy-label task; the leading family differed by dataset.	Empirically supported with qualifications	No uncertainty or paired superiority tests; target validity and provenance are incomplete.
SRQ-4	Exports are normalized to probability/binary pairs; final-run votes combine one rule source with four prediction sources.	Implementation supported	Voting is not clinical consensus and can preserve shared error.
SRQ-5	Source-coded DAGs yield exposure-specific observed adjustment sets.	Conditional on assumptions	Graph correctness, temporal ordering, and identifiability are unresolved.

Question	Evidence-based answer	Strength of answer	Principal boundary
SRQ-6	Primary, secondary, exploratory, matching, and diagnostic analyses show broad directional triangulation with explicit exceptions.	Empirically supported with qualifications	Agreement is not estimator equivalence or causal validation.
SRQ-7	Construct, identification, support, modeling, transport, provenance, and clinical-use limitations materially bound all findings.	Partially supported	Several limitations require new clinical review, provenance, or validation evidence.

11.2 Limitations and Threats to Validity

11.2.1 Construct and Measurement Validity

The principal construct limitation is that clinically inspired rules approximate conditions but were not evaluated against chart-adjudicated diagnoses. Available evidence consists of active rule code, binary artifact schemas, and limited descriptive validation summaries. This makes the constructs transparent but does not establish their sensitivity, specificity, or equivalence across datasets. Prediction can add classification error, and the mixed-source majority vote can preserve shared systematic error among models trained on the same rules. Existing mitigations are deterministic definitions, source-code traceability, dataset-specific naming, and explicit proxy terminology; the residual risk remains high because clinical review and reference-standard validation are absent.

Binary exposure states also compress severity, duration, and timing. A positive tag may combine clinically distinct trajectories, while a negative tag can reflect missing measurement rather than absence of a condition. Proxy onset relative to covariates and mortality is not fully established, so exposure misclassification and temporal ambiguity can bias both prediction and adjusted analyses. The processed outcome, `in_hospital_mortality`, is consistently required by the software, but its producing-artifact provenance is incomplete, and no competing-risk or post-discharge mortality analysis was performed.

The observation process is itself informative. Measurement frequency may reflect clinical concern, resource use, or care pathways; models that encode masks or time gaps can exploit this signal without removing its causal implications. Preprocessing and missingness-aware representation are therefore not equivalent to adjustment for the decision to measure. This concern can affect proxy construction, predictive comparisons, and effect estimation simultaneously.

11.2.2 Causal Identification and Estimand Validity

These are the central constraints on causal interpretation. Illness-state proxies are not necessarily well-defined interventions, so consistency and the meaning of contrasting exposure states may be difficult to specify. No target-trial emulation was performed, and the pipeline does not implement a method for time-varying treatment–confounder feedback. Exchangeability depends on a project-specified DAG, correct temporal ordering, and measured variables; topology alone cannot establish patient-record timing. Some archived observed adjustment sets are empty, every checked identifiability field remains unresolved, and unmeasured or misspecified confounding remains possible.

Positivity was assumed rather than established. Matching availability, failures, pair counts, and distances provide indirect evidence, but the archive lacks dedicated propensity distributions, common-support plots, and pre/post-match balance summaries. Interference is also assumed, not tested, and repeated admissions or shared ICU processes may complicate unit-level independence.

Estimand language remains deliberately restricted. The mean model-estimated CATE over the analyzed sample is an aggregate of model outputs, not automatically a population ATE. The descriptive matched-pair outcome difference is not automatically an ATT. Normalized CATE was omitted because scaling by the sample outcome rate is unstable across populations and does not create a risk ratio or another standard clinical estimand. These boundaries mitigate overstatement but do not resolve the underlying intervention-definition and identification problems.

11.2.3 Statistical, Modeling, and Support Limitations

Many displayed main estimates lack uncertainty intervals, no formal multiple-comparison strategy was documented, and this thesis did not add significance tests. Absence of an interval cannot be interpreted as either significance or insignificance. Model selection, nuisance-model adequacy, regularization, hyperparameters, and cross-fitting behavior remain assumptions whose impact was not exhaustively tested. CausalForestDML, LinearDML, and CausalPFN do not provide interchangeable uncertainty, and directional agreement can coexist with material magnitude and ordering differences.

Support evidence is incomplete and representation dependent. Some matching runs failed, some successful matches required larger Hamming distances, and some carried insufficient-pair warnings. Sensitivity outputs were partial or failed for several exposure families; reconstructed or fallback diagnostics have different status from estimator-native outputs. Permutation aggregates are limited sanity checks rather than formal inferential tests. CausalPFN lacks the corresponding archived downstream diagnostic family. The multi-estimator and diagnostic hierarchy exposes these risks, but it cannot remove them.

11.2.4 External Validity and Generalizability

MIMIC-III and PhysioNet differ in size, era, variables, measurement practices, and patient composition. Their proxy ontologies and DAGs are dataset specific, and nominally similar states are not necessarily equivalent. Numerical estimates were therefore not pooled. Operation of the workflow on both sources is positive evidence of engineering and methodological portability, especially given the inclusion of several predictive and causal model families. It is not evidence that the estimates transport to other hospitals, countries, care practices, EHR systems, or time periods.

No prospective validation, independent external clinical validation, or formal transport analysis was performed. Subgroup performance and effect stability by age, gender, ICU type, or other characteristics were not evaluated as a fairness study. Consequently, the analysis does not establish clinical generalizability or subgroup equity.

11.2.5 Reproducibility and Provenance

The validated result manifest, source tables, source paths, and checksums provide strong numerical-transcription support for the values reported in Chapter 10. They also improve artifact traceability by distinguishing canonical sources, duplicates, and excluded artifacts. This is not the same as computational rerun reproducibility. The `final-results/` archive is ignored or untracked in the main repository; exact numbered causal configurations are unavailable; result-generating source versions and copy histories are incomplete; and the nested causal repository contains local modifications.

Predictive split provenance, checkpoint-to-export mapping, and copied-output lineage remain incomplete. Raw and processed datasets are external, exact processed-pickle hashes and producing commands are unavailable, and some paths are absolute to the producing machine. Final software, environment, and hardware records are also incomplete. Thus the available records support archived numerical transcription and materially improve artifact traceability, while a clean-checkout rerun is not currently demonstrated. Scientific reproducibility—replication of findings under independently reconstructed data, definitions, and analysis—is weaker still and requires new evidence rather than additional formatting of the archive.

11.2.6 LLM-Assisted Design Layer

The LLM-assisted design layer contributed structured proposals for proxy concepts, binary decision rules, missingness considerations, and dataset-specific DAGs. Potential advantages are the systematic elicitation of domain concepts, scalable generation of candidate proxy definitions and rules, parallel dataset-specific proposals, explicit prompt artifacts that improve design provenance, and conversion of otherwise unstructured reasoning into inspectable source-coded artifacts. These advantages concern organization, traceability, and design productivity; the thesis did not compare them with clinician-only construction or measure whether LLM assistance improved clinical or causal correctness.

The main design risks include hallucinated or unsupported clinical assertions; sensitivity to prompt wording, supplied context, model version, and hosted-model updates; unknown system, browsing, and decoding settings; incomplete reproducibility of hosted-model behavior; and possible dependence on hidden training data. Project selection can introduce anchoring and confirmation bias, especially when plausible generated explanations are mistaken for independent evidence. The prompt record does not establish complete accepted/rejected decisions, and only high-level rather than line-by-line prompt-output-to-code traceability is available.

Validation gaps remain substantial. There is no documented clinician-panel adjudication of every proxy definition or graph edge, no chart-level proxy validation, no comparison of LLM-assisted with clinician-only or hybrid construction, no replication with another LLM family, and no prompt-perturbation or model-version sensitivity experiment. Error can therefore propagate from an unsupported design proposal into a deterministic label, from that label into a prediction and aggregate, and from a misspecified exposure or graph into adjustment and effect estimation. Preserved prompts and explicit source-code authority make the chain inspectable but do not validate it.

Predictive agreement with LLM-assisted rule-derived labels validates neither the clinical construct nor the LLM-generated design. Sensitivity and permutation diagnostics do not validate the proxy ontology or DAG. No claim is made that an LLM provides authoritative clinical judgment, discovered the true causal graph, or can replace a clinician or causal expert.

11.2.7 Ethical and Clinical-Deployment Considerations

Misclassification can misrepresent a record's clinical condition, amplify measurement bias, and generate misleading exposure contrasts. Groups with different testing intensity or access to measurements may be affected disproportionately. Automation bias is therefore a direct risk: users could overread model probabilities as diagnoses, proxy-state rankings as clinical priorities, adjusted estimates as treatment recommendations, or estimator agreement as causal certainty.

No dedicated fairness audit was performed. The presence of age, gender, ICU type, or other variables in an adjustment set does not constitute fairness validation, and subgroup predictive performance and effect stability remain untested. The repository does not establish the project-specific institutional approval, consent or waiver procedure, data-use agreement, or governance determination. These facts must be supplied from authoritative institutional records and must not be inferred from dataset access, de-identification, or repository contents.

The deployment boundary is unambiguous: *the system is a retrospective research workflow and is not suitable for patient-level clinical deployment or treatment recommendation in its current form.* No prospective safety, workflow, impact, or human-factors evaluation was performed. This limitation is not cured by predictive accuracy, estimator agreement, or artifact traceability.

Table 11.2: Study-specific limitations and threats to validity.

Validity domain	Repository-specific limitation	Potential impact	Existing mitigation	Residual risk
Proxy construct validity	Rule-defined states lack chart-adjudicated reference labels.	Construct error can distort prediction targets and exposure contrasts.	Explicit rules, proxy terminology, and source traceability.	High; clinical review and validation remain absent.
Prediction-label validity	Models learn rule-derived targets and voters may share error.	Performance can be high against an imperfect label source.	Separate rule, prediction, and vote terminology.	High; voting cannot remove correlated systematic error.
Exposure definition	Binary illness-state proxies are not necessarily manipulable interventions.	Consistency and causal meaning may be ill defined.	Exposure wording and no treatment recommendation.	High; intervention definition remains unresolved.
Temporal ordering	Proxy, covariate, and outcome timing is not established by graph topology.	Reverse ordering or post-exposure adjustment can bias estimates.	DAG descendant filtering and explicit caveats.	High; record-level timing needs validation.
DAG validity	Project graphs were not learned or empirically validated.	Misspecified arrows can select inadequate adjustment sets.	Source-coded transparency and prompt/code traceability.	High; expert review is incomplete.
Unmeasured confounding	Available variables may not block real-world backdoor paths.	Adjusted estimates may retain bias.	DML adjustment and layered sensitivity diagnostics.	High; sensitivity does not eliminate omitted-variable bias.
Overlap and support	Dedicated propensity and balance diagnostics are absent; matching failures occur.	Weakly supported contrasts may be unstable or nontransportable.	Pair, rate, distance, warning, and failure fields.	High; positivity is unestablished.
Estimator/model dependence	Estimators differ in structure, magnitude, ranking, and diagnostics.	Conclusions may depend on model form or tuning.	Prespecified hierarchy and multi-estimator triangulation.	Medium–high; agreement is mainly directional.
Uncertainty and comparisons	Intervals are unavailable for many displayed means; no multiplicity plan exists.	Precision and false-discovery risk cannot be fully assessed.	No new significance claims or tests.	High for exposure-specific inference.
Outcome downsampling	Sampling changes outcome prevalence and the empirical population.	Means no longer target an automatically identical population quantity.	Robustness-only role, no pooling, explicit sign-change report.	Medium–high; some conclusions are sampling sensitive.
Cross-dataset transport	Datasets differ in era, size, variables, proxies, DAGs, and populations.	Local patterns may not transport or be construct equivalent.	Dataset-specific reporting and no numerical pooling.	High; no external transport validation.
Missing result configurations	Numbered producing causal configs are unavailable.	Exact reruns and setting verification are blocked.	Run summaries and result manifest.	High for computational reproduction.
Predictive provenance	Split, checkpoint-to-export, and copy lineage are incomplete.	Final export and out-of-sample claims remain qualified.	Schema checks, paired logs, and wrappers.	High until per-export lineage is recovered.
Ignored result archive	`final-results/` is ignored or untracked.	Clean-checkout artifact recovery is not guaranteed.	Checksums, source packet, and canonical-source register.	Medium–high; external archival record is needed.
LLM-assisted design	Prompt/system settings, model sensitivity, accepted/rejected decisions, and line-level code traceability are incomplete.	Hallucination, anchoring, hidden-data dependence, and propagated rule or graph error are possible.	Preserved prompts, explicit source-code authority, and conservative claim boundaries.	High until blinded clinical review, replication, and sensitivity studies are completed.
Clinical validation	No prospective, chart-review, or external clinical validation was performed.	Clinical accuracy and utility are unknown.	Retrospective, hypothesis-generating use boundary.	High; bedside use is inappropriate.

Validity domain	Repository-specific limitation	Potential impact	Existing mitigation	Residual risk
Fairness and deployment	No subgroup fairness, safety, impact, or human-factors audit was performed.	Unequal errors and automation harm may go undetected.	Explicit non-deployment boundary.	High; dedicated evaluation is required.

Chapter 12

Conclusions and Future Work

12.1 Summary of the Study

This thesis addressed the challenge of connecting irregular ICU time-series data to transparent retrospective analysis without concealing the transformations required along the way. The motivating problem is not only that clinical measurements are sparse, asynchronous, and shaped by the measurement process, but also that design assumptions, prediction, and observational analysis can become disconnected when their labels, identifiers, preprocessing contracts, and interpretation boundaries are not explicit. The study therefore developed and evaluated an evidence-tracked workflow across PhysioNet 2012 and MIMIC-III that combines LLM-assisted domain-knowledge construction, deterministic rule-derived proxy labeling, deep proxy-state prediction, normalized and mixed-source aggregated proxy labels, LLM-assisted project-specified DAGs, matching, heterogeneous-effect estimation, and available diagnostics.

The resulting framework is a retrospective research workflow. Its proxy states are rule-derived analytical constructs rather than verified clinical conditions, and its adjusted estimates remain conditional on the measurement, graph, intervention-definition, temporal-ordering, support, and modeling assumptions stated throughout the thesis. This distinction is central to the contribution: the workflow makes its handoffs, evidence sources, and unresolved requirements visible instead of presenting an apparently seamless output as a clinical fact.

12.2 Main Findings and Contributions

The strongest contribution is the integrated, evidence-tracked connection between LLM-assisted proxy and DAG design, irregular-time-series representation, deterministic proxy-label construction, deep predictive modeling, algorithmic aggregation, and downstream observational analysis. The shared proxy-state interface provides a traceable route from selected source-code rules to rule-derived prediction targets, binary predicted voter inputs, a mixed-source vote combining one rule-derived source with four prediction sources, and dataset-specific exposure analyses. Separate causal and split-aware predictive contracts further make identifier, cohort, and export boundaries inspectable. Checked result tables, manifests, checksums, source packets, and explicit deferred issues complement this workflow by improving numerical traceability, even though they do not yet provide a complete clean-checkout reproduction record.

The empirical predictive comparison covered the four learned models STraTS, GRU, GRU-D, and TCN. The leading model differed by dataset: STraTS led the archived proxy-label test metrics in MIMIC-III, whereas GRU-D led them in PhysioNet (Chapter 10, Section 10.2). This result supports a dataset-specific view of representation choice. It does not establish a universal model ordering, statistical superiority, or clinical validity of the rule-derived targets.

In the primary original-cohort analysis, CausalForestDML produced positive mean model-estimated CATE summaries for all nine MIMIC-III exposures and for nine of ten PhysioNet exposures; the PhysioNet shock summary was negative (Chapter 10). These estimates document systematic variation in the constructed proxy-state analyses after the implemented adjustment procedure; they do not establish that a proxy state caused mortality or that modifying it would change an outcome. Matching supplied a descriptive empirical-support baseline, not independent confirmation, and its failures and weak-support flags remain part of the evidence.

The comparison between CausalForestDML and LinearDML adds useful but bounded triangulation: their mean summaries had the same direction in all 19 original-cohort dataset–exposure comparisons, while magnitudes and within-dataset rankings still differed (Chapter 10). Thus agreement is informative about the observed sign pattern but not a guarantee of estimator correctness, equal uncertainty, or valid identification. CausalPFN also agreed in direction with both DML estimators in 18 of 19 comparisons, with PhysioNet shock as the exception (Chapter 10). This is an exploratory complementary finding only: its primary method source is unresolved in the approved corpus, and its sensitivity and permutation stages were intentionally not part of the archived diagnostic workflow.

Together, these results support the thesis's methodological objective: LLM-assisted knowledge construction, deterministic proxy labeling, deep proxy-state prediction, and DAG-guided causal-effect estimation can be connected while preserving dataset-specific definitions, estimator hierarchy, and evidence boundaries. The contribution is the transparent integration and qualification of these components, not proof that the LLM-generated proposals, proxy constructs, graphs, or effect estimates are clinically correct.

12.3 Limitations of the Conclusions

The conclusions are constrained first by construct validity. The proxy states originate in LLM-assisted, project-selected rule designs encoded as deterministic source; they are not chart-adjudicated labels, and the two datasets differ in measured variables, proxy ontologies, and timing. Unsupported design assertions can enter the rules, prediction can add error, and mixed-source majority voting can preserve shared rule and measurement error; a binary tag can compress distinct trajectories, and a negative tag can reflect unavailable measurement rather than absence of a condition. Clinical review, reference-standard evaluation, complete prompt-to-code traceability, and exact support for several rule families remain incomplete.

Second, the observational analyses do not remove identification uncertainty. Illness-state proxy exposures are not necessarily well-defined interventions, the LLM-assisted project-specified DAGs are encoded assumptions rather than empirically established structures, temporal ordering is incomplete, and unmeasured confounding can remain. An LLM-assisted graph is not intrinsically more correct than a manually proposed graph. Matching rates, distances, failures, sensitivity analyses, and permutation checks qualify particular risks but do not establish global overlap, balance, exchangeability, proxy or graph validity, or causal certainty. Estimator agreement must therefore be read alongside estimator disagreement, support gaps, incomplete uncertainty reporting, and the different diagnostic coverage of CausalPFN.

Finally, the present evidence does not establish external clinical utility or evaluate the LLM as a design method against alternatives. The two datasets support application of the workflow in distinct settings but do not justify numerical pooling or transport claims. No external or prospective clinical validation, subgroup fairness analysis, safety study, impact evaluation, patient-level deployment evaluation, clinician-only comparison, alternative-LLM replication, or prompt-sensitivity study was performed. Authoritative institutional ethics and data-governance documentation also remains required. Reproducibility is incomplete: raw and processed data artifacts, exact numbered causal configurations, predictive split and checkpoint lineage, producing source versions, archive-copy history, and full LLM prompt-run and human-review records remain unresolved.

12.4 Future Work

The first priority is clinical and construct validation. Future work should freeze the proxy rules and conduct blinded clinician review of every proxy definition and graph edge, with multiple reviewers and reported inter-reviewer agreement. Rule-derived and predicted states should be compared with chart-adjudicated reference labels, including dataset-specific assessment of timing, missingness, thresholds, and differences among rule-derived labels, individual predicted labels, and mixed-source aggregated proxy labels. A controlled comparison of LLM-assisted, clinician-only, and hybrid rule and DAG construction would test whether the design layer adds value rather than merely plausibility.

The second priority is a stronger observational and measurement design. A future study should specify a target-trial-aligned question with eligibility, time zero, intervention strategies, follow-up, outcome, estimand, and an explicit plan for time-varying confounding. It should add prespecified overlap and balance diagnostics, support-aware estimands, uncertainty and multiplicity procedures, and richer sensitivity analyses. Rule ablation and threshold-sensitivity experiments, formal label-noise modeling, and explicit measurement-error analyses should quantify how proxy construction, prediction, and aggregation affect downstream estimates. DAG review should proceed edge by edge and include alternative-graph and adjustment-set analyses rather than treating one encoded graph as fixed truth.

The third priority is reproducibility and external evaluation. Signed per-run manifests should link raw-data access records, processed-artifact hashes, split identifiers, checkpoints, commands, resolved configurations, software environments, source versions, and archived outputs. LLM design manifests should record complete prompts, system and decoding settings, model versions, follow-up interactions, reviewer decisions, and prompt-output-to-code mappings. Prompt perturbations, model-version sensitivity studies, and replication across multiple LLM families should be performed. Frozen definitions should then be evaluated on temporal and external cohorts and in prospective studies of clinically justified subgroups before any safety or impact study is considered. CausalPFN should remain exploratory until its primary source and implementation version are verified and an appropriate diagnostic and uncertainty plan is available. Authoritative ethics and data-governance wording should also be completed before prospective or deployment-related evaluation.

12.5 Closing Perspective

The thesis shows the value of treating integration itself as an object of evidence. LLM-assisted knowledge construction, irregular ICU records, deterministic proxy labels, deep predictive outputs, graph-guided adjustment, and model-estimated contrasts can be connected in a transparent workflow, but none of those handoffs removes the need for clinical review, careful observational design, diagnostic scrutiny, or reproducible provenance. An LLM cannot replace a clinician or causal expert on the evidence presented here. The appropriate conclusion is therefore measured: transparent integration can make retrospective evidence more inspectable and more useful for hypothesis generation, while the boundaries that prevent stronger clinical or causal claims remain visible and actionable for future research.

Appendix A

Reproducibility and Evidence Boundary

This appendix records the scope of the reproducibility evidence available for the thesis. The accompanying project archive contains checked result tables, a result manifest, checksums, a source packet, and a decision register. The compact reproducibility package indexes the causal-run, predictive-run, data, environment, source-version, archive-copy, and unresolved-gap records without duplicating the detailed manifest. Together, those records support the numerical traceability of the displayed results and preserve distinctions between admitted evidence, excluded artifacts, and unresolved limitations.

The available records do not establish a complete clean-checkout reproduction. The exact raw and processed data lineage, numbered producing configurations, predictive split and checkpoint-to-export lineage, producing source versions, and archive-copy history are incomplete. A repository requirements file or current wrapper can describe an implementation interface, but is not evidence that it was the exact environment or command that produced an archived result.

The two external ICU datasets also remain subject to their own access and governance conditions. The repository does not establish institutional approval numbers, an ethics-board determination, consent basis, or the final data-governance wording required for submission or any prospective use. These gaps do not negate the checked numerical record, but they limit reproducibility, administrative compliance, and any stronger clinical interpretation.

For these reasons, the thesis treats the available materials as an evidence and provenance boundary: they make the reported workflow more inspectable while leaving the recovery of complete per-run lineage, qualified clinical review, and authoritative institutional documentation as necessary future work.

Bibliography

[1] Chenxi Sun et al. "A Review of Deep Learning Methods for Irregularly Sampled Medical Time Series Data." In: *Health Data Science* (2026), hds.0456. DOI: 10.34133/hds.0456. URL: https://doi.org/10.34133/hds.0456.

[2] Zachary C. Lipton, David C. Kale, and Randall Wetzel. "Directly Modeling Missing Data in Sequences with RNNs: Improved Classification of Clinical Time Series." In: *Proceedings of the 1st Machine Learning for Healthcare Conference*. Ed. by Finale Doshi-Velez et al. Vol. 56. Proceedings of Machine Learning Research. Northeastern University, Boston, MA, USA: PMLR, 2016, pp. 253–270. arXiv: 1606.04130. URL: https://proceedings.mlr.press/v56/Lipton16.html.

[3] Sindhu Tipirneni and Chandan K. Reddy. "Self-Supervised Transformer for Sparse and Irregularly Sampled Multivariate Clinical Time-Series." In: *ACM Transactions on Knowledge Discovery from Data* (2022). DOI: 10.1145/3516367. URL: https://doi.org/10.1145/3516367.

[4] Zhengping Che et al. "Recurrent Neural Networks for Multivariate Time Series with Missing Values." In: *Scientific Reports* 8 (2018), p. 6085. DOI: 10.1038/s41598-018-24271-9. URL: https://doi.org/10.1038/s41598-018-24271-9.

[5] Juan M. Banda et al. "Advances in Electronic Phenotyping: From Rule-Based Definitions to Machine Learning Models." In: *Annual Review of Biomedical Data Science* 1 (2018), pp. 53–68. DOI: 10.1146/annurev-biodatasci-080917-013315. URL: https://doi.org/10.1146/annurev-biodatasci-080917-013315.

[6] Alexander Ratner et al. "Snorkel: rapid training data creation with weak supervision." In: *The VLDB Journal* 29.2–3 (2020), pp. 709–730. DOI: 10.1007/s00778-019-00552-1. URL: https://doi.org/10.1007/s00778-019-00552-1.

[7] Judea Pearl. "Causal Diagrams for Empirical Research." In: *Biometrika* 82.4 (1995), pp. 669–688. DOI: 10.1093/biomet/82.4.669. URL: https://doi.org/10.1093/biomet/82.4.669.

[8] Miguel A. Hernan and James M. Robins. "Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available." In: *American Journal of Epidemiology* 183.8 (2016), pp. 758–764. DOI: 10.1093/aje/kwv254. URL: https://doi.org/10.1093/aje/kwv254.

[9] Miguel A. Hernan and Sarah L. Taubman. "Does Obesity Shorten Life? The Importance of Well-Defined Interventions to Answer Causal Questions." In: *International Journal of Obesity* 32.Suppl 3 (2008). Author/academic repost used because publisher PDF was not directly downloadable in this environment., S8–S14. DOI: 10.1038/ijo.2008.82. URL: https://doi.org/10.1038/ijo.2008.82.

[10] Ikaro Silva et al. "Predicting In-Hospital Mortality of ICU Patients: The PhysioNet/Computing in Cardiology Challenge 2012." In: *Computing in Cardiology*. Vol. 39. 2012, pp. 245–248. URL: https://physionet.org/content/challenge-2012/1.0.0/.

[11] Alistair E. W. Johnson et al. "MIMIC-III, a freely accessible critical care database." In: *Scientific Data* 3 (2016), p. 160035. DOI: 10.1038/sdata.2016.35. URL: https://doi.org/10.1038/sdata.2016.35.

[12] Alexander J. Ratner et al. "Data Programming: Creating Large Training Sets, Quickly." In: *Advances in Neural Information Processing Systems*. Ed. by Daniel D. Lee et al. Vol. 29. The local PDF is the extended arXiv v3 copy revised 8 January 2017. Curran Associates, Inc., 2016, pp. 3567–3575. arXiv: 1605.07723 [stat.ML]. URL: https://proceedings.neurips.cc/paper/2016/hash/6709e8d64a5f47269ed5cea9f625f7ab-Abstract.html.

[13] Karan Singhal et al. "Large language models encode clinical knowledge." In: *Nature* 620.7972 (2023). Corrected open-access publisher version; the publisher correction amended reference numbering., pp. 172–180. DOI: 10.1038/s41586-023-06291-2. URL: https://doi.org/10.1038/s41586-023-06291-2.

[14] Victor-Alexandru Darvariu, Stephen Hailes, and Mirco Musolesi. *Large Language Models are Effective Priors for Causal Graph Discovery*. Version 1; preprint under review. 2024. DOI: 10.48550/arXiv.2405.13551. arXiv: 2405.13551 [cs.LG]. URL: https://arxiv.org/abs/2405.13551.

[15] Victor Chernozhukov et al. "Double/Debiased Machine Learning for Treatment and Structural Parameters." In: *The Econometrics Journal* 21.1 (2018), pp. C1–C68. DOI: 10.1111/ectj.12097. URL: https://doi.org/10.1111/ectj.12097.

[16] Stefan Wager and Susan Athey. "Estimation and Inference of Heterogeneous Treatment Effects using Random Forests." In: *Journal of the American Statistical Association* 113.523 (2018), pp. 1228–1242. DOI: 10.1080/01621459.2017.1319839. URL: https://doi.org/10.1080/01621459.2017.1319839.

[17] Susan Athey, Julie Tibshirani, and Stefan Wager. "Generalized Random Forests." In: *The Annals of Statistics* 47.2 (2019), pp. 1148–1178. DOI: 10.1214/18-AOS1709. URL: https://doi.org/10.1214/18-AOS1709.

[18] J. M. Smit et al. "Causal Inference Using Observational Intensive Care Unit Data: A Scoping Review and Recommendations for Future Practice." In: *npj Digital Medicine* 6.1 (2023), p. 221. DOI: 10.1038/s41746-023-00961-1. URL: https://doi.org/10.1038/s41746-023-00961-1.

[19] Ioana Bica et al. "From Real-World Patient Data to Individualized Treatment Effects Using Machine Learning: Current and Future Methods to Address Underlying Challenges." In: *Clinical Pharmacology & Therapeutics* 109.1 (2021), pp. 87–100. DOI: 10.1002/cpt.1907. URL: https://doi.org/10.1002/cpt.1907.

[20] Hrayr Harutyunyan et al. "Multitask Learning and Benchmarking with Clinical Time Series Data." In: *Scientific Data* 6.1 (2019), p. 96. DOI: 10.1038/s41597-019-0103-9. URL: https://doi.org/10.1038/s41597-019-0103-9.

[21] Kyunghyun Cho et al. "Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation." In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2014, pp. 1724–1734. DOI: 10.3115/v1/D14-1179. arXiv: 1406.1078. URL: https://aclanthology.org/D14-1179/.

[22] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling." In: *arXiv* (2018). DOI: 10.48550/arXiv.1803.01271. arXiv: 1803.01271 [cs.LG]. URL: https://arxiv.org/abs/1803.01271.

[23] Patrick Essay, Jarrod Mosier, and Vignesh Subbian. "Rule-Based Cohort Definitions for Acute Respiratory Failure: Electronic Phenotyping Algorithm." In: *JMIR Medical Informatics* 8.4 (2020), e18402. DOI: 10.2196/18402. URL: https://doi.org/10.2196/18402.

[24] Mervyn Singer et al. "The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3)." In: *JAMA* 315.8 (2016), pp. 801–810. DOI: 10.1001/jama.2016.0287. URL: https://doi.org/10.1001/jama.2016.0287.

[25] Kidney Disease: Improving Global Outcomes Acute Kidney Injury Work Group. *KDIGO Clinical Practice Guideline for Acute Kidney Injury*. Tech. rep. 1. Kidney Disease: Improving Global Outcomes, 2012, pp. 1–138. URL: https://kdigo.org/guidelines/acute-kidney-injury/.

[26] F. B. Taylor Jr. et al. "Towards Definition, Clinical and Laboratory Criteria, and a Scoring System for Disseminated Intravascular Coagulation." In: *Thrombosis and Haemostasis* 86.5 (2001). Thieme publisher page exposed a PDF URL but returned HTML; The Blood Project public educational PDF matched title/authors/year and was used., pp. 1327–1330. DOI: 10.1055/s-0037-1616068. URL: https://doi.org/10.1055/s-0037-1616068.

[27] Jean-Louis Vincent et al. “The SOFA (Sepsis-related Organ Failure Assessment) Score to Describe Organ Dysfunction/Failure.” In: *Intensive Care Medicine* 22.7 (1996). No lawful public PDF was downloaded: Springer/Ovid routes returned HTML rather than a PDF; metadata retained with missing PDF status., pp. 707–710. DOI: 10.1007/BF01709751. URL: https://doi.org/10.1007/BF01709751.

[28] ARDS Definition Task Force et al. “Acute Respiratory Distress Syndrome: The Berlin Definition.” In: *JAMA* 307.23 (2012). No lawful public PDF was downloaded: JAMA/CDN routes returned 403 and Ovid returned HTML rather than a PDF; metadata retained with missing PDF status., pp. 2526–2533. DOI: 10.1001/jama.2012.5669. URL: https://doi.org/10.1001/jama.2012.5669.

[29] Miruna Oprescu et al. “EconML: A Machine Learning Library for Estimating Heterogeneous Treatment Effects.” In: *NeurIPS 2019 Workshop on Causal Machine Learning*. 2019, pp. 1–6. URL: https://tripods.cis.cornell.edu/neurips19_causalml/.

[30] Ilya Lipkovich et al. “Modern Approaches for Evaluating Treatment Effect Heterogeneity from Clinical Trials and Observational Data.” In: *Statistics in Medicine* 43.22 (2024), pp. 4388–4436. DOI: 10.1002/sim.10167. URL: https://doi.org/10.1002/sim.10167.

[31] Alicia Curth et al. “Using Machine Learning to Individualize Treatment Effect Estimation: Challenges and Opportunities.” In: *Clinical Pharmacology & Therapeutics* 115.4 (2024), pp. 710–719. DOI: 10.1002/cpt.3159. URL: https://doi.org/10.1002/cpt.3159.

[32] Theodore J. Iwashyna et al. “Implications of Heterogeneity of Treatment Effect for Reporting and Analysis of Randomized Trials in Critical Care.” In: *American Journal of Respiratory and Critical Care Medicine* 192.9 (2015), pp. 1045–1051. DOI: 10.1164/rccm.201411-2125CP. URL: https://doi.org/10.1164/rccm.201411-2125CP.

[33] Richard K. Crump et al. “Dealing with Limited Overlap in Estimation of Average Treatment Effects.” In: *Biometrika* 96.1 (2009). Duke university author PDF used., pp. 187–199. DOI: 10.1093/biomet/asn055. URL: https://doi.org/10.1093/biomet/asn055.

[34] Carlos Cinelli and Chad Hazlett. “Making Sense of Sensitivity: Extending Omitted Variable Bias.” In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 82.1 (2020). Author PDF used; first page states forthcoming JRSSB version and DOI., pp. 39–67. DOI: 10.1111/rssb.12348. URL: https://doi.org/10.1111/rssb.12348.

[35] Victor Chernozhukov et al. “Long Story Short: Omitted Variable Bias in Causal Machine Learning.” In: *Review of Economics and Statistics* (2026). Final publisher landing page exists, but publisher PDF returned 403/HTML; official arXiv author version was used. DOI: 10.1162/REST.a.1705. arXiv: 2112.13398. URL: https://doi.org/10.1162/REST.a.1705.

[36] Amit Sharma and Emre Kiciman. *DoWhy: An End-to-End Library for Causal Inference*. 2020. DOI: 10.48550/arXiv.2011.04216. arXiv: 2011.04216 [cs.AI]. URL: https://arxiv.org/abs/2011.04216.

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CliniCause: מסגרת מונחית מודל שפה גדול לחיזוי מצבי פרוקסי בלמידה עמוקה ולאמידת השפעות סיבתיות בסדרות זמן רב־משתנויות ולא־סדירות מטיפול נמרץ

מוגש לשם קבלת תואר "מגיסטר" בפקולטה להנדסה

קובי קנזי

בהנחיית: ד"ר דן וילנצ'יק